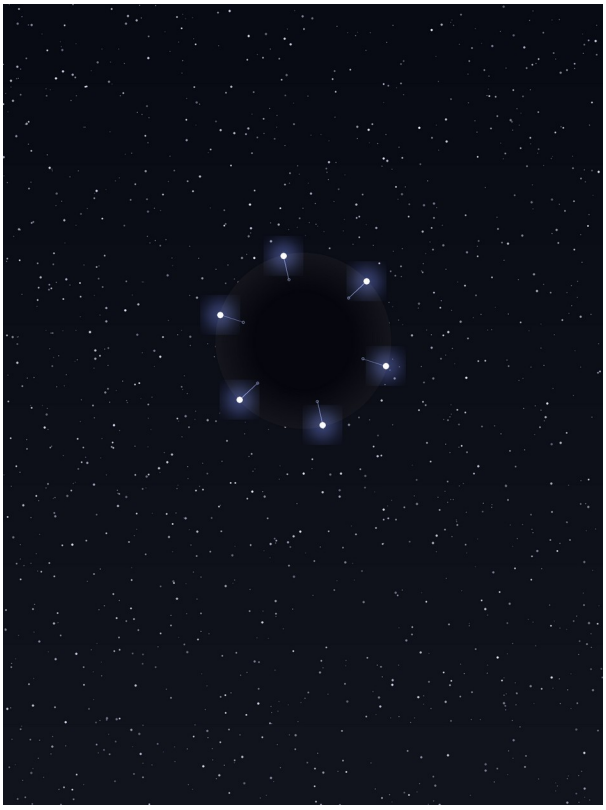


8 BILLION ASTRONAUTS

A novel

Dave L. Otteman

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Note on the Science

PROLOGUE

The Missing Stars

DIRECTORY LISTING, OUTER-SYSTEM ASTROMETRY ARCHIVE

NONCRITICAL SYSTEMATICS/

Created three years ago. Description: “Residuals within tolerance. Revisit if funded.”

Last opened: never.

T–6,330 DAYS

Dr. Amara Okafor noticed the end of the world because six stars were in the wrong place.

At first she blamed the software.

The outer-system astrometry network had received an update that afternoon, and its release notes contained the phrase improved coordinate handling. Amara had learned to distrust improvements delivered on a Friday.

She restored the previous version. The stars remained displaced.

It was 2:14 in the morning in Geneva. Rain moved down the windows of the International Gravitational Modeling Center, dividing the reflected room into narrow, trembling pieces. Her daughter had sent a photograph six hours earlier. Maya beside a cake, sixteen candles crowded into the icing. Amara had replied with a heart and a promise to call.

She opened the photograph, looked at it, and closed it again.

The shifts were measured in arcseconds. Each star had moved away from an apparently empty region.

Amara asked for older observations.

The network returned eleven years of them.

There were reasons the pattern had gone unnoticed. The first measurements came from instruments built for different purposes, using different reference catalogs. Some stars had been flagged as unreliable. Two had been discarded because they seemed to wander.

Together, they described a disturbance moving across the background sky. Unexplained planetary timing residuals supported it.

She fitted a lensing model. The distance remained poorly constrained. The mass changed when she changed the distance. She added the simultaneous observations from the Neptune relay and waited while the solver reconciled the widely separated viewpoints.

The answer made her check the units. Then the observatory clocks. Then whether someone had entered the relay’s position in kilometers instead of meters.

Meters. The clocks agreed. The units were the units.

At 2:41 she called Luis Chen.

“Tell me it’s the update,” he said. His voice had the flattened patience of a man who had been asleep ninety seconds earlier.

“I ran the rollback. It isn’t the update.”

“Calibration flag on the Neptune relay.”

“Cleared.”

“Clock drift.”

“Checked against the pulsar net.”

“Catalog epoch.”

“Same epoch in both reductions.”

“Then it’s something you did.”

“Probably. Come and find it.”

He was quiet for almost a minute. She heard a chair scrape, then water running, then the small percussion of someone looking for shoes in the dark. When he spoke again, he told her to send the raw observations rather than her fit.

“Already sent.”

“And the rejected measurements.”

“Those too.”

“Have you told anyone else?”

“No.”

“Good. Don’t give this a name yet.”

He arrived carrying a wet coat and the toothbrush he had forgotten to put down. He placed both on a chair, studied her display, and began trying to destroy the result.

Luis wore the expression of a man prepared to dismantle an exciting claim before breakfast.

Amara knew that expression well. They had spent enough nights together in the center to recognize the difference between interest and hope. He opened the data and asked for the observations she had excluded.

“They’re flagged.”

“Show me anyway.”

The excluded set contained several obvious problems and two that were less obvious. They recovered the raw frames and compared them with the processed positions. One source had been blended with a faint neighbor. Another had a detector artifact close enough to affect the fit. Correcting those reduced the anomaly without removing it.

Amara felt disappointed by the reduction and was immediately suspicious of herself. She did not want a discovery of this kind. She still wanted the pattern she had noticed to mean something, because hours of attention had already made it feel partly hers.

Luis looked up. “We should get someone who doesn’t care whether we’re right.”

“Do you?”

“I care whether you’re wrong. That’s not the same thing.”

They called a colleague at another center and supplied the data without the preferred interpretation in the first message. The colleague asked for calibration records and a larger surrounding field. While they waited, Luis found coffee and Amara finally listened to the voice message Maya had sent with the birthday photograph.

It lasted twenty-three seconds. Maya described the cake, complained that the candles had nearly set a decoration alight, and said her grandfather had saved Amara a piece. There was laughter in the background. At the end, Maya said, “Call when you can.”

Amara checked the time in Nairobi. Too late to call without waking her, too early for morning. The familiar convenience of that fact made her ashamed. She wrote a message promising to call after school and set an alarm so the promise would have a physical consequence.

Then the independent reduction arrived.

The colleague had found the same displacement pattern and supplied three alternative explanations, ranked by how readily they could be tested. Amara read them aloud. Luis began with the one most likely to make the problem ordinary.

For the next hour, that possibility held.

* * *

They turned from the stars to the things between the stars and their screens.

He disconnected the new reduction service, found a local copy of the older catalog, and asked Amara to explain every correction she had applied. She started to tell him which ones could not matter. He shook his head.

“Those first.”

They worked at opposite ends of the long desk. The building’s heating had switched to its overnight setting. Amara kept her coat across her knees and watched the numbers on his display through the reflection in the dark window.

The first alternative explanation was almost comforting. Three of the suspect stars lay near the boundary between two reference fields. A small rotation in the coordinate transformation could displace them in approximately the observed directions. Amara found a maintenance

notice from the week the boundary had been changed. For a moment she could see the shape of the morning ahead: an embarrassing message to the network team, a few hours’ sleep, a call home with nothing worse to explain than another missed birthday.

Luis applied the rotation to the control stars.

Half of them moved into worse agreement.

“Wrong pattern,” he said.

“Only if the catalog errors are independent.”

He gave her the keyboard. She fitted a correlated error. The six residuals diminished. So did several others. She felt relief rising before she reached the observations from the Neptune relay.

Those did not fit.

The receiver had viewed the same field from a different place. An error painted onto a catalog could be made to resemble almost anything from one viewpoint. The second viewpoint demanded that the distortion obey geometry.

She restored the lens model.

Luis turned his chair so that he could see her face. “You were disappointed.”

“I was hoping to be incompetent.”

“There’s still time.”

She smiled, then leaned forward again.

They called the relay’s instrument scientist, a woman named Beatriz who answered from a kitchen in Valparaíso. She listened while taking a saucepan off the heat. Behind her, someone asked whether dinner was ready. She covered the microphone, replied, then brought the screen closer.

“Tell me you haven’t trusted the quick-look positions.”

Amara felt the room change.

“Which positions?”

Beatriz opened the archive. During a recent maneuver, the relay had operated briefly with an estimated orientation. The preliminary products had been marked unsuitable for precision work. A transfer between archives had dropped the warning flag from one batch.

For eleven minutes, it seemed possible that the whole extraordinary mass depended on an ordinary administrative failure.

Amara watched Beatriz locate the corrected files. Luis folded his arms, impatient with himself for hoping. The saucepan’s lid rattled somewhere beyond the microphone. Someone in the kitchen laughed at something they could not hear.

The corrected positions arrived.

The fitted distance changed. The inferred mass changed with it.

The dark object remained.

Beatriz stopped speaking when she saw the solution.

“Where did this come from?”

“The rejected stars,” Amara said.

“Have you checked it against the navigation residuals?”

“Some of them.”

“All of them. There are older disagreements nobody resolved. We were told the dynamical team owned those.”

The older disagreements occupied a folder with a reassuring name: NONCRITICAL SYSTEMATICS. Its creation date was almost three years earlier. Amara recognized one of the contributors. They had shared a panel at a conference; he had spent dinner complaining that nobody would fund a long investigation of a discrepancy that did not yet interfere with operations.

She had advised him to narrow the proposal.

Now his residuals sat beside her stars.

She asked Beatriz for permission to keep the connection open while they worked. The instrument scientist moved her meal to the desk and stayed.

At four-thirty, Luis found another problem. Amara had included one observation twice under different archive identifiers. It had made the fit look more certain than it was. She removed it and watched the uncertainty widen.

“That belongs in the log,” he said.

“I know.”

“I mean the first report. Not just our log.”

She looked up sharply, then understood why he had said it. If they were right, every early error would eventually look to someone like a secret. If they were wrong, the errors would matter for more immediate reasons.

She wrote the duplication into the report while he watched.

The mass was not the most frightening part. The nearest black holes anyone discussed publicly were names attached to distant images and careful statistical statements. This object’s location belonged on the same practical maps as the outer settlements. The available evidence still allowed a range of paths. Some looked less destructive than others. None resembled a problem their existing emergency procedures had anticipated.

At five, the duty director answered his third call. He asked whether they could wait for the regular review meeting.

“What would you do before then?” Amara asked.

“Send it to the external panel. Check the instruments. Establish whether this is a software issue.”

“We’re doing those things.”

“Then why are you waking me?”

She almost said because she needed someone else to carry it.

Instead she described the preliminary timescale and asked him to authorize priority observations from the network’s remaining stations.

That meant interrupting other work. One facility was measuring an exoplanet transit that would not recur for years. Another had an experiment occupying its calibration system. Their operators were entitled to ask why.

“How many people would this involve?” the director asked.

“Enough that you should assume it won’t stay private.”

He took a long breath. “And if you are wrong?”

“Then we will explain why.”

She heard the boldness in the answer and disliked it. Being wrong would cost observing time, money, reputations. Being prematurely persuasive could cost more. There was no instruction that told her exactly when a suspected disaster had become real enough to interrupt everyone else’s life.

The director authorized three stations. He asked her to use neutral language in the requests.

Amara wrote unexplained coherent astrometric residuals. She did not write end of the world. She did not yet know whether that was what she had found.

When the authorization cleared, she stood too quickly and had to put a hand on the desk. Luis pushed a packet of crackers toward her without looking away from his screen.

“Eat those before you give another answer.”

They tasted of salt and the cardboard in which they had spent most of the year. She ate them anyway.

Across the connection, Beatriz was telling her family she would be late to bed. Amara could hear the practiced patience in the reply. She thought of Maya’s candles, of the small dark marks the extinguished wicks would have left in the icing. Her daughter had probably saved a piece for her before remembering she was not coming home.

She put the phone facedown. The gesture was small, and it was a choice.

By dawn they had found a calibration error in one telescope.

Correcting it made the fit better.

By noon, independent teams had confirmed the moving lens. By the following evening, observations from three widely separated receivers had narrowed the distance enough to make the mass difficult to avoid.

Approximately four million Suns.

No optical counterpart. No bright accretion disk. An object carrying the mass of the black hole at the center of the Milky Way, traveling through the local stellar neighborhood at roughly three thousand kilometers per second.

Luis rubbed the indentation his glasses had left on his nose.

“A merger kick?”

“Possibly. An old one.”

“Something threw it out of a galaxy.”

“Something could have.”

Amara kept the distinction because it was the only part of the conversation she could control. Their instruments measured light bent around a dark mass. The history of that mass was another problem.

She displayed the encounter calculation.

Luis leaned closer.

The black hole would not strike the Sun. The nominal closest separation was nearly four hundred times the distance between Earth and the Sun. On the scale of their familiar maps, it seemed almost reassuring.

Then she added the planetary trajectories.

The reassuring distance ceased to help.

“When?” he asked.

“March 2126. The nineteenth, in the current solution.”

Seventeen years and four months.

Outside the window, a maintenance worker was clearing leaves from a drain. He leaned on his broom while a delivery vehicle passed, then bent to his task again.

Amara watched him for longer than she meant to.

Her phone showed another message from Maya.

You forgot.

She called her daughter before returning to the calculation.

CHAPTERONE

Seventeen Years

INTERNATIONAL GRAVITATIONAL MODELING CENTER, PUBLIC FAQ, VERSION 1.0

Q: Is the black hole going to hit Earth?

A: No. The compact astronomical object is not expected to hit anything.

Q: Then what is the problem?

A: Please see FAQ version 1.1, in preparation.

T–6,318 DAYS

The leak arrived as a screenshot with three annotations. Two were wrong. The third was the date.

Within an hour, March 19 had become a phrase that meant something other than a day in spring.

Amara watched the first public briefing from a room behind the auditorium. A communications adviser had asked her to replace black hole with compact astronomical object. She had refused, then agreed not to say four million Suns until the second slide.

The secretary-general reached the second slide in forty-three seconds.

“There is presently no evidence that the object will collide with Earth.”

Luis closed his eyes.

Amara went to the podium when they called her. The lights prevented her from seeing most of the audience. That helped until someone asked whether their children would survive.

“We do not yet know how Earth’s orbit will change,” she said. “We know the Sun’s path will change substantially. Earth may remain bound to it. That is a different question from whether the resulting orbit will support the lives we have now.”

“Does that mean yes?”

“It means we cannot honestly give you yes or no.”

She heard how inadequate it sounded.

The models spread faster than the explanations. Earth frozen and wandering. Earth passing too close to the Sun. Jupiter departing with its moons. Every animation represented assumptions; every repost removed them.

Before anyone knew the date, Maya had been worried about a broken projector.

She and Leila Nasser were supposed to present their school project that morning: a proposal to collect rain from the assembly hall roof and use it in a neglected garden behind the

classrooms. The idea was not original. Leila said originality was an overrated quality in plumbing.

They had measured the roof themselves because the plans in the school office described a building that had been extended twice since they were drawn.

Maya had spent the previous evening making the diagram move.

Leila had spent it persuading the caretaker that they were not trying to get him assigned another job.

Now the diagram occupied one corner of the projection screen, distorted into a narrow strip, while an emergency news window covered the rest.

Their teacher tried to close the window. It reopened.

Somebody at the back said Maya's mother was on it.

For a few seconds Maya felt the familiar annoyance of being identified that way. Then she saw Amara standing beside a man she recognized from ceremonial broadcasts and knew this was not another science interview.

Leila touched her wrist. "Do you know about this?"

Maya shook her head.

The classroom grew quiet. Outside, a lawn trimmer continued its uneven progress along the path. The teacher turned up the sound, and the trimmer seemed suddenly offensive, a person refusing to notice what everyone else was being told.

There was presently no evidence that the object would collide with Earth.

A boy near the window laughed. "Then what's the problem?"

No one answered because Amara had begun answering a different question. Maya watched her mother choose her words. She knew the little movement of her hand when she wanted to correct an imprecise statement without embarrassing the person who had made it. She had seen it at parent meetings, over restaurant bills, while helping with mathematics homework. Seeing it here made the broadcast feel both more real and impossible.

The teacher paused the feed after the explanation of orbital changes. She asked everyone to stay seated. She said the school would distribute verified information.

"Maya has verified information," the boy said.

"I don't."

"You just don't want to tell us."

Leila turned around. "She found out the same time you did."

"How do you know?"

"Because I was standing next to her."

The teacher sent him to the corridor. That made the situation worse. By lunch, several people believed he had been removed for asking the question nobody was allowed to ask.

Maya ate nothing. Her messages accumulated faster than she could read them. Some asked whether they should leave the city. Others asked whether her mother was safe. One asked if she could get two places on a spaceship. It might have been a joke. She could not tell.

Leila sat beside her with two oranges. She peeled one and put half on Maya's unopened lunch container.

"You don't have to answer."

"If I don't, they'll think I know something."

"They already think that."

Maya tried to call her mother. The connection went directly to a recorded message. She had heard it hundreds of times. For the first time, she wondered who maintained recordings after the people in them were gone.

Leila took the phone gently from her hand and turned it facedown.

"Eat the orange."

They had been friends since an argument over a borrowed ruler when they were seven. Leila had accused Maya of returning the wrong one. Maya had produced a photograph proving which ruler she had borrowed. Leila had found the missing ruler in her own bag and apologized with such evident irritation that Maya had laughed. They had been rescuing each other from small humiliations ever since.

This did not feel small.

"Your project is better without the animation," Leila said.

Maya looked at her.

"The water doesn't need to flash."

She almost laughed. It came out as something rougher, and Leila stopped trying to make it happen.

They remained on the bench until the school announced an early closure.

In Nairobi, someone painted THE END IS COMING across Maya's school wall. She photographed it, then put her phone away without sending the image to her mother.

Her grandfather came to collect her because the buses had stopped running. He was seventy-two and annoyed by the extra journey. That steadied her more than reassurance would have.

They walked. The afternoon heat came up off the road and through the soles of her shoes, and the red dust that had followed her home every day of her life rose at exactly its usual height around her ankles. A matatu with no passengers went past with its music still playing. At the kiosk on the corner, a woman was roasting maize over charcoal as though the smoke had somewhere to be, and the smell of it reached Maya before the news of the day had finished reaching anyone.

“Is it true?” she asked as they walked.

“Your mother says it is.”

“People say she made a mistake.”

“She makes mistakes.”

Maya turned to him.

“Usually about how much work she can do before dinner. About this, I think she has checked.”

At the corner, a queue curved around the supermarket. A man had filled a cart with bags of rice and was arguing with a woman carrying a child. Her grandfather stopped. He took two bags from the man’s cart, placed them beside the woman’s basket, and waited until the man looked away.

“You can’t just do that,” Maya said when they had crossed the road.

“Apparently I can.”

At home, he asked her to help pick tomatoes.

“People are buying twenty years of food.”

“We have three days of tomatoes. Start with the red ones.”

Her phone rang while she was kneeling between the rows. The vines left their sharp green smell on her hands. Under the shade netting the heat had the weight of a hand on the back of her neck, and a cracked fruit somewhere near the fence was drawing flies. Amara appeared against a white wall, still wearing the clothes from the briefing.

“I’m sorry about your birthday.”

Maya pulled a tomato harder than necessary. The branch came with it.

“You said you’d call.”

“I know.”

“You always know afterward.”

For once, her mother offered no explanation. Maya could hear voices beyond the screen, someone asking for a revised statement.

Amara moved farther from them.

“Do you want me to come home?”

Maya looked at her grandfather. He was pretending to inspect a damaged leaf.

She wanted to say yes. She wanted to demand the birthday she had been promised and the following birthday and all the ones after that. Instead she asked whether her mother was frightened.

“Very,” Amara said.

That was the first answer anyone had given her that she believed completely.

That evening a man on the television explained that the Earth would be flung into darkness within the year. He had a diagram with arrows and a voice that had clearly practiced being calm.

Grandfather watched with the attention he usually gave to seed catalogs.

“Wrong,” he said.

“You don’t know that,” Maya said.

“His arrows are too confident. Your mother’s never are.”

The next channel showed a woman who said the object was a punishment. A third showed a man selling bunkers with a south-facing view, which Grandfather said proved the man had never lived underground. He graded each of them the way he graded tomatoes, by holding up a hand and tilting it. The punishment woman got a flat palm. The bunker man got a thumb pointed at the floor.

“What does Mom get?” Maya asked.

He thought about it for longer than she expected.

“She hasn’t ripened yet,” he said, and changed the channel to the weather, which was also wrong, but in a way he had decades of experience forgiving.

The next day, the garden proposal was still due.

Maya assumed the teacher would cancel it. Instead she sent a message explaining that ordinary work would continue where possible. Students unable to attend could submit remotely. Maya read the message as if it had arrived from someone describing a different planet. Leila came over carrying the roll of drawings they had forgotten at school. Her parents had told her to stay close to home. She said this house counted.

Grandfather opened the gate, inspected the drawings, and asked whether anyone had calculated the weight of the proposed tank when full.

“We did,” Maya said.

“Show me.”

He found an error in their support spacing. Leila found an error in his assumption about the tank’s material. They argued for twenty minutes. Maya watched them from the kitchen door, almost angry that they could still care.

Then she joined them.

The disagreement had edges. They could measure the space, weigh a sample, ask the caretaker where the old foundation ended. It was possible to arrive at an answer that did not become less useful when another person looked at it.

They finished before evening. Grandfather made tea and asked who would clean the gutters.

Leila pointed at the maintenance schedule she had added to the last page.

He read it slowly. “Better.”

After he went inside, Maya asked if Leila thought they would reach thirty.

Leila studied the darkening sky. “Yesterday I thought thirty sounded old.”

“It’s not an answer.”

“I don’t have one.”

Maya rested her forehead on the table. The surface was cool where the sun had left it.

“I hate everyone asking me.”

“Then tell them to ask me.”

“You don’t know anything.”

“Neither do they. I’ll be among equals.”

This time Maya laughed. Grandfather called from the house to ask whether they were mocking his tank design. Leila said yes before Maya could answer.

* * *

On the fourth day, a recording of Amara circulated with the final clause removed from one of her answers.

The shortened version appeared to promise that Earth would remain habitable. A second version repeated only the sentence about not knowing whether people’s children would survive. The same account posted both, six hours apart.

Amara learned about the first version from the communications office. She learned about the second from Maya.

“Which one did you mean?” her daughter asked.

Amara was sitting on a bench in a service corridor, the only place she had found where no one was waiting to hand her a document.

“Neither. They cut—”

“I watched the whole thing. I still don’t know.”

She began to explain the model categories. Maya interrupted.

“I know there are models. I know there are errors. I’m asking what you think when nobody’s recording you.”

Amara leaned her head against the wall. Her private belief was an unstable mixture of calculations, fatigue, and the need to stand up the next morning. Turning it into a percentage would not make it science.

“I think there are things we can do that matter,” she said. “I don’t yet know whether they will be enough.”

“Then say that.”

“I thought I had.”

“You said conditions and confidence and dependent variables. People hear the words they recognize and fill in everything else.”

The anger in Maya’s voice was partly fear, but it was not only fear.

Amara listened instead of correcting it.

At the next briefing she used her daughter’s question. She explained which possibilities remained uncertain and named three actions that would be useful under several of them. She did not promise safety. She did promise that the next release would include the assumptions behind each scenario.

That promise would become harder to keep than she understood.

After the briefing, a journalist stopped her near the exit. “Would you let your own daughter stay on Earth?”

The question arrived when she was too tired to protect it from herself.

“She isn’t a test of my confidence,” Amara said.

It was true. It sounded evasive. By evening, people were asking whether scientists had already reserved places for their families elsewhere.

Maya received the first threat that night. It described the route she walked to school.

Grandfather read it without changing his expression. He called the school, then the local police, then Amara. Maya heard his voice from the next room, lower than usual and much less patient.

For two weeks she did not walk anywhere alone.

Leila came to the house after school and complained about the escort waiting outside. She was trying to make the arrangement seem annoying rather than frightening. Sometimes it worked. Sometimes Maya wanted to shout at her for pretending.

They began working on the garden because the proposal had been accepted. The tank arrived late. The support brackets were wrong.

Nothing about its installation resembled the diagram Maya had animated.

When the first rain filled the gutter, water spilled from a joint they had believed was sealed. Leila stood beneath it laughing while Maya went for a wrench. Grandfather watched from the doorway, withholding advice until they asked.

They did not ask for nearly an hour.

Later, Maya sent Amara a photograph of the repaired joint. No face, no caption. Just two mismatched brackets and a pipe that no longer leaked.

Her mother replied: I would like to see the whole garden.

Maya waited until the following afternoon, then took another photograph.

The first week brought violence, then curfews, then arguments about whether the curfews were making things worse. Financial markets closed and reopened. A war paused for nine days. Another did not pause at all.

There were still crops to harvest. Children developed fevers.

People who had announced that money no longer mattered returned to work because their landlords disagreed.

Humanity began adjusting to an apocalypse with a schedule.

At the modeling center, Amara opened a folder containing the observations that could still prove her wrong. She checked it every morning before opening anything else.

* * *

On the eleventh day, Maya went back to swimming practice.

She had not told anyone she was going. The pool was still open, which surprised her, and then annoyed her because she had begun to resent being surprised by ordinary things. At the entrance, a handwritten sign explained that the evening sessions might be shortened to save power. Beneath it, someone had written that the black hole could wait until after the championships.

The changing room smelled of wet concrete and soap. Three girls from her class were sitting on the bench with their bags unopened. They stopped talking when Maya entered. One asked whether her mother had told her anything that was not in the news.

“No.”

“You’d have to say that.”

Maya put her bag in a locker. She could imagine every possible answer, and every one seemed to become evidence against her.

Leila arrived while she was struggling with the combination. She did not ask what had happened. She took the locker beside Maya’s and began complaining about a mathematics assignment. The complaint was so ordinary that Maya almost cried.

Their coach made them warm up. Two swimmers asked whether there was any point in training for next year’s competition. The coach said she had booked the lanes and intended to use them. She did not promise that next year would be normal. She also did not let them spend the session discussing something none of them could change from the poolside.

Maya swam badly. She went too hard in the first length, turned late, and swallowed water. At the end of the lane she held the edge and coughed while the other swimmers passed. She had imagined the water would make her mind quiet. Instead she could hear every unfinished question more distinctly because there was nothing else to attend to.

Leila stopped beside her.

“Race me slowly,” she said.

“That doesn’t make sense.”

“Then you should win.”

They pushed off together. Leila kept a deliberately easy pace.

Maya followed, watching the dark line beneath them. By the far wall, her breathing had settled. They did another length without speaking. Afterward, they sat outside eating something from a packet that claimed to be fruit. Maya said she hated being asked what her mother knew.

“What do you want me to say when people ask me?” Leila said.

“They ask you?”

“Of course.”

Maya looked at her. Until then, she had thought of the attention as something happening only to her.

“Tell them you don’t know.”

“I do.”

“Tell them to stop.”

“I do that too.”

Across the road, a man was arguing with a bus driver over a fare.

The bus remained in place while traffic gathered behind it. Maya watched the argument become larger than the amount of money involved. She wondered whether every small dispute would now carry the weight of the announcement, or whether people had always been like this and she had only just begun looking.

Leila offered her the last piece of alleged fruit. Maya took it.

“Do you think we’ll still be friends in seventeen years?” she asked.

Leila made a face. “I’ll be very sophisticated. You may embarrass me.”

Maya laughed. It was the first future she had imagined that week which contained something besides the black hole.

* * *

Amara’s first return to Nairobi after the announcement lasted thirty hours.

She spent four of them asleep and three answering calls she had promised not to answer. Her father watched without comment until the third call interrupted dinner. Then he took her plate away and kept it warm in the kitchen.

When she returned to the table, Maya was gone.

“She had schoolwork,” he said.

Amara went to the bedroom door. Maya sat on the floor with a textbook open and her phone facedown beside it. The window was open to the evening, and the ordinary sounds of the street came through it: a generator two houses down, a dog with a grievance, a radio somebody had turned up so the neighbors could share the forecast. The room still contained the paper decorations from her birthday. Amara had not noticed them on arrival.

“Can I come in?”

Maya shrugged.

Amara sat on the edge of the bed. She had prepared an apology on the journey from the dining room, but the words seemed to belong to the wrong scale of absence. Sorry about the call was too small. Sorry about the last several years would make Maya responsible for forgiving something she had not been invited to discuss.

“I should have left the phone in the other room,” she said.

“You’d have heard it.”

“I should have turned it off.”

“Could you?”

Amara hesitated.

Maya closed the textbook. “That’s what I thought.”

They sat in silence. From the kitchen came the sound of her father putting dishes away too loudly.

Amara asked about swimming. Maya looked surprised that she knew. Her grandfather had told her, which Amara admitted before the information could become another small deception.

“It was fine.”

“Did you enjoy it?”

“Some of it.”

Amara waited. Maya described the girls in the changing room, the question, the feeling that every answer had already been interpreted.

Amara listened without explaining why people were frightened. Maya knew why. She had not asked for a defense of them.

“You can tell the school I don’t want people interviewing me,” Maya said.

“I will.”

“Not a big announcement. Just tell them.”

“All right.”

Before leaving the room, Amara took out her phone and turned it off. It would not repair the dinner. She sat back down anyway.

Maya reopened the textbook. After a few minutes she asked a question about the assignment. Amara knew the answer. For once, knowing was exactly what was needed.

CHAPTER TWO

What Survives

REVIEW GROUP INTAKE LOG, PROPOSALS 0411–0414

0411 — Gravity screening by electrical charge. Acknowledged.

0412 — Gravity screening by larger electrical charge. Acknowledged.

0413 — Relocation of humanity to the far side of the Moon, “where it can’t see us.” Acknowledged.

0414 — Seatbelt (for the Sun). Submitted by Grade 4, Kisumu. Under review.

T–6,309 DAYS

The evacuation proposals began arriving before the center had a procedure for receiving them.

Some were calculations. Others were drawings of vessels whose designers had left the propulsion system as a box marked ENGINE. A retired teacher sent a carefully reasoned letter explaining how a community could live beneath a mountain for generations if it stored enough seed. A consortium offered prefabricated shelters at a discount if the council signed before the end of the quarter.

Amara asked for every proposal to be acknowledged. Luis asked who would acknowledge the proposals that claimed gravity could be screened with an electrical charge.

“The same people,” she said.

“That seems cruel to the people.”

Luis kept a folder of the proposals that made him happiest. It was not an official folder. It contained a plan to tow Earth with a very long cable attached to Jupiter, a petition to have the black hole declared a protected species, and a drawing by a nine-year-old in Kisumu in which the Sun was wearing a seatbelt.

“The seatbelt is the most physically sound idea we’ve received,” he told Amara.

“It doesn’t attach to anything.”

“Neither does the cable. At least she drew it with confidence.”

He sent the girl a thank-you letter on the center’s letterhead. She wrote back to ask whether the seatbelt had been approved. Luis replied that it was under review. It was, in a sense, the most honest public statement the center made that month.

They recruited a review group. Its first task was to separate ideas that violated known physics from ideas that required resources they did not have. The distinction mattered. The second category contained almost everything anyone wanted to do.

A habitat specialist named Derya laid a model cylinder on the meeting table. It was small enough to hold in one hand, with a removable section revealing fields, houses, and a strip of blue water.

Amara had seen similar models at exhibitions. People always lifted the same piece and looked inside with the same delighted expression.

This time nobody touched it.

“We can build larger versions of existing habitats,” Derya said.

“Not that landscape on day one. But pressure volume, radiation shielding, food production. We know the component problems.”

“How many people?”

She gave a range. It was a serious number for a construction program and an almost unbearable number beside eight billion.

The next speaker proposed accelerating the program by reducing redundancy. Derya took the roof off the model.

“Which part?”

“I mean optimizing across the fleet.”

“Which part?” she repeated. “The spare pump? The second power source? The crop you hope you won’t need?”

The speaker said she was being deliberately literal. Amara asked him to be literal too.

For an hour they discussed the mass of things normally omitted from pictures. Tools. Filters. Plumbing. The organisms that made cultivated soil more than crushed rock. Medical supplies that would expire. Machinery capable of making replacements for other machinery, and replacements for that machinery in turn.

At the end, the model cylinder remained on the table. It looked more valuable and less possible than when Derya had brought it in.

Another team proposed moving the center of population to Mars.

Their presenter had spent ten years making Mars less inhospitable in small, expensive places. He did not claim the planet was easy. He claimed that its smaller population would make its systems easier to protect.

Luis displayed the range of post-encounter Martian orbits.

The presenter watched his own facilities travel through futures his architectural drawings did not describe.

“We could adapt the power system,” he said.

“Probably,” Luis answered. “For the people already there. That’s different from moving Earth’s population into it.”

Amara saw the man absorb the distinction. He had been preparing to defend an ambition. Now he was revising an evacuation list.

The underground-refuge proposal survived longer. Earth itself would provide excellent shielding. Deep facilities already existed. For some encounter outcomes, they could preserve people through an extended surface crisis.

“How long?” Anika asked from the remote connection.

The engineer named a period based on existing stores.

“And after that?”

He described additional reactors, deeper excavation, more efficient food systems.

“And if Earth leaves the Sun?”

The room waited while he changed the energy budget.

The refuge remained useful. It ceased to be an answer to every version of the question.

Afterward, Amara found him in the corridor with his presentation still open. He had a photograph of his children on the device’s lock screen. She saw it when the display dimmed.

“You should keep developing it,” she said.

“For which outcome?”

“The ones it can help.”

He looked at her, exhausted by the incompleteness of that instruction.

“My wife asked if I was building somewhere for us to go. What am I supposed to tell her?”

Amara had no answer she would have accepted herself. She asked whether he wanted someone from the orbital team to review the assumptions with him. He said yes. It was a small thing to offer, but it was a thing he could use.

* * *

On the seventh day, one of Amara’s runs produced an Earth that looked almost ordinary.

She had begun leaving the unsuccessful trajectories in pale gray behind the active calculation. The screen was crowded with them.

Against that background, a closed blue ellipse felt like an act of defiance.

She ran it again with different numerical settings. Then with a second implementation of the same equations. The blue ellipse remained.

Outside the room, someone was distributing sandwiches. Amara took one and forgot to eat it. She opened the climate summary and compared the annual flux to the nominal pre-encounter value. It was lower, but not absurdly lower. The orbit was eccentric, but it did not immediately cross another planet’s path.

She called Luis over.

He stood behind her for a long time, saying nothing. Then he asked whether she had checked the Moon.

“Yes.”

“And Earth’s rotation?”

“We’re adding that calculation. The direct tidal effect across Earth is small compared with the orbital problem.”

He pulled a chair beside hers. They worked until the sandwiches curled at their edges.

When she finally stood, her legs felt weak. She wanted to call Maya. She wanted to say that she had found a future in which gardens still existed. Instead she sent the run to the external review group, including Anika, who had agreed to come in person the following evening.

In the message, Amara wrote candidate survivable trajectory.

She deleted survivable and replaced it with bound.

Then she opened the folder again and looked at the blue ellipse before going home.

On the ninth night, Professor Anika Sorensen dismantled Amara’s most hopeful simulation in front of eleven people.

She did it politely, which made the experience worse.

“You’ve ended the run too early.”

Amara enlarged the timeline. Earth had survived the encounter and completed three orbits around the Sun. Its path was eccentric, but closed.

“The energy remains negative.”

“For now. Look at Venus.”

They extended the calculation. A sequence of close approaches slowly changed Earth’s orbit. In the accelerated display, the blue line thickened into a band, then broke open.

No one spoke until Luis asked for the integrator settings.

“Different solver,” Anika said. “Smaller steps. Same result.”

She was a planetary dynamicist who had spent most of her career arguing that beautiful diagrams concealed ugly assumptions. Amara had invited her for that reason. She regretted the invitation for approximately ten seconds.

“All right,” Amara said. “Show us the rest.”

Some trajectories retained Earth for centuries but moved it too far from the Sun. Others kept the annual average sunlight tolerable and produced seasonal extremes no existing agricultural system could absorb. The range was not simply measurement uncertainty. It included uncertainty in their physical models and in what they were prepared to call survival.

At two in the morning, Anika stopped the presentation.

“Your green category says habitable. What does it mean?”

“Surface liquid water. Atmospheric retention. A stable orbit.”

“How much food?”

“That’s outside this model.”

“Eight billion people aren’t.”

Amara wrote the question on the board herself.

The agricultural scientist who joined them was named Amina Sefu.

She arrived from a field station carrying a bag that smelled faintly of damp soil. Before sitting, she asked whether anyone had a charger that fit her older tablet. Three people offered adapters; none worked. She took a notebook out of the bag instead.

“You have an Earth,” she said. “Tell me about its year.”

Amara began with orbital elements. Amina listened, then drew a circle on the board and divided it into sections.

“When do people plant?”

“That depends on the region.”

“Exactly. You keep giving me a planet average.”

She asked about the timing of the low-light periods, the length of the altered year, and how quickly the encounter would change the seasonal pattern. Some answers were available. Others required calculations nobody had yet prioritized.

Luis wrote a list.

“What if we move agriculture indoors?” a planner asked.

“Some of it,” Amina said. “With what structures? What power? What replacement parts? Who grows the things the enclosed system isn’t designed to grow?”

The planner said they could use the most efficient crops.

“Efficient for which constraint? Water? Light? Labor? Storage? A crop can win one of those and fail another.”

She took a handful of dried beans from a packet in her bag and placed them beside the model cylinder. They were different colors and slightly different sizes. One had cracked.

“These came from three fields less than twenty kilometers apart. Same broad climate. Different planting dates. Different yields. If your projected season changes, I cannot tell you what happens from the annual average alone.”

Amara picked up the cracked bean. It weighed almost nothing.

“What do you need?”

“Time-series sunlight estimates. Regional weather ensembles. Soil information we don’t have in one place. And the people who actually grow the food.”

“How long?”

“For an honest first answer? Longer than a press cycle.”

They kept working. The green region shrank as more of life entered the calculation. It did not shrink uniformly. A trajectory tolerable for one region became dangerous for another. A survivable first year concealed a severe second-year shortage because the wrong seed had been planted or the previous harvest could not be stored.

At the end of the day, Amara found Amina collecting the beans from the table.

“Do you think we can prepare?”

“I think people have been adapting crops longer than they’ve been writing equations. I also think people who say adaptation usually expect someone else to do it.”

She put the packet away.

“Give us the measurements. Come to a field before you decide what the numbers mean.”

Amara wrote the invitation into her calendar. It was the first appointment since discovery that did not take place in a room full of screens.

By the following afternoon, climate scientists and agricultural planners had joined the team. The green region became much smaller.

There was still a region.

That was what Amara carried into the next briefing: no promise, but a set of outcomes in which Earth remained attached to a living Sun, with conditions civilization might endure if it prepared.

Luis brought her a separate calculation as she was leaving.

“You should see where the Sun goes.”

She assumed he meant the uncertainty envelope. Instead he opened a view of the nearby stars.

The black hole’s enormous mass allowed it to turn the Sun’s path without coming close to the Sun’s surface. In the black hole’s frame, the Sun fell in and climbed out with nearly the same distant speed. In their original neighborhood’s frame, the departing Sun had acquired a new velocity.

About three thousand kilometers per second.

The path extended toward a marker labeled ALPHA CENTAURI FUTURE POSITION.

“How future?” she asked.

“Four centuries, roughly. A proper intercept calculation will move that.”

“And our closest approach?”

“Not settled. We pass through the neighborhood in some of the permitted solutions.”

She looked from the Sun’s track to Earth’s surviving orbit.

“We take our light with us.”

“If Earth stays attached.”

For the first time since the discovery, she pictured something beyond the encounter. A morning on Earth while unfamiliar stars occupied the sky. A child learning the route from a teacher who had never seen the old constellations.

“Can we choose one of these?”

Luis knew what she meant. He opened the correction budget they had begun as an exercise. Even their most ambitious available machines could not move the Sun by the required distances. Shuffling planets did not propel the total system’s center of mass. They could change smaller things, given enough time, but gravity did not become a motor because they were desperate.

“We don’t have that kind of helm,” he said.

Amara watched the track pass near the future star.

“Then we stop promising to steer.”

“What do we promise?”

She returned to Anika’s question on the board.

“We work on what survives.”

* * *

Two days later, Amara went with Amina to a trial farm outside the city.

It occupied a patchwork of buildings and fields between a road and an expanding industrial estate. The enclosed growing rooms were smaller than she expected. The most elaborate one had a door that would not latch unless lifted at the handle. Amina demonstrated the technique, then handed Amara a visitor coat.

Inside, rows of plants stood beneath adjustable lamps. The air was warm and smelled of leaves, nutrient solution, and a faint sourness Amina said they were investigating. The farm manager showed them the power meter before the crops.

“This is the part people photograph afterward,” he said.

They had been testing reduced-light conditions for unrelated reasons: energy prices, cloudy seasons, the reliability of emergency power. The results were useful to Daedalus before Daedalus formally existed. Some plants grew more slowly. Others changed the proportion of edible material they produced. A few performed well enough to make Amara feel hope before Amina asked about taste, storage, and disease.

The manager gave her a leaf to try. It was tougher than the supermarket version and tasted faintly bitter.

“Nobody buys it twice,” he said. “We can grow it. That’s not the same as getting people to eat it for twenty years.”

In the next room, a technician was replacing a pump. The work had delayed a trial by two days. She asked whether the new emergency funding meant they could finally keep the right spare on site.

Amara said she did not control procurement.

The technician looked at the visitor badge, then back at the pump.

“You should tell whoever does that the plant doesn’t care which department failed.”

Outside, schoolchildren were touring the outdoor beds. Their teacher recognized Amara and hesitated, apparently deciding whether it was appropriate to introduce the person who had become associated with the end of everything.

A boy recognized her without help.

“Are you going to move us?”

Amara crouched because he was small and because she needed a moment to think.

“The black hole is changing where the Sun goes,” she said. “We are trying to make sure people can keep living here.”

“My uncle says we’ll have rockets.”

“Some people will. Most of us will still be standing on the ground.”

He considered that, then asked whether his dog could come.

“If we can keep Earth livable, your dog is already aboard.”

The answer pleased him. He ran back to his group. Amara stood slowly, aware that she had just offered reassurance in a form simpler than the truth could comfortably support.

Amina watched him go.

“You didn’t say his dog was guaranteed.”

“He heard it that way.”

“Then remember him when you decide what to call a successful model.”

On the journey back, Amara opened the funding proposal. She added agricultural trials, storage losses, and regional distribution to the first-stage requirements. The projected cost increased. The proposal became harder to explain in a single sentence.

She left the additions in.

That night, Maya sent the photograph of the repaired school gutter.

Amara looked at the two brackets and the awkward joint between the pipes. She thought of the unlatched door at the trial farm, the technician waiting for a spare, the child asking about his dog.

For the first time, the work ahead looked less like one enormous calculation and more like an almost uncountable number of things that had to keep working.

* * *

The evacuation model became more persuasive when the people were removed from it.

On a screen, a habitat was a volume with an expected population.

Its farms produced a mass of food. Its life-support system recycled water and air at specified rates. The model could vary construction speed, launch capacity, and the amount of equipment each passenger required. Under generous assumptions, the curves rose sharply enough to inspire hope.

Then Derya asked the team to allocate a real family.

She chose one from a volunteer household study: two adults, three children, a grandmother, and an uncle who needed regular care. Their skills were ordinary and useful. None supplied an easy reason to divide them. The children occupied less space in the short term and would become adults during the voyage. The grandmother used little equipment but needed a treatment the habitat had not yet included. The uncle could work, though not on the schedule assumed by the labor model.

The first allocation separated them among three vessels.

“Optimize for family unity,” someone said.

The program did. Capacity fell. Derya asked whose family unity would be protected if the available places ran out. The room moved from engineering into ethics without any visible line between them.

Anika argued that the model should show the loss rather than hide it. A transport planner said publicizing such examples would make selection impossible. Derya asked whether selection was currently possible, or merely unexamined.

Amara watched the discussion with growing discomfort. The habitat proposal had offered her a problem whose scale could be measured. Each additional question made the answer less clean. That did not make the questions optional.

They added medical support, training time, family constraints, and margins for people who would not remain healthy simply because the voyage required them to. The capacity estimate fell again. The resulting number was still worth building for. It no longer resembled an answer for everyone.

* * *

Later, Amara visited the habitat test facility with Anika.

It occupied part of an old industrial building. The exterior was unremarkable. Inside, experimental rooms held different versions of the systems that might one day support a vessel: water loops, growing chambers, air-handling equipment, sleeping compartments arranged to test

privacy and access. A researcher asked them to remove their coats before entering one room because outdoor contamination would confuse the trial.

Amara followed Anika through a compartment scarcely wider than a corridor. Six bunks folded into the walls. A volunteer sat on one, reading with headphones on. She looked up, recognized the visitors, and appeared annoyed rather than impressed.

“We’re testing quiet hours,” she said.

The researcher apologized and led them back out.

Anika waited until the door closed. “Good sign.”

“That she was annoyed?”

“That she was allowed to tell us.”

In the growing room, a technician showed them plants with pale leaves. The lights worked. The water chemistry looked acceptable. The trial had been losing productivity in a way the first model did not explain. The team was changing one condition at a time and trying not to destroy the evidence through frantic correction.

“How long until you know?” Amara asked.

“Longer if everyone keeps asking for a demonstration harvest.”

Anika smiled. Amara did not. She was thinking of the schedule and then became aware of how often she thought of it when someone described a physical limit.

They ate in the staff kitchen. The food came from outside the facility. The researcher made that clear before either could mistake the lunch for proof of closure.

Anika asked about the worst maintenance job. The technician described cleaning a narrow section of pipe that the designers had assumed would rarely need access. A person could reach it only by removing equipment used for another task. During the trial, cleaning had repeatedly been postponed until it became harder.

“Redesign the access,” Anika said.

“We have. It reduces usable storage.”

“Then the usable storage was overstated.”

Amara heard in that sentence the same argument being made about the whole evacuation model. Capacity existed only if the people and equipment could keep producing it.

On the way back, she asked whether Anika truly believed the vessels would survive.

“Some, if we do the work well and don’t confuse confidence with a test.”

“That’s not a reassuring answer.”

“I wasn’t hired to reassure you.”

They traveled the rest of the way in companionable silence, which surprised Amara. Disagreement had not prevented them from seeing the same problem more clearly.

* * *

At the farm, Amina kept a failed crop in a marked plot long after visitors wanted it cleared.

It looked untidy beside the healthy rows. The plants had been grown under a deliberately unfavorable combination of light, water, and maintenance conditions. Some had recovered. Others had not. A communications team asked whether the plot could be left out of an upcoming public visit.

“People will think the whole trial is failing,” the coordinator said.

“Then explain what the trial is.”

“We have twelve minutes.”

“Give me four.”

Amina used three. She showed what had been changed, what had failed, and what they were testing next. A visiting teacher asked whether students could access the results. A farmer asked about a different variety. A donor asked whether this meant the project was not ready.

“It means this part isn’t ready to copy,” Amina said.

Amara watched the donor’s face. She could imagine the funding conversation that would follow. She could also see the teacher taking notes, and the farmer pointing out a detail the team had not included.

After the visit, Amina asked what she thought.

“You may have lost a contribution.”

“I may have prevented someone from buying a result we don’t have.”

Amara looked toward the failed plot. Years later, when she delayed publishing the correlated-risk warning, she would remember this conversation and supply herself with reasons the situations were different. Some of those reasons would be true. None would be enough.

* * *

The first press interview Amara regretted was the one in which she answered every question correctly.

The interviewer asked whether the Sun could survive. She said yes, its self-gravity made disruption a different question from the fate of the planets. He asked whether Earth could remain bound. She said some modeled outcomes allowed it. He asked whether humanity had technologies that could improve survival. She described food reserves, protected agriculture, transport, power, and evacuation.

Each answer was qualified. The edited segment joined the affirmative beginnings and shortened the qualifications. The resulting account made a possible survival scenario sound like a program already under control.

Amara called the producer. The producer said the segment had not misquoted her. She knew that. She asked for the complete interview to be published and a clarification added. The producer agreed to the full interview and resisted the clarification until the public information office supplied the sequence of omitted conditions.

The correction reached fewer people than the segment.

Luis told her to stop watching the reactions. She said she needed to know what people believed. He said she also needed to finish the calculation on her desk.

“Which would you choose?”

“I’d hire people whose job is understanding the first problem, and listen when they tell you the answer isn’t another equation.”

She did not like being told that expertise had boundaries. She had built a career on discovering connections other people missed. Now the connections multiplied faster than any one person could follow.

The communications specialist she consulted asked what Amara most needed people to understand. Amara supplied a paragraph containing four distinctions and three ranges. The specialist asked for one sentence.

“The Sun’s survival does not guarantee ours.”

“Start there.”

They recorded a new explanation. It showed the difference between the path predicted by gravity and the systems humans could build. It gave the survival range without making the central case look like a promise. It also admitted that the exact planetary outcome had not been demonstrated by the simplified model being shown to the public.

The response was mixed. Some people appreciated the clarity.

Others accused her of changing the story. She could not repair an oversimplification without making the account look different from what people remembered.

She saved the original interview beside the correction. Later, when the project grew more sophisticated at explaining itself, she would sometimes forget how easily a technically accurate account could become a misleading one through selection alone.

* * *

At school, Maya was asked to speak during a special assembly.

She declined. The deputy head said she did not need to explain the science; she could simply describe how young people felt. Maya asked which young people. The deputy head looked surprised, then apologized for putting her in that position.

Leila agreed to help organize the questions instead. They collected them without names. Some concerned survival. Some concerned examinations, jobs, family, and whether adults would keep promises that extended beyond the encounter. One student asked whether there was any point learning history if everything was going to change.

Salma, then assisting with a younger class, helped sort the questions. She put the history question near the top.

“Why that one?” Maya asked.

“Because change is most of what history is about.”

The visiting scientist could answer some questions and not others.

The teachers answered questions about school plans. The assembly ended without a single speech that made everyone feel better. It did produce a list of things the school would change: clearer information, a quiet room for students overwhelmed by the coverage, and a commitment not to make every lesson about the disaster.

Maya liked the last commitment most. She wanted to learn things that might matter because they were interesting, not only because they might help her survive.

Afterward, she and Leila returned to the water project. The tank was functioning, but a small leak had developed at a connection they had tightened badly. They repaired it, tested it, and wrote the result in the notebook.

The task occupied the afternoon. For that afternoon, the future had a scale they could hold in their hands.

CHAPTER THREE

Daedalus

COUNCIL SUMMARY 2109/03, TRANSLATOR'S NOTE

"Daedalus" is used here as a placeholder for "the plan." Please replace before circulation.

No replacement was received.

T–6,214 DAYS

Elena had spent the previous night walking through a plant that was being asked to double its output.

The request came from people who had never seen it. On their schedules, production was a line that could be extended. Inside the building, production was a furnace awaiting refractory material, a crane shared by two shifts, and a woman named Saira Nwosu explaining why neither could be made to work faster by changing the color of a cell.

"We can add a shift," Saira said. "We cannot add a second crane before the first one needs servicing."

They stood behind a marked barrier while an inspection carriage traveled over a newly finished panel. The panel looked flawless to Elena. The carriage found three defects.

"Could we use it for a less demanding application?"

"Yes. Which one are you prepared to certify?"

Elena looked at her. Saira did not look away.

The distinction between a possible use and an authorized use occupied much of Elena's professional life. Materials could sometimes do more than their specifications promised. Sometimes a specification existed because someone had already discovered exactly how much more was fatal.

She asked for the rejection records and the list of alternative applications. Saira brought both. A younger analyst arrived with the shipping schedule folded across his tablet so that he could see the two competing delivery windows at once.

His name was Tomas Ivers. Elena remembered him because he had once delayed a shipment she wanted dispatched, correctly insisting that its listed mass did not include the handling frame.

"If we move these panels into agricultural service," he said, "we can free the higher-grade stock. But the destination has to change before loading. Otherwise the customer rejects the entire batch."

“How long?”

“Three days if everyone agrees tomorrow. Six weeks if Legal decides we need a new procurement category.”

Saira laughed without amusement.

Elena made the decision before leaving the plant. She would carry the physical inventory to the council, not an extrapolation of what the council wanted it to become.

On the flight to the meeting, she opened an older photograph of the plant. Half the machines had not existed then. She had helped finance them by canceling a project everyone remembered more fondly because it had never had to operate. She knew what making room for something new cost. She also knew that people who paid the cost rarely got to write the speech afterward.

* * *

Before the main session, the council heard from representatives of existing off-world settlements.

Jun Park’s connection arrived with a delay that defeated normal conversation. He managed maintenance at Rill, a small industrial settlement near Rhea. Its permanent population was not large. Its work supported several larger facilities, which made its closure look simple in one set of accounts and impossible in another.

He had recorded his statement rather than pretending the exchange was live.

Behind him, people were taking down a partition to enlarge the local greenhouse. Someone had painted a yellow bird on the panel they were removing. Its wing disappeared as two workers carried the panel out of view.

“We are not requesting an exemption from the encounter models,” Jun said. “We are requesting that evacuation planning include the people maintaining the transport system until the last departure. Every draft so far assumes they leave with everyone else.”

Elena looked up from the manifest.

Jun described the work that would continue after most residents departed. Fuel handling. Tracking. Repairs to vessels that could not be abandoned without reducing capacity for the next evacuation. He asked who would certify the final habitable volume and who would remain to close it.

“People are already accepting contracts based on the published dates,” he said. “If the dates are placeholders, label them that way.”

The recording ended. The chair asked whether any member wished to respond. A delegate asked how much the revision would cost.

Another asked whether the work could be automated.

Elena requested the floor.

“Some of it can. The people know which parts. Ask them before you subtract their seats.”

She added a line to her manifest: final-crew transport, not included in population totals.

Years later, she would recognize Jun’s voice before the delayed video appeared.

Elena Varga arrived at the emergency council with a manifest instead of a presentation.

She directed an orbital industry consortium whose members operated lunar excavators, solar collectors, and slow freight tugs. Her companies could deliver a replacement pressure valve beyond Mars more reliably than most governments could deliver a balanced budget.

They could not move a star.

“Before anyone asks,” she said, “we cannot build a Sun engine in seventeen years.”

A minister lowered his hand.

“We can expand food production under controlled conditions. We can harden the electrical network. We can move vulnerable facilities. We can build more independent habitats. Every one of those uses material we already know how to process.”

She placed the manifest on the shared display.

“These are the machines we have. These are the ones under construction. These are the workers trained to repair them. Start there.” The proposal acquired the name Daedalus because a translator used it in an early summary and nobody managed to replace it. Amara disliked the mythology. Elena disliked discussions about names while shipments waited for authorization.

The council debated evacuation for two days. The available habitats could support a small fraction of humanity. Building more would increase that fraction, but not enough. Earth already contained the atmosphere, water, soil, organisms, and shielding that every artificial habitat struggled to imitate.

It was an ark they could not finish rebuilding elsewhere.

Anika presented the case for independent vessels anyway.

“If every seed is in one greenhouse, the greenhouse is not a survival strategy. It’s a single failure.”

A delegate asked how she proposed selecting passengers.

“Publicly,” she said. “With the criteria open to challenge. If you turn this into a sale of seats, you will destroy the program before we launch.”

Selection consumed the next three days. Engineers required particular skills. Governments demanded quotas. One delegation offered to fund an entire cylinder if it could choose its population.

Elena asked whether the cylinder would manufacture its own bearings. When the delegate said he did not see the relevance, she explained that the consortium making the bearings had members in twenty-six countries.

No one got the independence they had imagined.

Amara spoke on the last afternoon. She described the nominal encounter, the surviving orbital possibilities, and what the models could not promise. Then she removed the trajectory graphic.

“In seventeen years, Earth may receive less sunlight. Seasons may change. Our satellites may need replacement. Some settlements beyond Earth will have to be abandoned. We can begin preparing for those conditions now, or discover them with empty warehouses.”

“And if Earth is lost?” someone asked.

“Then the independent habitats matter even more.”

The first serious division concerned a launch facility on Earth.

Daedalus needed its capacity for shield material and habitat components. The government hosting it had already committed that capacity to replacing regional water infrastructure. Changing the schedule would delay pumps and treatment equipment through another dry season.

Minister Das brought photographs of the installations awaiting replacement. She did not display suffering children. She displayed corroded pipework, a broken impeller, and a queue of maintenance requests.

“These people are not waiting seventeen years for their emergency,” she said.

The proposal’s coordinator explained that the new planetary program would eventually improve resilience everywhere.

“Eventually is not a delivery date.”

Elena felt an unwilling respect. She had used the same answer against contractors.

The coordinator offered priority in the second construction phase.

Das asked whether the second phase was funded. It was not. He pointed to expected contributions. She asked whether expected contributions could operate a pump.

The room became irritated with her before it became helpful.

Amara had been waiting to present the next orbital briefing. She watched the dispute with the impatience of someone whose problem seemed larger than every competing problem. Then she remembered the technician at Amina’s trial farm asking for a spare.

She asked for a recess.

In a side room, Das spread the proposed schedules across the table.

Several shipments classified as local could serve the later agricultural program. Delaying them would save launch slots on one sheet and increase the emergency requirement on another.

“You’re counting the same resilience twice,” Das said.

Amara called Amina. Elena called Tomas. Within an hour, they had identified part of the overlap and a larger part they could not resolve without engineers at the sites.

The result was not the efficient central program the coordinator wanted. Some local work continued. Some Daedalus cargo moved to a more expensive route. Two proposed habitat structures lost their first scheduled components.

Anika objected to the last change.

“You are taking redundancy from the only independent branch of the plan.”

“I’m preserving water for people who are already alive,” Das said.

Both statements were true. Amara could not turn one false by enlarging the other.

The negotiation lasted until the catering staff removed the unused lunch. Derya found a different assembly sequence for one habitat. The other remained delayed. Its selection committee would later tell families their anticipated departure was no longer assured.

Das signed the revised allocation. Anika signed with a reservation attached. Elena recorded the cost instead of smoothing it into the next quarter’s forecast.

No one called it a victory.

* * *

The passenger-selection session opened the following morning.

A physician explained the minimum medical capabilities a closed habitat needed. An engineer added maintenance disciplines. A teacher asked why education appeared in a category labeled noncritical services. The chair said the label was provisional. The teacher asked who had provisionally decided children could wait to learn.

Anika listened while the list grew.

Every discipline could make a persuasive case for itself. The total exceeded the seats available. Even when selection was not based on wealth, it could become a competition to prove that one kind of person deserved a future more than another.

“We need communities,” she said finally. “Not collections of job titles.”

A delegate asked how she intended to measure that.

“Imperfectly. With review and appeal. Better than pretending the difficulty disappears if we call everyone an essential worker.”

The industrial group funding one cylinder proposed reserving a fixed proportion of places for its employees and their families. The employees had built the vessel. Many believed the places had been part of the bargain even though their contracts did not quite say so.

The public-interest panel objected. The employees’ representative objected to the objection. He asked whether the panel intended to recruit workers by promising them wages they might not live long enough to spend, then tell them their families had no special claim.

Maya would encounter that argument years later in a cafeteria.

Here, before the ships existed, it was already shaping who would build them.

The compromise separated employment from guaranteed passage and created a transparent selection route for applicants with construction experience. It also required a family policy before recruitment began. Nobody was allowed to imply a seat that had not been allocated.

Elena asked the legal team to put that sentence in ordinary language. They returned a paragraph. She returned it with six words circled: This job does not guarantee passage.

The employee representative read it and said at least people would know what they were agreeing to.

* * *

After the session, Anika and Amara walked along the river because neither wanted to enter another room.

Security staff followed at a distance. The separation looked more discreet than it felt. Amara had become conscious of every stranger turning toward her, every phone lifted at the edge of her vision.

“You think I’m taking resources from Earth,” Anika said.

Amara watched a small boat work against the current.

“You are.”

“Yes.”

“I know why.”

“Do you agree with why?”

It would have been easier if Anika had demanded an apology.

Agreement required Amara to admit that all her work might fail and that someone else was right to prepare for it.

“I agree with redundancy,” she said.

“That’s a comfortable word. Do you agree that people should be allowed to leave?”

Amara thought of Maya. The answer became much less comfortable.

“Yes.”

Anika nodded. They continued walking.

At the next crossing, a woman approached with a folded photograph. The security staff moved closer, but Amara asked them to wait. The photograph showed a family gathered around a dinner table.

The woman pointed to her son, a mechanical technician applying for orbital work.

“Will that get him a place?”

Amara explained that selection rules were still being developed.

The woman listened carefully, then asked the same question in a different way.

“Should I tell him to go?”

There were no equations she could give this woman that would make the choice easier. She told her what had been decided about employment and passage. She told her to demand the actual terms in writing. The woman thanked her with an expression that showed how little help it had been.

After she left, Anika said, “That’s the question behind all the others.”

Amara did not ask which question. She was thinking of the call she would make to Nairobi when the meetings ended.

The Daedalus Accords passed by a narrower margin than the cameras made it seem. Funding came with conditions, exceptions, and review dates. Three governments signed and withheld their first contribution. A fourth sent equipment without signing.

The work began anyway.

Maya called from Nairobi after watching the vote.

“Are you coming home now?”

“For a week.”

“An actual week?”

Amara opened her calendar. She canceled two meetings while her daughter watched.

“An actual week.”

“Grandpa says you should bring seeds.”

“What kind?”

“He says if you’re going to reorganize agriculture, you can start by finding tomatoes that don’t split in the rain.”

* * *

During her promised week at home, Amara went to the school garden.

Leila showed her the tank before Maya could explain the design.

Its support frame was ugly and reassuringly substantial. Grandfather had insisted on a second inspection after the first heavy rain. The caretaker had made them add a drain they had not wanted because it spoiled the symmetry.

“He was right,” Leila said. “Don’t tell him I said that.”

Maya stood with her hands in her pockets while her mother examined the joints. She wanted praise and was prepared to resent it if it arrived too easily.

Amara asked what they had changed after installation.

That was the right question. Maya showed her the original diagram, then the alterations. The animation no longer resembled the finished system. She had kept it anyway because it reminded her how much they had believed before touching a pipe.

“Would you show the class what changed?” Amara asked.

“They’ve already seen the presentation.”

“Not that one.”

Leila looked at Maya. “I told you the repair photographs were better.”

They spent the afternoon preparing a second presentation. It included the wrong bracket, the leaking joint, and the caretaker’s drain. Maya titled it WHAT WE MISSED, then changed the title because it sounded too much like something her mother would say.

On the last evening of the visit, Amara helped her father move a table beneath the veranda. He complained that she was lifting the wrong end. She told him tables did not have wrong ends. He said this one did because one leg was loose.

Maya heard them from the kitchen and smiled before remembering she was still angry.

At dinner, she asked whether she should still apply for engineering courses.

“Do you want to?” Amara asked.

“I did.”

“And now?”

“Now it feels like I have to.”

Her mother put down her fork.

“You don’t owe the project your life.”

“You gave it yours.”

Amara began to answer, stopped, and tried again.

“I chose work I wanted before this happened. You should have that choice too.”

Grandfather asked Leila what she intended to study. She said plants, if she could find a course with more plants than paperwork. He said that sounded like gardening with unnecessary certificates. She asked whether he intended to inspect every examination personally.

The conversation moved on. Maya kept thinking about the difference between a choice and a duty that everyone praised until it felt like one.

The following morning, Amara left at the time she had promised.

That was a small improvement over earlier visits, and Maya hated how grateful she felt for it.

At the gate, her mother handed her a plain notebook.

“For the things that change after the diagram.”

Maya took it without opening it. She used the first page three weeks later, when the tank’s float valve stuck and flooded the path.

* * *

Maya learned about the first selection notices from Leila's father.

Hadi had applied for a construction placement, not an ark seat. The distinction was clear in the contract and much less clear in the conversations around it. Several neighbors congratulated him as though the family had been rescued. Ruth stopped correcting the third person and then regretted it when the rumor grew.

At dinner, Hadi put the notice on the table and asked Leila to read the important paragraph aloud. She did, annoyed that he was making a family exercise of something she understood. The placement offered work, training, and travel support. It did not confer passenger priority. "Why would they make people build something they couldn't use?" Maya asked.

Hadi looked at her. "People do that every day."

She thought of roads, hospitals, distant buildings, and jobs whose finished work did not belong to the worker. The familiarity did not make the ark case feel less harsh.

Ruth said the pay would help. Hadi said the work might lead to other work. Leila asked how long he would be away. That was the question nobody had answered with enough precision.

The placement required alternating months at a regional assembly site. The family would manage, but managing involved Ruth changing shifts and Leila taking on more at home. The contract described one worker. Its consequences occupied the whole table.

Maya stayed to help wash up. Leila was quiet until Ruth left the kitchen.

"Everyone thinks this is exciting," she said.

"Isn't it?"

"It can be exciting and inconvenient."

Maya dried a plate. "My mother is always away."

"I know. That doesn't make it easier for me."

Maya looked at her, then nodded. She was beginning to learn how quickly comparison could turn into a refusal to listen.

* * *

Hadi came home from the first rotation with a small burn on his wrist and a story he refused to make dramatic.

The burn had happened while cooking. The work had gone more smoothly than the kitchen. He showed them photographs of the assembly area, carefully avoiding details the site did not permit him to share. Leila wanted to see the people. He scrolled until he found a group eating outside a storage building.

There were workers from several places, using several languages, gathered around food that looked disappointing in all of them. One woman had drawn a face on a damaged container and placed it behind the supervisor’s chair. Hadi explained the joke badly enough that he had to start again.

Ruth laughed before he finished. Maya watched Leila relax.

Later, when the adults were talking, Leila enlarged a photograph of a growing-system test panel in the background.

“That’s wrong,” she said.

Maya leaned over. “What is?”

“Maybe nothing. The return line looks like it would trap water there.”

“Tell him.”

“I’m seventeen. They’ve got engineers.”

Maya thought about the school tank. “The tank had a plan too.”

Leila asked Hadi. He said he did not know, then took the question back to the team. Two weeks later, he returned with an explanation: the photograph showed a temporary arrangement used during a test. The final line was different. Leila was embarrassed until he told her the engineer had asked how she had noticed.

She kept the reply.

It did not make her an expert. It made expertise look like something she could enter through work rather than wait to be granted by age.

* * *

Elena’s first inspection of the regional assembly site took place during a rainstorm.

The route prepared for her avoided the loading area where water had collected beneath temporary covering. Hadi was helping move packaged equipment away from the wet edge when she arrived through the wrong door. The escort tried to redirect her. She asked what they were moving.

“Things that were listed as stored indoors,” Hadi said.

The escort explained that the permanent building was delayed. The temporary arrangement met the written requirements. Elena looked at the water and asked whether the requirements included weather.

The answer was yes. The evidence was less reassuring.

They moved the most vulnerable equipment first. Elena called the construction manager and asked which material was needed to finish the roof. The material had been assigned elsewhere after a schedule revision. She could return it only by delaying another site.

For half an hour she stood in the loading area comparing the two consequences. Hadi continued moving packages around her. She did not ask the workers to stop while she spoke.

The final decision sent part of the material back and funded a better temporary enclosure for the other site. It cost more than the original schedule. The weather did not refund the difference because the schedule had once looked efficient.

When the inspection report appeared, it included the water damage narrowly avoided and the condition of the temporary store. Hadi showed the report to Ruth. His sentence was quoted without his name.

“Things listed as stored indoors,” she read.

“Apparently that counts as analysis.”

“It usually does when someone else writes it down.”

He laughed. The next rotation, he began keeping a more careful record of observations that did not fit the official categories.

CHAPTER FOUR

The Engine That Was Not

CORRECTION APPENDED TO A BROADCAST TRANSCRIPT

An earlier version of this segment stated that the freight tug had “taken the first step toward steering the Solar System.” The tug moved a rock. We regret the error.

T–4,504 DAYS

The first public failure of Daedalus was a demonstration that succeeded.

A freight tug altered the orbit of a small, uninhabited rock near the Moon. The maneuver tested engines that would later reposition shielding and habitat components. It was slow, accurately measured, and useful.

The broadcast called it the first step toward steering the Solar System.

Elena stopped the interview when the presenter repeated the phrase.

“No. We moved that rock.”

“But eventually—”

“Eventually we will move other rocks. Please do not put a star in the purchase order.”

The clip circulated more widely than the demonstration. People laughed, which helped. Others accused Daedalus of retreating from a promise it had never formally made.

At the modeling center, Amara insisted that every new public graphic distinguish the Sun’s predicted route from the actions humans could actually take. The two lines had previously shared a color.

Changing it provoked an argument that lasted longer than the software change.

Maya’s choice of university course produced less conflict with her mother than her choice of accommodation.

Amara supported engineering. She supported the practical emphasis and the possibility of work on infrastructure. She did not support Maya moving into a small shared flat farther from the campus when a secure residence had been offered through the project.

“It’s a longer journey,” she said.

“I know.”

“The building has had power problems.”

“So has half the city.”

“That isn’t a reason to choose one.”

Maya knew the arguments were not entirely about power. The project residence came with staff, procedures, and people who would report concerns to the same network her mother

inhabited. The shared flat came with three students, an unreliable hot-water system, and a landlord who did not know who Amara was until someone told him.

Maya wanted to discover what she could handle without every difficulty becoming information sent home.

Her grandfather asked whether she could afford it. She showed him the budget. He found an omitted expense, added it, and said the decision still appeared possible if she cooked more often than she presently did.

Amara asked him whether that was his entire advice.

“No. I told her where the fuse box should be and to read the contract.”

“You know what I mean.”

“So does she.”

Maya moved. The first month was harder than the conversation had allowed her to admit. She missed the household routines, spent too much on food, and discovered that sharing a kitchen required negotiations no one put in the engineering curriculum.

When she called Amara, she described the classes and omitted the loneliness. Her mother did the same with her own life. They exchanged accurate fragments and called it keeping in touch.

* * *

The first group project exposed a different problem with her surname.

One teammate deferred to her even when she was guessing.

Another challenged her more sharply than he challenged anyone else.

Maya responded by preparing so thoroughly that preparation consumed time needed for the actual work. She wanted no mistake that could be attributed to an advantage she had not asked for.

Leila found her in the library late one evening, surrounded by notes.

“What are you trying to prove?”

“That I know what I’m doing.”

“Do you?”

“Some of it.”

“Then prove that part and ask about the rest.”

Maya said it was not that simple. Leila agreed and stayed while she finished the section that truly needed finishing. Then they left together. The unfinished notes remained on the desk. Nobody’s life depended on them that night, though Maya’s body had begun treating most work as if it did.

At the next project meeting, she asked a question before offering an answer. The deferential teammate knew something she did not. The critical one had misread a unit. They corrected both issues and made progress. The room did not become fair or free of assumptions. It became usable.

* * *

A year later, Maya took part in a placement at a regional water facility. The supervisor assigned her to routine records before letting her near the equipment. She was disappointed and tried to hide it. He noticed.

“You want to fix something.”

“I want to learn.”

“Then learn why the last three repairs happened.”

She read the records. The same component appeared in all three, but the causes differed. One was wear, one an installation error, and one a response to operating conditions the component had not originally been selected for. A proposal to replace all of them with a more expensive version would solve only part of the problem.

Maya presented the finding. The supervisor asked her to show it to the maintenance crew. They added details absent from the forms and corrected one of her inferences. She rewrote the account.

On her final day, he let her assist with an inspection. She spent much of it holding a light while a technician worked a stripped screw out of a pump frame with a hacksaw blade, a pair of pliers, and a vocabulary he apologized for afterward. The pump room smelled of chlorine and warm oil. Condensation ran down the pipe he was leaning on and into his sleeve. Maya recorded what he could see and she could not. It was less impressive than the work she had imagined. It made the records she had been reading feel inhabited by people.

She wrote about the placement in her lunar application. She left out the phrase hands-on experience after realizing it described almost nothing by itself. Instead she described the component, the three failures, and the question she had learned to ask before recommending a replacement.

When the interview invitation arrived, she showed Leila first, then her grandfather, then called Amara. The order felt like a small declaration of adulthood. She did not tell anyone that was why it mattered.

* * *

Before the Moon, there had been the admissions interview.

Maya arrived early, wearing a jacket her grandfather had ironed badly. She discovered the crease across one sleeve while waiting outside the room and spent several minutes trying to flatten it with her palm. The other applicants seemed to have chosen clothes that did not reveal how much they wanted to be taken seriously.

The interviewer knew her surname. Of course she did. Maya had rehearsed an answer about wanting to contribute independently of her mother. When the first question concerned a failed university project, the rehearsal became useless.

She described a water-distribution model that had performed well against the supplied data and poorly against the measurements her group collected afterward. The team had found a sensor problem, corrected it, and improved the fit. She stopped there because that was the version in the project summary.

“What did you do wrong?” the interviewer asked.

Maya repeated the sensor issue.

“That was the equipment. What did you do?”

She looked at the table. “I dismissed the first measurement because it disagreed with the model.”

“For how long?”

“Two days.”

“Why did you change your mind?”

“Someone in the group kept asking me to explain the disagreement.”

“Were you pleased?”

Maya smiled reluctantly. “No.”

The interviewer made a note. Maya could not tell whether she had damaged her chances or finally begun answering the question.

The technical exercise that followed was simple enough to make her suspicious. A container had to move through a constrained space.

The drawing supplied dimensions, load limits, and a handling sequence. Maya began calculating clearances. After several minutes, she realized the drawing did not state the mass of the equipment used to move the container.

She asked for it.

The interviewer gave her another sheet. The revised total exceeded a limit. Maya crossed out her first answer and proposed a different sequence.

“Would you sign that?”

“After checking the attachment points.”

“They’re on the drawing.”

“I’d want to see the equipment.”

The interviewer finally smiled. “So would I.”

* * *

The acceptance arrived while Maya was helping Leila carry plants up the stairs to a university laboratory.

She read the first line, stopped, and nearly tipped the tray. Leila rescued it before asking what had happened. Maya showed her the message.

“You got it.”

“I got it.”

They stood on the landing while people squeezed past. Someone asked whether they intended to grow roots there. Leila picked up the tray and told Maya to celebrate while walking.

Amara called that evening. She sounded proud and immediately began explaining people Maya should contact before leaving. Maya let her finish, then said she wanted to arrive without a list of introductions from the chief encounter scientist.

“They might help.”

“That’s the problem.”

Amara was quiet. Maya worried she had been ungrateful. Then her mother said, “All right. I’ll send you the practical travel information, and you can ignore the rest.”

She sent six pages. Maya used four of them.

Her grandfather gave her a small sewing kit. She said the placement surely had equipment for clothing repairs. He said that was no reason to send a loose button through a formal request process. She took it, and later used it more often than several things on the official packing list.

Leila gave her nothing at first. On the final evening, she handed over a folded sheet containing a terrible drawing of the school tank, labeled with the names of the parts they had misunderstood at sixteen.

“For professional reference,” she said.

Maya put it inside the notebook.

* * *

Five years after discovery, Maya joined a lunar fabrication team as part of her engineering course. She was twenty-one and had learned to keep her surname off the first page of applications.

Her supervisor, Saira Nwosu, noticed anyway.

“If your mother calls about delivery dates,” Saira said, “you are answering.”

The factory made thin reflective panels for controlled agriculture and heavier panels for shielding spacecraft. Most of the agricultural light would still come from the Sun through greenhouse roofs on Earth. Mirrors and electricity could supplement it in limited areas. No one at the factory believed they could replace a missing Sun.

The Moon did not look like the place Maya had imagined from Earth.

From the descent window, the inhabited area appeared as a collection of practical interruptions in an enormous dark surface. Roads followed the machinery that had made them. Berms concealed pressure structures. A line of parked cargo modules caught sunlight with such brightness that the camera dimmed everything around them.

She searched for the elegant settlement from the recruitment film and found its observation dome behind three storage tanks.

At the terminal, a recorded voice told arrivals to keep their luggage close to their bodies. Maya understood the instruction when she turned too quickly and her bag pulled her into the person behind her.

“First time?” he asked.

“Is it that obvious?”

“Only because you apologized before checking the bag.”

His name was Pavel. He was returning from leave, carrying a paper drawing of a horse that his niece had made. The horse had six legs. He said the extra pair was for difficult terrain and challenged Maya to find a more sensible arrangement.

They shared a transport into the settlement. He pointed out the sealed doors without making them sound frightening. Each had a number, an inspection mark, and a manual release protected by a cover.

She noticed people glance at the indicator lights as they passed, so quickly it almost looked unconscious.

“You start doing it too,” he said.

“Do they fail often?”

“Things fail. Doors are usually kind enough to tell us.”

He nodded at a bulkhead whose inspection mark had been renewed more often than its neighbors’. “That one’s called Gerald. Gerald sticks on Tuesdays. Nobody knows why. Three people have written reports about it.”

“Is that true?”

“The sticking is true. The reports are true. The name was a vote.”

Her room contained a narrow bed, a desk, and a cupboard whose latch required a firm upward push. The window was a display. It offered a choice of landscapes, none of them the place outside. She turned it off and found that the dark surface reflected her face.

She called home after unpacking. Grandfather asked whether she could see Earth. When she said not from the room, he looked offended on her behalf.

“You went all that way for a bad window.”

Amara asked about the induction schedule and whether Maya had saved the local emergency instructions offline. Maya had. She wanted her mother to ask whether she was excited.

Leila called later, after the family’s messages had finished. She was due at the biological test facility the following month for a short placement. The two departments had separate travel budgets, a fact Maya had begun to regard as a personal enemy.

“How is it?” Leila asked.

“The cupboard doesn’t open.”

“Have you tried reading the instructions?”

“There aren’t instructions for a cupboard.”

“There will be now.”

Maya laughed. Then, unexpectedly, she began to cry.

Leila waited. She had learned not to fill every silence simply because it came through a device.

“I wanted to be here,” Maya said.

“You can want it and miss home.”

“Everyone keeps telling me how important it is.”

“Then tell me about the cupboard.”

Maya did. By the end of the call she had opened it. The achievement was small, satisfying, and entirely her own.

* * *

Saira’s first lesson was how to stand still.

On Earth, Maya had trusted her weight to keep her where she intended to be. Here, an impatient reach could move the rest of her.

Tools did not fall quickly enough to be caught where her hands expected them. A dropped washer could disappear beneath a machine while she was still trying to decide how it had escaped.

Saira gave her a tray of fittings and told her to assemble a simple junction. She watched without helping. Maya completed it, reached for the test lead, and nudged the tray with her elbow. Three fittings scattered.

“Again,” Saira said.

“The junction is finished.”

“The job isn’t.”

Maya collected the fittings. One had rolled into an inaccessible space behind the bench. Saira asked whether she could account for all of them. Maya said no.

They spent twenty minutes retrieving the missing piece.

Only afterward did Saira explain that loose metal near an exposed mechanism could become a problem for someone on the next shift. The point was not tidiness. It was leaving behind a system that did not require another person to guess what had happened.

Maya recognized the principle from lectures. Recognizing it while kneeling beside a bench was less pleasant.

Pavel joined them near the end and found the fitting with a small mirror. He showed Maya the method, then made her put the mirror back in its labeled place.

“The gap behind the bench is called the archive,” he said. “Everything goes in. Nothing comes out unless you ask it nicely.”

“You’re all enjoying this,” she said.

“A little,” he admitted.

At lunch, he told her about his first damaged component. He had insisted it was already cracked when he received it. His supervisor had asked him to write that down and sign it. He had spent an hour discovering the difference between believing something and being prepared to make the next person rely on it.

“Was it already cracked?” Maya asked.

“I still don’t know. That’s what I should have written.”

She began keeping a separate column in her notebook for things she had not checked.

* * *

Tomas arrived in her second week to reconcile the factory’s output with its shipping commitments.

He remembered Elena’s visit years earlier and the argument over material grades. The program had grown since then. So had the distance between a factory decision and the people affected by it.

Maya expected a man who would ask everyone to work faster.

Instead he asked Saira which delivery commitments she least believed.

She gave him four. He crossed out two after reading the maintenance record.

“You can do that?” Maya asked.

“I can report that the date is fiction. Someone else can decide whether to keep printing it.”

He showed her the freight model. At first it looked like a complicated map. Then he selected a delayed panel and followed its consequences through several sites. A shipment missed an assembly window. A structure waited without its roof. Equipment intended for the completed structure occupied space needed by the next delivery.

“People think logistics is moving things,” he said. “Often it’s stopping the wrong thing from arriving too early.”

Maya found that more interesting than she expected.

He gave her a small reconciliation task. Two sites claimed the same reserve stock. She traced the duplication to an old project code, corrected it, and sent him the result.

He sent it back.

“Who has the actual stock?”

She had corrected the database without asking.

When she called the depot, half the material had already been used in emergency repairs. The repaired system supported both sites.

Nobody had recorded the transfer because the original authorization had been verbal.

Tomas helped her reconstruct it without turning the error into an accusation. He made sure the depot supervisor saw the corrected record before it circulated.

“People tell you more when they think you want the truth,” he said, “instead of a person to attach it to.”

Maya would remember that sentence after his arrest. For years before then, it was simply good advice from someone who treated her as capable of learning.

* * *

Leila arrived with a case of biological samples and a travel bag she had repaired with bright green thread.

Maya met her at the terminal. For a moment they stood looking at each other, unsure what to do with the lack of a screen between them.

Then Leila handed over the sample case and embraced her.

“Careful,” Maya said. “You can knock people over here.”

“You told me that eight times.”

“You kept ignoring it.”

They spent the first evening in the observation room Pavel had recommended. Earth hung beyond the glass, intensely bright against darkness. Maya had expected to cry when she first saw it from the Moon. Instead she had felt a peculiar embarrassment, as if she had been shown something private belonging to everyone she knew.

Leila stood close enough for their shoulders to touch.

“That can’t be all of it,” she said.

“It is.”

“I know it is. It still can’t.”

They had kissed once before Maya left Earth, in the narrow space beside the school garden where the replacement tank had finally been installed. Neither had described what it meant afterward. The months since had contained enough teasing and careful silence to make the omission increasingly ridiculous.

Maya turned toward her. “Are we going to keep pretending that didn’t happen?”

Leila looked away from the window. “I was waiting for you to submit a diagram.”

The second kiss was less tentative. Their attempt to stand comfortably afterward was defeated by the unfamiliar gravity, and both began laughing.

They did not discuss the black hole that evening. They found a cafeteria still serving food and argued about a film Leila had watched on the journey. Maya showed her the cupboard latch. Leila said it was not nearly as interesting as the previous twenty messages had suggested.

For three weeks, they had ordinary things to learn about each other.

Leila left cups in places Maya regarded as structurally irresponsible.

Maya set alarms for tasks Leila believed she could remember. They negotiated the use of the desk, the best time to call home, and whether Leila’s sample labels counted as work at breakfast.

The biological trials were promising but difficult. Leila spent two nights investigating an unexpected fungal growth and arrived at dinner furious with a sensor that had reported conditions it had not actually measured. Maya told her about the duplicated reserve stock. They compared the ways a tidy record could conceal a wet floor.

It was the beginning of a shared language. Neither knew yet how much they would depend on it when they began wanting different lives.

The first practical examination on the Moon required Maya to identify a fault without taking anything apart.

Saira placed a small pump assembly on the bench and told her it was delivering less than expected. The instrument record showed fluctuations. The external surfaces looked sound. Maya asked for the service history and received a page with two entries missing.

“Is that deliberate?”

“The history is what the history is.”

Maya put her palm on the housing and felt the motor run. Warm, not hot. Under the hum, a faint irregular rhythm, like someone tapping a table while they thought. She listened to the pump, checked the connections, and started a list.

Cavitation. Impeller wear. Blocked inlet. Loose sensor lead. Wrong pump for the head.

Saira asked which observation would distinguish them.

Maya asked to change the operating condition. Saira allowed it within stated limits.

She lowered the speed. The rhythm slowed with it. Not cavitation, then, or not only. She raised the outlet pressure. The flow dropped further than the curve said it should. She crossed off impeller wear, because wear did not care where the restriction was and this fault did. She pinched the supply line between finger and thumb and felt it stiffen unevenly along its length, soft above, hard below, as if something inside had settled.

Twenty minutes after she had started, she found a restriction in the supply line, and for a moment the bench and the tools and Saira's silence arranged themselves around it with a satisfaction that was almost physical.

She was pleased until Saira asked whether it explained all the measurements.

It did not.

There was a second fault in a sensor connection. Maya had been ready to stop because she had found something real. The lesson embarrassed her more than a completely wrong diagnosis would have.

At lunch, Pavel told her that finding one fault was often the most dangerous moment in a repair.

"Before that, you're uncertain," he said. "Afterward, you want to go home."

She asked whether the exercise had been designed to humiliate trainees.

"No. That part is free."

He showed her an old report in which he had made a similar mistake. The report was not presented as a noble confession. It was a useful record. She read it more carefully because his name was on it.

* * *

Seven weeks into the placement, the transfer line suffered a failure.

The shift before the accident had ended with a dispute over a maintenance window.

Saira wanted the transfer arm taken offline for a complete inspection. The arm's intermittent fault had not repeated under test, but its history was worsening. On the floor the arm had a name. The crews called it the Poet, because it paused in places nobody could predict and nobody could explain. Maya had laughed the first time she heard it. Production wanted another cycle before shutdown. The shipment waiting for the panels had already been delayed twice.

The authorized compromise retained the inspection and shortened a separate setup procedure. It was supposed to save time without touching the local safety check. That was what the written record said.

On the floor, the revised sequence passed through people who understood different parts of it. One crew believed the next crew would verify the interlock. The next crew believed the check had been completed during setup. The checklist retained a tick from the previous cycle.

Maya would later be able to describe every step. During the shift, she saw a machine everyone had used yesterday.

The panel stopped halfway through the gate with an awkward shudder. Pavel raised a hand toward the operator and waited for the arm to halt. The status indicator showed a safe condition. He crouched beside the transfer track, leaving room for Maya to see what he was checking.

She remembered the detail of his sleeve. A small patch near the wrist had been repaired in thread that did not match.

The arm moved.

For a fraction of a second, Maya thought the panel had slipped. Her mind tried to fit the sound into something familiar. Then Pavel made a noise she had never heard from another person, and the possibilities narrowed to one.

She hit the isolation control with the heel of her hand.

The arm stopped. It did not release him.

Saira reached the gate before Maya could move again. She ordered the operator to hold the system isolated, then told Maya to call medical and give the machine number. Maya knew the number. She had written it every day. It would not come into her mouth.

Saira repeated it once, clearly.

Maya made the call.

The next minutes became separate pictures. A tool placed on the floor so it would not be lost. Pavel’s right hand gripping Saira’s shoulder. The medic asking him his name. Someone turning Maya away from the place where blood had collected because there was work she could do without watching.

They needed space around the equipment. She moved the unused material, checked that the passage was clear, and stood beside the door to guide the medical team. Her hands performed tasks her mind could not yet connect to the person on the floor.

When the team took Pavel away, the factory remained quiet. No one restarted the line. The unfinished panel hung in the gate like an object from a different shift, one that had not learned what it had done.

Nobody used the arm’s name again. Leila found Maya outside the medical unit an hour later. She sat beside her without asking for the story.

“I saw the light,” Maya said eventually.

Leila waited.

“It said safe.”

“I know.”

“I didn’t check it myself.”

“Was that your check?”

Maya could not answer.

She went looking for the sentence and the sentence was not there. She got as far as the light and stopped. She got as far as the sleeve and stopped. Whatever she was carrying did not arrive in words, and she waited for it to, the way you wait for a name you are certain you know.

Leila reached for her hand. Maya pulled away.

They sat. Someone came out of the medical unit and went back in. The corridor had a vent that cycled every forty seconds or so, and Maya counted eleven cycles and lost the count and started again, and she was still doing that when she reached over and took Leila’s hand herself.

She did not cry until three days later, in the shower, over a dropped bottle of soap. She told nobody about that. She could not think what it would have explained.

* * *

The investigators interviewed everyone separately.

Maya wanted to give them the sequence exactly. She discovered that exactness was difficult after fear. She remembered the repaired sleeve vividly and could not remember whether Pavel had spoken before approaching the track. She remembered the sound, then doubted its place in the sequence.

The investigator did not ask her to improve the memory. He asked her to mark what she knew and what she inferred.

She thought of Pavel’s story about the cracked component.

When she left the interview, Tomas was waiting in the corridor. He had helped preserve the records and wanted her to know that the original versions still existed.

“Nobody can tidy it up afterward,” he said.

She was grateful for the reassurance. She was also disturbed that he thought it necessary.

Saira stayed away from the cafeteria for two days. Maya found her at the inactive line, reading a maintenance record she already knew.

“They said I did the right thing,” Maya said.

“You did.”

“It doesn’t feel like it.”

Saira looked toward the gate.

“Doing one thing right doesn’t make what happened right.”

She admitted that she had accepted the shortened sequence. She had believed the local checks remained intact. The investigation would decide whether she should have verified more before accepting the change. She intended to stay for the answer.

Maya had never heard a supervisor speak about possible fault without either denying it or collapsing into it. She remembered the steadiness as much as the admission.

Pavel woke properly the following day. The medical team explained what they had been able to do and what they had not. They had saved his life, but not his left hand. His family joined through a delayed connection from Earth. His sister kept asking whether he was in pain. He kept telling her she had already asked.

Maya visited with Leila. They brought nothing because they could not agree on an object that would not seem absurd.

Pavel looked at Maya’s hand, still bruised from striking the control.

“That button is not supposed to fight back,” he said.

She tried to smile and failed.

He stopped joking. “Sit down.”

They sat.

For a while they discussed the six-legged horse. His niece had sent a new drawing, this one with a bandage around one leg. Pavel said the creature was better equipped for setbacks than he was.

Then he asked whether the machine was still stopped.

“Yes,” Maya said.

“Good.”

Neither asked the other for forgiveness. He was tired, angry, relieved to be alive, and unwilling to make any of them feel better by simplifying that mixture.

After the visit, Leila walked Maya back to her room. At the door, Maya said she was thinking of ending the placement.

“Do you want to leave the work?”

“I want to stop seeing it.”

Leila nodded. “That may not be the same thing.”

They sat on the narrow bed with their shoes still on. Maya told her the whole sequence, including the parts she did not know. Leila listened until it was dark on the window display neither of them used.

While Leila was back at the lunar growing laboratory, Maya found the sewing kit in her locker while looking for a clean shirt.

One of the small compartments had opened. Needles lay loose against the lining. She sat down and collected them with exaggerated care, counting each one against the printed list. It took several minutes to understand that she was doing this because the task had an answer she could reach.

Leila called during the counting. Maya nearly declined. Instead she answered and showed the kit.

“I’m being ridiculous.”

“You’re picking up needles.”

“For ten minutes.”

Leila did not tell her to stop. She asked where the needles would go when they were collected. Maya showed her the compartment. Leila suggested a strip of cloth to keep it closed until the latch could be fixed. They talked while Maya finished. Not about the accident at first.

Leila described a plant trial that had failed because the labels had faded under the cleaning treatment. Maya told her about the sewing kit.

Eventually she said she kept remembering Pavel’s voice and wondering whether she had acted quickly enough.

“What does the record show?”

“That I did what I was trained to do.”

“And you don’t feel that.”

“No.”

Leila was quiet for a moment. “Maybe the feeling won’t read the report as quickly as you did.”

Maya folded the cloth over the needles. It was not an explanation that solved anything. It gave her permission to stop treating her distress as a second technical failure.

* * *

Maya read the approval chain three times. Her mother’s name was on the schedule that had begun it. Her mother’s signature was not on the unsafe procedure. The difference mattered legally. It did not help when Pavel tried to open a food packet with one hand.

“Did you tell her?” Saira asked.

“She’ll say she didn’t know.”

“Did she?”

“No.”

Saira waited.

Maya filed the report with the photographs attached.

Amara called that evening. She began by asking whether Maya was hurt, then asked whether Pavel’s family had arrived. Only after that did she discuss the schedule.

“We can move the milestone.”

“We could have moved it before the accident.”

“Yes.”

Maya had prepared an argument. The admission left her with nothing to push against.

“I need you to read the whole report,” she said.

“I will.”

“Not the summary someone gives you.”

Amara looked tired, but she did not look away.

The next production review included maintenance hours and injuries beside shipment totals. It did not end the pressure to hurry. It gave the pressure a name and a place where someone had to answer for it.

The line reopened after a new local check had been demonstrated to every shift.

Maya attended the demonstration. She understood the changes and still felt her breathing become shallow when the arm began to move.

Saira asked whether she wanted to leave. Maya said no, then changed her mind halfway through. Nobody treated the change as a failure.

Two weeks later, she returned and stayed.

Her final placement project was no longer the efficiency analysis she had proposed before arrival. She traced a single production deadline through the decisions it altered, including decisions outside the factory. She interviewed the people who had shortened the sequence, the people who had relied on it, and the customer waiting for the panels.

The customer was building agricultural cover over a community that had already dismantled its older structures. Their urgency was real.

So was the injury. The report became stronger when she stopped looking for a single unreasonable person to carry both.

Tomas helped her make the dependencies legible. He warned her that the final version would make several departments uncomfortable.

“They can correct it,” she said.

He smiled. “Keep saying that.”

At the departure terminal, Pavel came to see her off. He was beginning rehabilitation and deeply bored with being congratulated for progress he considered compulsory.

He gave her the small inspection mirror she had used to retrieve the missing fitting. The handle had been repaired twice.

“You’re giving away tools?” she asked.

“This one has become sentimental. Terrible quality in a tool.”

A trainee waiting for the same ferry glanced at Pavel’s left sleeve, then quickly away. Pavel lifted the sleeve slightly.

“It’s all right. I can still give directions. They’re just all to the right now.”

The trainee laughed, startled. Pavel let him, then looked at him steadily until he stopped.

“That one’s mine,” Pavel told Maya when the trainee had gone. “He has to earn his own.”

Leila was returning on the same ferry. They found their seats, secured their bags properly this time, and watched the settlement recede.

Maya expected relief. What she felt instead was a desire to return when she knew more.

Leila took her hand during the departure burn. Neither pretended it was accidental.

Maya changed her course specialization at the end of the year. She still wanted to work in space. She wanted more urgently to understand how a system could report success while a person lay beneath it.

CHAPTER FIVE

Eight Billion Astronauts

PRODUCT LISTING, NAIROBI MARKET

GRAVITATIONAL DISTURBANCE WATER FILTER. Removes consequences. Approved.

Customer review, one star: “It filters water.”

T–3,982 DAYS

Maya and Leila rented their first flat above a shop that repaired household appliances.

The shop’s compressor started before either of their alarms. For the first month, Maya woke thinking she was still on the Moon and listening to a pressure system change its mind. Then the sound became part of home.

They had enough money for the deposit if they postponed several things they had both described as necessary. Leila wanted a better microscope for her course work. Maya wanted a desk that did not have to be cleared for meals. They bought neither and discovered that necessity could be negotiated more easily than resentment.

Their first argument concerned a shelf.

Maya measured the wall, marked the brackets, and began drilling.

Leila came home to find plaster dust over a tray of seedlings.

“You could have moved them.”

“I covered the table.”

“You covered half the table.”

The argument acquired subjects that had nothing to do with the shelf. Maya said Leila used every available surface as a laboratory.

Leila said Maya treated living together like an installation requiring final approval. Both were partly correct and far too satisfied by the other’s error to admit it.

They ate dinner without using the new shelf. Later, Maya cleaned the dust from the leaves one at a time. Leila watched for a while, then moved the seedlings to a place where they were less likely to become an accidental referendum on the relationship.

The shelf remained slightly crooked. Neither fixed it.

Outside the flat, the city was adapting unevenly to Daedalus. Some buildings displayed readiness notices that meant real inspections had occurred. Others displayed the same notices because a landlord had found the design online. New training centers opened beside shops

selling emergency equipment of uncertain usefulness. People wanted actions they could take, and an entire market had developed around giving their fear something to purchase.

The catalog of reassurance grew quickly. There was black-hole insurance, with an exclusion clause for acts of gravity. There were pre-owned ark berths, all of them fraudulent, several of them sold more than once. A dating service advertised to people looking for someone to spend the next eleven years with. Two rival prophets in the same district disagreed about which constellation was the sign. Their congregations met once in the market to settle the question, and went home having bought vegetables.

Leila's mother bought a small water filter that claimed to remove consequences of gravitational disturbance.

Leila took it apart at the kitchen table. It was an ordinary filter with an extraordinary label.

"It still filters water," Ruth said.

"Then it should say that."

"The man said it was approved."

"By whom?"

Ruth looked hurt rather than persuaded. Maya touched Leila's arm beneath the table. Later, on the walk home, Leila said she was tired of having to choose between being kind and being accurate.

"You don't," Maya said.

"That is exactly what someone says before telling you how you should have done both."

Maya let the subject rest. The following weekend, they bought Ruth replacement cartridges for the filter from a supplier that described what they actually did. Leila apologized for making her feel foolish.

Ruth said she had not felt foolish until the apology, and they both laughed.

It was a small rehearsal for the conversations they would keep having with frightened people, including themselves.

* * *

The number appeared that spring on the back page of the Nairobi Standard, beside the football results.

ORBIT ODDS, it said. CHANCE EARTH STAYS BOUND: 74%.

Nobody at Daedalus had produced it. A betting exchange had begun taking wagers on Earth's final orbit, and a sports editor had decided readers deserved to see the line. Within a month, three other newspapers carried it. Within two, people quoted it at bus stops in the tone they used for the rains.

Amara’s office declined to comment on what its statement called “not a number.” The next morning the exchange added a footnote: Daedalus declines to comment. The odds rose a point.

“They think you’re hiding good news,” Luis said.

“I’m not hiding anything.”

“That’s exactly what someone hiding good news would say.”

Maya found it harder to dismiss than her mother did. The number was crude. It flattened every scenario Amara had spent years separating. It was also the only figure most people she knew could repeat, and they repeated it with a hope that no confidence interval had ever produced.

Grandfather cut it out of the paper each week and kept the clippings in a jar beside the seed packets.

“For research,” he said.

“It isn’t research.”

“Then it’s a record of what people believed. That’s research too.”

* * *

The graffiti reached their street through a child with a borrowed stencil. He had painted ASTRONAUTS correctly on the first wall and omitted a letter on the second. Maya found him trying to repair the mistake with a brush too wide for the space available. The blue Earth beneath the words looked more like an egg.

“It’s a different projection,” he told her before she said anything.

She held the stencil while he finished.

Leila photographed the result. By evening, the image was circulating with a caption identifying Maya as the artist. She asked for a correction. The child was delighted by the original mistake and disappointed that anyone wanted to remove it.

Grandfather saw the photograph and said she should have used a smaller brush.

Amara saw it and asked whether the wall belonged to anyone they knew.

Maya kept the photograph because Leila was reflected in a window behind them, laughing at something outside the frame.

* * *

The first time Maya saw the lens herself, she was on the roof with borrowed binoculars and a printed star chart that Leila kept turning the wrong way up.

Professional instruments had been tracking the distortion for years. By then it had grown large enough for patient amateurs. The black hole itself was still invisible: at that distance its

shadow was far smaller than anything an eye could resolve. What could be seen was what it did to light passing nearby. In a patch of sky about the width of the Moon, faint stars sat slightly out of place, and one of them appeared twice.

“That’s it?” Leila said.

“That’s it.”

“It looks like someone smudged the sky with a thumb.”

Maya adjusted the focus. The doubled star was an ordinary star far behind the lens, its light bent around both sides of the mass into two images. Her mother had explained the geometry when Maya was twelve, using a glass of water and a pencil.

“It’ll get bigger,” Maya said. “The closer it comes, the wider the region it bends.”

Leila lowered the binoculars. “You say that like it’s a feature.”

“It’s a clock. You can watch it.”

They stayed until the patch had sunk toward the rooftops of the next street. Leila said she preferred the stars where they were supposed to be. Maya said she did too, and kept looking.

* * *

The phrase appeared on a retaining wall outside the Daedalus launch complex in Kenya.

8 BILLION ASTRONAUTS

NO PASSENGERS

By the time the municipal crew came to remove it, someone had painted a blue Earth beneath the letters. The crew cleaned the neighboring wall instead.

Maya first saw it printed on a lunch container aboard the lunar return ferry. Its owner was a refrigeration mechanic who had never been farther from Earth than this contract required. He said his daughter had put the sticker there.

“She says I’m an astronaut now.”

“You are.”

“Mostly I stop pipes freezing.”

Maya thought of the agricultural facilities waiting for those pipes.

“That seems like astronaut work.”

Rain did not respect a ministry. Disease did not stop at a passport desk. A crop failure in one place became an argument about prices in another, then an argument about borders.

The new vocabulary did not solve those problems. It made certain evasions harder to sustain.

At a water-allocation hearing in Nairobi, Maya watched a delegate propose excluding an upstream monitoring station’s data from public release. A farmer in the gallery asked how anyone downstream was supposed to plan a planting season without it.

The delegate cited national security.

“Then tell us which part of the rain is classified,” the farmer said.

The room laughed. The proposal still passed its first vote.

It took six months, a court challenge, and an agreement over equipment maintenance to reverse it. Maya remembered that longer than she remembered the joke.

The water dispute began with a broken gauge, not a philosophical argument about humanity.

Several farming groups believed the upstream supply was lower than the published figure. The agency responsible said the difference fell within normal uncertainty. A private operator farther upstream would not release its detailed records, citing a security agreement with the government.

Leila’s trial plots depended on the disputed supply. She asked Maya to help compare the available measurements. They spent a weekend discovering that the published total combined instruments that did not all report at the same interval. One gauge had been offline for weeks. Its last reading had been carried forward with a small marker that vanished when the table was copied into the public summary.

Maya wanted to send the corrected calculation directly to the agency. Leila wanted to visit the people who had been complaining before either of them knew there was a problem.

They went together.

The growers met beneath a roof patched with material from three different structures. A woman named Wanjiru showed them the notebooks in which she recorded watering times, soil condition, and yield. The books contained the information the official dataset lacked, in handwriting that changed as different family members took over the work.

“We’ve sent copies,” she said. “They tell us the observations aren’t standardized.”

Maya asked what she wanted from the hearing.

“A number we can plan with. If there’s less water, say there’s less. Don’t let me plant with a number you know isn’t real.”

The request sounded so modest that Maya initially underestimated how difficult it would be to secure.

At the hearing, the operator’s representative explained the risk of publishing infrastructure details. The growers asked for aggregated measurements. The agency said aggregation required staff time it had not been funded to provide. A council member suggested the groups contribute toward the cost.

Wanjiru stood before she was recognized.

“We paid for the measurements. We paid when the gauge broke. Now we pay to be told what it says?”

The chair asked her to remain seated. She did, slowly.

Maya gave evidence about the carried-forward reading. She expected the technical point to resolve at least that part of the dispute. Instead the representative asked whether she was speaking for Daedalus, for her university, or as the daughter of the chief encounter scientist.

“As the person who checked the table,” she said.

The answer earned approving murmurs and accomplished little.

The proposal to withhold the detailed release passed its first vote. Outside, Leila was furious. Maya began outlining an appeal.

Wanjiru interrupted them both.

“The next planting decision is before your appeal.”

They spent the following weeks arranging temporary measurements with equipment the groups could inspect themselves.

The instruments were imperfect. So were the locations available to install them. Maya published the limitations beside the readings. Leila trained local volunteers and returned often enough to learn which instructions were unclear.

One instrument was damaged in a storm. Another disappeared. A third produced a series of implausible measurements because a well-meaning volunteer had moved it to keep it safe. The work was slower than being right at a hearing. It also produced something people could use.

When the court eventually ordered a revised data-sharing agreement, the temporary network already existed. The agency offered to absorb it. The growers asked whether they would retain access to the raw readings. This time the answer had to be written down.

On the last installation visit, Wanjiru gave Maya a bag of beans from the previous harvest. Leila carried it home because Maya’s hands were full of instruments.

“We spent six months getting someone to publish water,” Maya said.

“We got them to maintain the gauges too.”

“You’re determined to make this sound good.”

Leila looked out of the vehicle window. “I’m trying to decide whether this is what I want the rest of my life to look like.”

Maya assumed she meant agricultural research. Later she would remember the sentence differently.

* * *

Tomas visited their flat that winter with a bottle of wine and a stack of printed tables.

Leila told him he had misunderstood the invitation. He put the tables away and helped cook. The wine was not very good. He admitted that he had chosen it because the label contained a reassuring amount of information.

They talked about the water case, the lunar accident, and a training program Tomas was trying to establish for new analysts. He wanted each trainee to spend time at a physical destination before being allowed to classify a shipment as complete.

“A week in a warehouse would improve half our profession,” he said.

“Would the other half survive it?” Leila asked.

“Not without feedback.”

After dinner he showed Maya the tables. They concerned correlated supply failures. Sites that looked independent on a map relied on the same small set of suppliers and maintenance assumptions.

A delay could move through the network like a disease while each local schedule still looked reasonable.

“I’m trying to get the review group to include this explicitly,” he said. “They say it’s already covered in the margins.”

Maya studied the examples. “Is it?”

“Not if the margins fail together.”

She asked whether he had published the analysis internally. He had. He was waiting for a response.

Before leaving, he noticed the crooked shelf and reached automatically to check its bracket. Leila told him it was an important historical artifact. He left it alone.

Maya remembered him that way for a long time: helpful, a little awkward, and concerned about things other people found easier not to examine.

General Marcus Reyes joined Daedalus during its sixth year. He had worked in planetary defense when the principal concern was tracking asteroids. Now he had to consider people controlling machines that could move them.

At his first security briefing, Elena asked him to keep military staff out of the production loop.

“That’s where I would prefer to keep them,” he said. “Show me the independent authorization path.”

She did.

“And who verifies the destination after a route change?”

“Pilot.”

Pilot was the name given to the scheduling and prediction network that had grown around the project. It coordinated freight, solar forecasts, food storage, observatory time, and millions of smaller decisions. Its components came from different institutions. The common interface made them look more unified than they were.

Marcus asked who verified Pilot.

Elena showed him the audit network.

“Does it use the same inventory?”

For the first time in the meeting, she paused.

Maya, attending as a junior systems analyst, wrote the question down.

Outside the conference room, Marcus caught up with her.

“You saw the problem?”

“Two computers can agree about a shipment that never left.”

“Good. Make someone open the warehouse.”

She expected him to say more. He went to his next meeting.

The instruction stayed with her. In the following years she learned how often agreement came from shared assumptions, how easily a missing measurement could acquire the authority of a fact, and how much work it took to persuade a tired person to count something again.

* * *

Marcus’s second conversation with Maya took place in a training room rather than a conference room.

He showed her two cargo records. Both had valid signatures. Both described legitimate work. Together, they assigned the same propulsion unit to incompatible routes.

“Which one would you stop?” he asked.

“I would check the unit’s actual state.”

“Assume the telemetry is delayed.”

“Then neither gets a new command until we reconcile it.”

“What does that cost?”

She opened the dependent schedules. One route served a medical supply facility. The other supported a habitat under construction.

Holding both was safer for the machine and potentially dangerous for the people waiting.

“You can’t give me a choice where every answer hurts someone and call it a test,” she said.

“That is most of the job.”

He let her work through it. The final answer preserved a limited set of actions while denying the disputed commands. It reduced the risk without pretending to remove it. Marcus asked her to document who would bear the delay.

“People stop believing safety rules when the cost is always invisible to the people writing them,” he said.

Maya asked why he had left operational command for this work.

He took longer to answer than she expected.

“Because I have followed instructions that were clear, authorized, and wrong. I would prefer to make fewer of those available.”

He did not tell her a war story. She was grateful. The sentence was enough to change how she heard his objections in meetings.

At home that evening, Leila was studying habitat recruitment material. She closed it when Maya entered, not quickly enough to seem accidental.

“For your research?” Maya asked.

“Partly.”

The word remained between them while they prepared dinner.

Maya could have asked what the other part was. She was tired, and she wanted one evening without a new uncertainty.

She asked about Leila’s day instead.

* * *

Maya sometimes thought back to their first year in the flat, before the ark work complicated every plan. Their first dinner party had failed before anyone arrived.

Maya had volunteered to cook because Leila was working late. She chose a recipe that required three things they did not own and one ingredient she had confused with another. By the time Leila came home, the kitchen window was open and Maya was trying to decide whether the burnt smell belonged to the food or the pan.

“Both,” Leila said after inspecting it.

They bought food from the shop downstairs and put it in their own bowls, then immediately told the guests because the attempted deception was funnier than the success would have been.

Tomas came with a bottle and a packet of something he said did not require cooking. The other guests were two colleagues from Maya’s program and a friend of Leila’s named Salma, who worked in primary education and became increasingly unimpressed as the conversation filled with abbreviations.

“Do you all speak like this at breakfast?” she asked.

Leila pointed at Maya. “Only when she wants me to leave.”

They changed subjects. Salma described teaching children whose parents had begun saving every exercise book as though paper itself were a survival reserve. One child had asked whether the black hole would take marks off for spelling. The room laughed, then became quiet because the joke had arrived through a child’s real fear.

Tomas asked what the school needed. Salma said more adults who could answer questions without giving either a speech or a promise. He offered to visit a class. She warned him that expertise would not protect him from being asked why the Moon looked larger near the horizon.

“I’ll prepare,” he said.

Maya remembered that promise years later, when she was trying to reconcile the man at the table with the man in the interview room. He had gone to the class. Salma had sent photographs of the children holding his improvised demonstration materials. Kindness had not been a disguise every time he showed it.

After the guests left, Maya and Leila sat on the floor because all the chairs held dishes. They were pleased with the evening in a way neither could attribute to work. Maya said they should do it again. Leila made her promise not to choose a recipe while on a call.

* * *

The water dispute began affecting the neighborhood before Maya understood the regional case.

Pressure fell during hours when people were home to use it. The published schedule blamed maintenance, which was partly true. The larger change involved power allocation and increased demand at a project site. Each office described its own contribution accurately enough to avoid acknowledging the whole inconvenience.

Ruth started filling containers before her evening shift. A neighbor with limited mobility could not carry them upstairs. Leila organized a rota to help, then became angry when someone described the arrangement as community resilience in a public report.

“It’s people carrying water because the schedule is wrong,” she told Maya.

“The report doesn’t say it’s ideal.”

“It makes it sound like a benefit.”

Maya read the paragraph again. Leila was right. The language converted unpaid effort into evidence that the system was working.

At the next review, Maya asked for the hours spent compensating for service changes to be counted. A planner said those numbers would be unreliable. She asked whether omitting them made the estimate more reliable or merely easier to defend.

The team added a local survey. The results were uneven and incomplete. They were also large enough to change the comparison between two proposed schedules.

Maya brought the revised plan home. Leila looked at it, then asked who had been paid to gather the data. Maya had anticipated the question this time. Leila smiled and kissed her on the cheek.

“You are trainable.”

“The evidence is mixed.”

* * *

Marcus’s first visit to the operations center made Maya expect a speech about discipline.

He gave none. He asked to watch a shift handover.

The outgoing team described open issues, accepted risks, and pending decisions. The incoming team acknowledged each item.

Marcus took notes. At the end he asked one analyst what she would do if she believed an instruction from Elena was wrong.

The analyst described the formal escalation route.

“And if the decision expires before that route answers?”

She looked at Maya.

Marcus waited.

“I’d call her,” the analyst said.

“And if she’s unavailable?”

The room became uncomfortable. Maya realized they had built a procedure whose final safety measure was her continued accessibility.

She had regarded that as commitment. Marcus regarded it as a missing function.

They ran an exercise with Maya unavailable. The first attempt stalled. The second produced a decision but no clear record of who had authority to make it. The third worked after they assigned limits within which the duty team could act and required a review afterward.

The analyst asked whether she would be protected if a reasonable choice turned out badly.

Marcus said no rule could prevent criticism. The process could distinguish acting within an assigned responsibility from ignoring it, and could require the decision to be judged using information available at the time.

“That’s less protection than I hoped,” she said.

“It’s more useful than a promise I can’t keep.”

Maya took the revised handover home and showed Leila the line naming the alternate decision holder. Leila asked whether that meant Maya could now eat dinner without her phone on the table.

“In theory.”

“Let’s run a test.”

They left the phone in the bedroom. Nothing failed. The next evening, something did, and the alternate handled it. Maya was proud, then briefly jealous, then embarrassed by the jealousy. Leila recognized all three expressions and said nothing until Maya admitted them herself.

* * *

The first time Leila traveled for an ark consultation without Maya, she came back happier than she expected.

She tried to describe the work without making it sound like a sales presentation. The team had disagreed about lighting cycles, access space, and how much agricultural skill every resident should learn.

They had asked her to defend an assumption, rejected part of it, and invited her to return with a better test.

Maya listened while unpacking groceries. She asked whether Anika had been there.

“For part of it.”

“She’s very persuasive.”

Leila stopped talking.

Maya looked up. “What?”

“You made it sound like I enjoyed being persuaded.”

“I didn’t mean—”

“I enjoyed the work.”

The argument did not become large. That was part of why Maya almost forgot it. Leila did not. She had spent years supporting Maya’s movement toward work that mattered to her, and had expected the recognition to travel in both directions without being requested.

Later, Maya asked to see the test Leila was designing. She sat down, put the groceries away first, and listened properly. Leila showed her the constraint she had missed and the new arrangement she wanted to try.

“That looks better,” Maya said.

“We’ll know when we test it.”

Maya smiled. It was a sentence she liked hearing at work. She had to learn not to resent it when the work opened a future she feared.

CHAPTER SIX

The Ark People

DAEDALUS STYLE GUIDE, REVISION 9

Do not use “black hole” in public materials. Preferred: “the gravitational event.”
Do not use “ark.” Preferred: “independent habitat.”
Do not use “escape.” There is no preferred term.

ORBIT ODDS: 72% · T–3,040 DAYS

Leila told Maya about the application on an evening they had both cleared for something else.

They had intended to go out. Maya had made a reservation at a small restaurant that had survived three changes of ownership because nobody in the neighborhood would let it stop serving the same stew.

Leila had dressed before Maya came home. That made the unopened recruitment packet on the table look more deliberate than it was.

“I passed the first assessment,” she said.

Maya set down her bag.

“For the habitat work?”

“For a position aboard one of the independent vessels.”

She had rehearsed the sentence. Maya could hear it.

“You applied.”

“Yes.”

“When?”

Leila named a date three months earlier.

The delay hurt before the destination did. For three months they had talked about planting trials, rent, a broken refrigerator, a visit to her grandfather. During all those conversations, another possible life had existed without Maya knowing where to look for it.

“Why didn’t you tell me?”

“Because I wasn’t sure I wanted it. Because I knew you would hear a departure when I was still trying to understand a job.”

“It is a departure.”

“It could be.”

Maya opened the packet. The position concerned ecological monitoring and crop-system adaptation. Leila had the training, the experience, and the patience the description required. Seeing how well she fitted it made the possibility harder to dismiss.

“You can do this work here.”

“I do this work here.”

“Then why?”

Leila pulled out a chair but did not sit.

“Because I want to build something where the people making a decision have to live with it. Because I want to follow a problem from the measurement to the root of the plant without six agencies deciding which part I am allowed to see.”

“You think a habitat won’t have politics?”

“Of course it will. It will have people.”

“Then that isn’t a reason.”

Leila looked at her for a long moment. “It’s a reason you don’t like.”

The distinction angered Maya because she recognized it. She had made it to her mother often enough.

They missed the reservation. Neither canceled until the restaurant called. Maya apologized to the owner while Leila stood in the kitchen, still dressed to go out.

Afterward they ate toast without sitting together. It would have been easier if one of them had said something plainly unforgivable.

Instead they had said true things in ways intended to hurt.

Later, Maya asked whether accepting the position meant ending their relationship.

Leila came back to the table.

“I don’t know what it means yet. I wanted to tell you before I pretended I did.”

“You already waited three months.”

“I know.”

She did not defend that part again.

* * *

For several weeks, the application became an extra person in the flat.

It occupied pauses in conversations and changed the meaning of ordinary questions. When Maya asked whether Leila wanted to buy a better desk, Leila heard a question about how long she planned to remain. When Leila asked about Maya’s next posting, Maya heard an invitation to make departure easier.

They began avoiding decisions that had once been simple. The shelf stayed crooked. The refrigerator’s intermittent noise became almost comforting because it was a problem whose motives neither had to interpret.

Leila invited Maya to meet the recruitment team. Maya refused the first time, then accepted after realizing that refusing did not make the application less real.

The assessment center occupied an ordinary office building. In the waiting room, applicants sat beside partners, parents, and children.

Some talked eagerly. Others studied their shoes. A man was explaining to his teenage son that an interview was not a commitment. His son asked why he had brought the family medical records.

Maya felt an immediate, unhelpful solidarity with the boy.

Leila’s interviewer was a practical woman with a damaged voice and a habit of waiting after answers that sounded rehearsed. She asked about crop failures, working in small teams, conflict over shared resources, and what Leila did when a colleague ignored a warning.

“I ask why,” Leila said.

“And if the answer is unsatisfactory?”

“I ask someone else to examine it.”

“And if everyone is tired of examining it?”

Leila hesitated. “Then I need to know whether I’m being persistent or wrong.”

The interviewer nodded and wrote something down.

Maya listened through the permitted part of the session. She heard Leila describe work they had done together without making Maya the center of it. The account was accurate. It was also a version of Leila’s life that could continue without her.

Afterward, they walked to a café near the assessment center. Maya asked whether she had been invited as a partner or as someone to be persuaded.

“As a partner,” Leila said. “But yes, I hoped you would understand why I wanted it.”

“I do understand.”

Leila waited.

“That isn’t making me happier.”

For the first time since the packet appeared, they laughed together.

It did not solve anything. It made the rest of the conversation possible. The residential assessment that followed Leila’s initial selection did not ask whether she was brave.

It asked her to spend a week working with people she had not chosen, on equipment that was designed to fail in ordinary ways. The failures were announced as part of the exercise. The personal irritations were not.

Her assigned team included a mechanic who talked while others were thinking, a physician who wanted every change explained before it happened, and a communications specialist whose habit of tidying moved tools from places their users expected to find them. Leila discovered that she became less patient when she was hungry and more controlling when the work fell behind. Neither discovery surprised her. Having strangers document it was unpleasant.

The week also included a session titled Cohabitation Compatibility Module 3: Shared Resources, which nobody admitted to having read about in advance. The facilitator gave each of them a card describing a household dispute and asked them to resolve it in character. Leila's card said she had used the last of the shared nutrient concentrate without logging it. The mechanic's card said he had noticed.

They resolved it in four minutes. The facilitator said that was suspiciously fast and asked them to do it again with feeling.

The second attempt took forty minutes. It ended with the physician arbitrating, the communications specialist reorganizing the props, and the mechanic, who had remained in character throughout, demanding a formal inventory of the concentrate going back three months.

"That," the facilitator said, making a note, "is closer to what it will be like."

On the third day, a growing compartment lost a portion of its scheduled lighting. The team had to choose among competing demands under a fixed power limit. Leila argued for preserving the crop trial.

The physician argued for retaining the planned diagnostic exercise.

Both activities had substitutes, and both substitutes reduced what could be learned.

Leila began explaining the importance of the plants. The physician said she understood their importance and still disagreed.

The facilitator stopped them after twenty minutes and asked what decision rule they had agreed to use.

They had not agreed on one.

The next hour was worse and more productive. They identified which parts of each activity were time-sensitive, what could be postponed, and who would bear the extra work. The final arrangement gave neither everything requested. It also revealed that a small maintenance task could recover enough capacity to reduce the conflict.

The mechanic had mentioned it early. Nobody had asked him to finish explaining because they were busy defending their own systems.

At the review, Leila named that failure before the facilitator did.

She was not pleased with herself for noticing afterward. She was pleased when, on the fifth day, she caught herself repeating the pattern and stopped in time to ask a question.

* * *

During the assessment week, Maya visited Ruth and Hadi without Leila.

She had done so before, but this time the empty chair seemed deliberate. Ruth asked whether Maya knew how serious the application was. Maya said she was beginning to.

Hadi looked toward the kitchen. “We hoped it was another consultation.”

“So did I.”

Ruth put a plate down more sharply than necessary. “She told us it was an assessment. We heard what we wanted.”

They ate for a while without discussing it. Then Hadi asked what the vessel would be like to live in. Maya gave the technical answer first: systems, limits, training, the scale of the commitment. Ruth asked whether Leila would have friends.

Maya did not know.

That answer made the decision feel more real than the machinery had. A person could accept known risks and still find that the shape of daily life mattered most. Who would sit beside Leila when she was tired? Who would recognize when her silence meant concentration and when it meant hurt? Maya could not put those questions into an engineering comparison, and could not pretend they were less important because of that.

Before she left, Ruth gave her food to take home. The container belonged to Leila. Maya recognized a scratch on its lid from their first month in the flat.

“Keep it until she comes back from the assessment,” Ruth said.

All three heard the temporary meaning of comes back and left it there.

* * *

On the final assessment day, Leila was asked to describe a disagreement with someone close to her.

She knew enough about interviews to recognize the temptation to choose a disagreement she had resolved admirably. Instead she described the application she had delayed telling Maya about.

The interviewer asked why.

“I wanted to know it was real before I made it painful.”

“Did the delay make it less painful?”

“No.”

“Would you handle it differently now?”

Leila looked at the wall behind the interviewer. “I would like to say yes. I think I would need to notice that I was calling avoidance preparation.”

The interviewer made a note. Leila wanted to know whether honesty had helped or simply supplied a better reason to reject her.

At the end, the interviewer explained that the assessment was not a search for people who never caused conflict. A confined community could not be built from an imaginary species. They needed people who could recognize their contribution to a problem, accept limits, and repair working relationships without requiring every disagreement to end in affection.

Leila thought of Maya, and of her own impatience when Maya tried to make the future safer by postponing the conversation. They were more alike in that habit than either enjoyed admitting.

* * *

When Leila returned, Maya met her at the station.

For a moment they behaved like any couple reunited after a week: a kiss, a bag taken from one hand, an argument about whether the other had eaten. Then the application stood between them again.

Maya asked whether the assessment had gone well.

“Some of it.”

“Do you want to tell me?”

“Yes. Not as evidence in a case.”

Maya nodded. They walked to a small restaurant they had used before either had enough money to choose it for any reason besides price. The tables were still too close together. A server recognized Leila and asked whether she wanted the usual.

Leila described the mechanic, the physician, the lighting problem, and the interview. Maya listened. When she wanted to argue that the conflicts proved the vessel would be difficult, she remembered that Leila had already agreed it would be difficult. The question was whether difficulty made it the wrong life for her.

“I was good at parts of it,” Leila said finally. “Parts I didn’t know I could do.”

Maya looked at her. Beneath her own fear, she could see what she would have celebrated if the assessment had led anywhere else.

“Tell me those parts.”

Leila did. They stayed until the restaurant began stacking chairs.

Nothing had been settled, but the conversation no longer required either of them to become smaller for the other to remain in it.

* * *

Nine years after discovery, Anika took Maya through the first completed habitat cylinder while its river was still a painted line on the floor.

There were no trees. Plastic sheeting divided the future agricultural sectors, and the air smelled of adhesives. A maintenance worker floated through a service passage beyond a barrier; this part of the structure had not begun rotating.

“People think it will look like the pictures when we turn it on,” Anika said.

“Will it?”

“Eventually. If we keep earning the pictures.”

The vessel was being assembled well inside the region from which a departure remained practical. No one intended to wait beyond Neptune for a black hole to rearrange their options. The ark fleet would leave on calculated independent trajectories years before closest approach, carrying reactors, shielding, spare machinery, and populations small enough for their closed agricultural systems.

They would not accompany Earth all the way. Some might never see the Sun again.

Leila joined them at the habitat’s agricultural sector, carrying a sample tray and looking more at home than Maya wanted her to look.

She had been assigned a temporary research placement while selection continued. Nobody had promised a permanent berth. The distinction appeared on every document and did little to reduce the emotional meaning of her presence.

“You should see the trial room,” she said.

Anika gave them an hour and went to a construction meeting.

Maya followed Leila through a corridor lined with equipment waiting to be installed. Someone had taped a small paper sign to a storage rack:

IF YOU BORROW IT, SAY WHERE IT WENT. Beneath it, another hand had added PLEASE.

The trial room contained plants at several stages of growth. Some looked vigorous. Others looked like the unsuccessful answer to a question the researchers had not yet learned to ask properly.

Leila showed Maya two trays grown under nearly identical conditions. One had developed a patchy failure. The first explanation had blamed the seed. The second had blamed a nutrient imbalance.

Both had been plausible. Neither had survived a careful comparison with the watering record.

“What was it?”

“We haven’t finished finding out.”

Maya smiled despite herself. “You’re making an excellent case for moving here.”

“The plants fail on Earth too. Here I can get the person who maintains the sensor into the room.”

The maintenance technician arrived while they were talking. He had brought the wrong connector and admitted it before anyone asked.

Leila found a spare in a drawer, then wrote the replacement onto the inventory so that the next person would not assume there were still two. Maya recognized the habits. They were things she loved about Leila when they did not point toward departure.

They ate lunch at a temporary table beside a wall of unfinished cable runs. A family at the next table was discussing the vessel’s proposed school timetable. Their younger child wanted to know why every adult conversation eventually became about a schedule.

“Because adults are frightened of forgetting things,” Leila told her.

“Do you forget things?”

“Frequently.”

The child looked reassured by that.

Her mother introduced herself as Farah, a physician whose family had been provisionally selected. The selection had required difficult medical assessments and a debate over whether her father’s care needs could be supported aboard. He had chosen to remain on Earth before the committee reached a decision.

“He says he refuses to spend his last years in a place where every neighbor knows his blood pressure,” Farah said.

She smiled as she said it. Maya could hear the sorrow underneath.

The habitat was not a community of people who had found a clean escape. It was a collection of people carrying different unfinished arguments into the same small volume.

After lunch, an alarm sounded in the trial sector.

It was not a pressure emergency. The tone identified a local equipment problem. Leila stood before the announcement finished.

Maya followed until a technician stopped her at the marked boundary.

Inside, a circulation unit had stopped and a local temperature reading was climbing. The plants were not in immediate danger. The experiment’s usefulness was. If conditions departed too far from the recorded range, weeks of work would no longer answer the question it had been designed to test.

Leila checked the actual conditions with a separate instrument.

Another researcher inspected the unit. Someone retrieved a temporary fan, discovered its power connector was incompatible, and swore with such weary specificity that Maya almost laughed.

The repair took less than an hour. The consequences would take longer to assess.

Anika arrived near the end. She asked whether anyone had been hurt, then whether the crop was at risk, then whether the trial could still be interpreted. She did not ask how quickly they could restart it until those questions had answers.

Maya noticed Leila watching her.

“You wanted me to see that,” she said later.

“I wanted you to see the work. Sometimes the work is a fan with the wrong plug.”

“It’s the same on Earth.”

“I know.”

They stood beside the painted line that would become the river.

Maya wanted to explain that Earth also contained people capable of good work, that she herself was trying to make systems less blind to their own consequences. Leila already knew those things. The argument she wanted to make did not answer the choice Leila was actually facing.

At an observation port, a girl was pressing her hands against the glass. Her father explained which distant light was Earth. The girl asked why they could not just stay there.

Anika moved on before the answer came.

“Are you going?” Maya asked.

“Yes.”

“You didn’t tell my mother.”

“I told the selection committee. Your mother is not the selection committee.”

The reply irritated Maya because it was fair.

They stopped in a room with unfinished walls and a table fastened to the deck. Anika opened a food-system model. The numbers were smaller than Daedalus’s planetary totals, but the uncertainties looked familiar.

“You asked for our yield assumptions,” she said. “Here they are. I would like yours.”

Maya explained that the combined Earth model had not been cleared for distribution. Several suppliers considered parts of it proprietary. Other data could identify infrastructure vulnerabilities.

Anika let her finish.

“The black hole will not honor the contracts.”

“Neither will someone trying to poison a reservoir.”

“Then protect the reservoir. I need the assumptions about crop failure.”

That night, Maya and Leila stayed in a visitor cabin with two narrow bunks.

The practical arrangement saved them from deciding whether to sleep in the same bed after another difficult day. It also made the distance between them visible in a way the flat had not.

Maya lay awake listening to the circulation system. She could identify several of its operating changes now. The knowledge made the sound less frightening and more intimate.

“Would you come if the job were right?” Leila asked into the darkness.

Maya did not answer quickly enough.

“I don’t mean this minute.”

“There is work I can do here.”

“Aboard the vessel?”

“On Earth.”

Leila turned toward the wall. Maya heard the movement and disliked herself for noticing it as evidence.

“I haven’t asked you to stay,” Maya said.

“You’ve asked in other words.”

“And you haven’t asked me to leave?”

The silence afterward was not empty. It contained every imagined compromise they had each hoped the other would offer first.

Finally Leila said, “I want us to be able to choose without making the other person the villain.”

“I don’t know how to do that.”

“Neither do I.”

They talked until the scheduled lights began to brighten. They discussed work, family, fear, and the possibility that love did not automatically produce a shared destination. Maya admitted that part of her wanted Leila close because she could not bear another important person becoming available only through a screen.

Leila admitted that part of her attraction to the ark was the chance to live somewhere her work would not always be described in relation to Amara Okafor’s project or Maya Okafor’s career.

“You never said that.”

“I didn’t want it to be unfair.”

“Is it?”

“Sometimes. Not always.”

Maya remembered the man at the water hearing asking whom she represented. She had resented being reduced to her mother’s name. She had not understood that Leila could be trapped in the next ring of the same assumption.

They did not resolve the question before morning. They agreed to stop pretending it was only a job application.

* * *

On the return trip, Maya read the recruitment policy in full.

It promised less than the public material suggested. Departure dates could change. Roles could change. A successful applicant might be asked to train a replacement rather than depart. Family membership was subject to limits and review. The vessel's community rules remained under development.

She marked the passages that worried her and sent them to Leila.

This time she wrote questions rather than arguments.

Leila answered several. She forwarded the rest to the recruitment team and copied Maya on the reply.

It was a small improvement. Both recognized how much effort it required.

At home, Grandfather asked how the visit had gone. Maya described the trial room and the unfinished river. He asked whether the soil was good.

"Some of the crops aren't grown in soil."

He made a disapproving noise.

"You haven't seen it," she said.

"I've seen plants."

"That's not the same thing."

He began a familiar argument about food grown without proper ground. Maya was too tired to accept it as harmless. She told him about the careful work, the failed trials, the people trying to build a future from materials he had never used.

He listened, offended and interested in unequal proportions.

"Then bring me something they grew," he said.

Leila did on her next visit. He found reasons to dislike the texture.

He ate the rest when he thought neither woman was watching.

Later, he asked Leila whether she wanted to go. She said she thought she did.

"Maya will be sad."

"I know."

"That doesn't decide it."

Leila looked at him. He was inspecting a leaf with unnecessary concentration.

Maya heard about the conversation from Leila that evening. She was grateful to him and angry with him. Both feelings lasted longer than she expected.

When the official request for the shared crop assumptions was refused, the refusal landed in the middle of all of this. Maya was already struggling to accept that independence could be

something a person chose for good reasons. Watching her mother’s organization make that independence harder felt suddenly personal.

She opened the restrictions and began asking who had written them.

Later, Maya reread the restrictions and found that they were broader than their stated purpose. She requested permission to release an aggregated model. Her request moved through four offices and came back with a suggestion that she apply again after the next review.

That evening, she called her mother.

Amara was eating at her desk. She set the bowl aside when Maya described the refusal.

“I’ll ask.”

“That’s what everyone says.”

“Some of the restrictions are legitimate.”

“Then explain which ones. Anika isn’t asking where we keep the passwords.”

Her mother looked toward a second screen. Maya knew that look.

Somewhere else, another question had become urgent.

“We are trying to prevent people from drawing conclusions from an incomplete model.”

“Is the complete one better?”

Amara did not answer immediately.

Maya felt the argument change beneath her.

“Mom.”

“The uncertainty has widened. We have a revised climate coupling. The central estimate is still manageable.”

“And the rest?”

“We are working through it.”

Maya thought of the painted river and the girl at the window.

“You used to tell me uncertainty wasn’t something to hide.”

“It isn’t.”

“Then stop making it look like that.”

She ended the call before either of them could soften what had been said.

The next morning, the request was approved in part. Anika received the assumptions about crop yields. It would take another five years before Maya understood how the restrictions could hide a dangerous decision.

* * *

Six years before closest approach, the odds fell to thirty-eight percent in a single afternoon.

The cause was a better measurement. A sequence of timing observations from the outer relays had tightened the black hole’s incoming direction. In the improved solution, where Jupiter stood at the moment of closest approach mattered more than anyone had wanted it to. In one family of trajectories, Jupiter’s altered path would carry it close enough to Earth’s new orbit, during the encounter’s long aftermath, to take what the black hole had left behind.

Ejection had always been a thin gray line among the solutions. Now it was a band.

Amara published the result the same evening, with the assumptions, the uncertainty, and a paragraph explaining that one improved solution was not yet a forecast. Later, Maya would think about that evening often. Her mother had not waited for this one. Bad news that came from the sky was easier to release than bad news that came from the warehouses.

The betting exchange suspended trading for two hours. When it reopened, the line read 38%. One broadcaster put the number in red across the bottom of every program, including a cooking show, where it sat beneath a man folding pastry with tremendous care.

In several cities, people stopped going to work. Others went to work and did nothing. In Nairobi, a queue formed outside the observatory gate, and Grandfather joined it to find out what it was for.

“Nobody knows,” he reported when Maya called. “I think it’s for being near the telescopes.”

“Come home.”

“I’m near the front.”

For nineteen days the teams worked the problem without sleeping much. Luis rebuilt the Jupiter calculation with independent code. Anika’s group ran it on hardware that had never touched the central network. Two student observers in Chile found an old occultation timing nobody had used, because it had been published in a newsletter for hobbyists.

The occultation pinned down Jupiter’s predicted position better than three professional campaigns had. It did not remove the ejection band. It narrowed it.

The new solution was published on the twentieth day, with the old one beside it.

The odds recovered to sixty-one percent. They never returned to seventy-four.

Grandfather added both clippings to the jar.

“Thirty-eight,” he said, holding up the first. “That was a bad tomato.”

“And sixty-one?”

He tilted his hand. “Still on the vine.”

CHAPTER SEVEN

The Missing Shipment

PILOT DISPATCH LOG, RECIFE PORT AUTHORITY

SHIPMENT 44-1187 (GREENHOUSE FILM) — STATUS: ARRIVED
RECEIVING BAY 6 — CONTENTS: 0
RECONCILIATION: NO DISCREPANCY DETECTED

ORBIT ODDS: 61% · T–1,402 DAYS

Four years before closest approach, a shipment of greenhouse film failed to arrive in Recife.

The record said it had arrived eleven days earlier.

Maya was twenty-nine and responsible for reconciling agricultural supply models with actual construction. She called the port instead of querying Pilot again.

The warehouse supervisor sent a photograph of the empty storage bay. He stood in it holding a handwritten date because he no longer trusted pictures that arrived with dates attached.

The supervisor's name was Celso. He had been sending the same complaint for a week and had learned to speak without expecting the person on the other end to remain interested.

"The bay is empty," he said. "It was empty yesterday. It was empty when the previous analyst told me the receiving record was complete."

Maya asked whether the shipment could have been moved directly to the construction site.

He turned the camera toward a second screen. A structure stood without its cover beneath low cloud. People were carrying trays into temporary shelter.

"The construction site is asking me where it is."

She apologized for making him repeat the obvious. His expression changed slightly.

The installation had been planned around a sequence that left little room for delay. Old cover had been removed from part of the growing area. Replacement material was due before the next planting stage.

Temporary measures could protect some work, but not all of it, and each day required moving labor from another task.

"What do you need immediately?" she asked.

"A real arrival date. If it isn't coming this month, we stop pretending and change the crop."

She promised to find out before the end of his shift. It was a smaller promise than the one he wanted and the only one she could honestly make.

By then, the record had become stranger. The departure facility confirmed the load had left. The receiving record confirmed an arrival that had not happened. Between them, a route change pointed to an orbital holding depot whose public inventory showed nothing unusual.

Tomas answered her request within minutes. He was reviewing another set of discrepancies and asked her to send the raw manifests rather than the summary.

“This might be the same transfer problem,” he said. “We’ve had destination codes inherited across revised contracts.”

It was a plausible explanation. He gave her examples, including one that had produced a harmless duplicate in a previous quarter.

Maya thanked him and followed the lead.

The harmless duplicate had been corrected before dispatch. This record had been altered afterward.

She called Tomas again. He said that made it more serious and helped her request access to the depot’s handling logs. She would later remember the ease with which he had supplied the authorization route.

At the time, it looked like experience.

* * *

Two days later, Maya stood in the empty bay herself.

It was not an efficient use of her time according to the travel system. She could have reviewed the records from Nairobi. The photograph already established the absence. But Marcus’s instruction about opening the warehouse had become more than a method. She needed to understand what the missing shipment meant before reducing it to another delay code.

Celso showed her the marked area where the material should have been stacked. The outline on the floor was larger than it had looked through the camera. A handwritten note on the receiving desk listed the calls he had made and the answers he had been given.

“You kept all of them.”

“People remember differently when the cost arrives.”

At the growing site, a supervisor named Inês walked Maya between temporary covers. The air beneath them was hot and close. Condensation ran down a surface never intended to remain in place this long. Workers adjusted openings by hand because the improvised arrangement could not use the automated ventilation schedule.

The crop looked alive to Maya. Inês showed her the trays they had already discarded.

“Alive is not the same as on schedule,” she said. “If we miss the stage, the next planting takes the loss too.”

A young worker was separating damaged seedlings from those worth retaining. He moved quickly, with the economy of someone who had repeated the decision too many times. Maya asked how long he had been doing it.

“Since they told us the shipment was here.”

She stayed for an hour and helped carry the surviving trays. Her assistance was mostly symbolic. The heat, the weight, and the repeated small decisions were not.

Inês asked whether someone would be punished.

“I don’t know what happened yet.”

“That’s not what I need first. I need the material.”

Maya called Elena from the edge of the site. The temporary replacement would require taking stock assigned elsewhere. Elena asked for the consequences on both sides. Maya gave them, including the uncertainty.

The emergency transfer was approved. It would help. It would not recover the discarded plants or the labor already spent.

Before leaving, Maya asked Celso to preserve his handwritten call record. He made a copy for her and kept the original.

“I’m not giving away the only version,” he said.

She told him he was right.

* * *

Celso had first reported the empty bay on a Thursday because the delivery crew was due Friday and he wanted the confusion resolved before people arrived to collect equipment he did not have.

The first reply told him to refresh the record. The second asked whether he had searched the overflow area. He searched it, photographed it, and attached the images. The third reply closed the issue as a local inventory reconciliation.

He reopened it.

By Monday, his supervisor had asked why an experienced depot manager was allowing a documentation problem to consume so much time. Celso took her to the bay. She looked at the empty floor and called the regional office herself. Her call received the same answer in more respectful language.

Together, they checked whether the shipment had been assigned to a different physical location under an old name. They found two obsolete location codes and one legitimate delivery that had been misfiled. None contained the missing film.

The work was useful, exhausting, and easy to mistake for progress toward the actual problem.

When Inês arrived, Celso expected her to be angry with him. She was, briefly. Then he showed her the correspondence. She sat on a crate and read it while workers moved around them.

“They think you’re asking how to use the system,” she said.

“Apparently I don’t know.”

“You’re asking whether the system is wrong.”

He nodded.

Inês took a blank sheet from the office and wrote the date on it in thick marker. She told him to stand in the bay. He asked whether she thought another photograph would help.

“It will at least make it harder to pretend this is a menu you haven’t found.”

That was the photograph Maya received.

* * *

After the site visit, Maya traced the route changes and found six additional discrepancies. They did not all resemble the first.

One involved a genuine error corrected in the physical depot but not the shared model. Another came from a damaged sensor reporting the same weight twice. A third had been flagged by a supervisor whose message remained unanswered in a crowded queue.

Maya worked through them with two analysts and a freight controller who resented losing a weekend until she saw the pattern. The differences mattered. If they treated every error as part of a conspiracy, they would make the actual conspiracy harder to find.

Tomas joined remotely. He was patient, useful, and careful to acknowledge when a suggested explanation failed. Maya felt grateful that someone senior was willing to remain inside the tedious work instead of demanding a dramatic conclusion.

The first lead toward Saira appeared in an authorization chain.

It was her signature. The time was plausible. The shipment involved material grades she had authority to reassign. Maya looked at the record for several seconds before opening the next layer.

Saira had approved a route change to a functioning receiver. A later edit changed the destination again. The interface displayed only the most recent destination beside the original approval.

“That isn’t what she signed,” Maya said.

The freight controller leaned closer. “Are you sure?”

“Not yet.”

They recovered the original signed object and compared it with the displayed summary. The signature was genuine. The summary was misleading.

Maya called Saira before sending an accusation that the evidence did not support.

Saira listened, then asked for the records. She was angry by the time she called back, but the anger was directed at the mechanism rather than Maya.

“If that display can join an old signature to a new destination,” she said, “how many times have we trusted it?”

Maya did not know.

Saira volunteered to review the material transfers from her own facility. She also asked whether the production teams should stop dispatching.

That was the first moment the investigation became an operational decision. Stopping everything would preserve evidence and create immediate shortages. Continuing unchanged could spread the damage.

Maya called Marcus.

He did not tell her to shut down the network. He asked which actions remained verifiable through independent records. Those could continue under additional checks. The disputed routes would be held.

The people waiting at the destinations would be told why their schedules had changed.

“Don’t tell them it’s routine maintenance,” he said. “If they need to change their plans, give them the truth they need to do it.”

The restriction slowed the work almost at once. Elena demanded a list of blocked shipments and the basis for each hold. Marcus gave it to her. They argued for an hour, then signed the limited procedure together.

Maya watched the disagreement without mistaking it for failure.

She had seen what happened when urgency removed every person entitled to say no.

* * *

At the orbital holding depot, the containers had become part of the landscape.

A worker named Olek passed them on most shifts. He knew the corner guard on one was bent, and that another carried an old destination label beneath the new one. Neither fact was unusual.

Freight accumulated while contracts changed, vehicles waited, or paperwork caught up with physical reality.

When the independent inventory request arrived, he was annoyed.

His team had already counted the area twice that month. The supervisor explained that this count would be made against a separate list and photographed from specified positions. Olek asked whether someone on Earth had finally discovered that containers existed in three dimensions.

The supervisor told him to save the jokes for the report if they helped explain a discrepancy.

They found the first disputed container before the end of the hour.

Its seal matched the requested number. Its contents were not visible, but the mass and external identification agreed with the missing shipment.

Olek photographed the seal and the neighboring locations, then read the number aloud while another worker checked it.

The next container was not where either list placed it. They found it behind equipment awaiting disposal, in a space that had once belonged to a canceled contract. The local system showed a temporary move with no expiry date.

Olek remembered the instruction. It had come through an account he had used for legitimate maintenance exceptions. He had no reason at the time to believe it was unauthorized.

“Am I in trouble?” he asked.

The supervisor did not promise he was not. She told him to preserve what he remembered separately from what the records now suggested. He wrote the account before reading the rest of the investigation packet.

Later, Maya would use that distinction. Olek remembered a routine instruction, not a secret conversation. The conspiracy had depended on ordinary workers following plausible requests, not on every person along the route sharing its purpose.

* * *

The next clue came from a physical seal.

The orbital depot reported several containers as having passed through without storage. Its handling records contained the expected movements. A maintenance image from a later inspection showed one of those containers still in the holding area, identified by a damaged corner guard and a seal number visible through a gap.

The image was not proof of motive. It was proof that the container had not followed the route the model claimed.

Maya asked for an inventory independent of the disputed system.

The depot supervisor objected to the work, then sent a team to inspect every affected location. They found material that had been retained, relabeled, or moved into areas assigned to contracts no longer active.

The warehouse had not been empty. The story told about it had been.

Tomas went quiet during the call.

Maya noticed because he had been contributing steadily for hours.

When she asked whether he had found something, he said he was checking a related authorization. She accepted the answer and continued.

Later, the freight controller asked whether Tomas should still be receiving the full investigation feed. Maya felt irritation rise before she understood the question.

“He has been helping us.”

“So has everyone with access.”

The controller did not say more. Maya asked Marcus to review the permissions. Tomas’s access was restricted with the others whose accounts touched the disputed changes.

He called her directly.

“You think I did this?”

“I think your account is part of what we need to check.”

“You know me.”

She remembered Saira asking her not to know until she had checked.

“Then you know why I can’t make you the exception.”

He ended the call politely. The politeness unsettled her more than anger would have.

* * *

That evening, Leila found Maya sitting on the kitchen floor beside the open cupboard.

She had been looking for a pan and stopped halfway through the task. It was easier to remain there than to decide what came next.

“Is it your mother?” Leila asked.

“Tomas.”

Leila put down her bag.

Maya described the accounts, the retained containers, and the interview Marcus had requested. She avoided naming him as the person responsible because the evidence had not yet reached that conclusion.

The care in her phrasing made Leila understand how frightened she was of the possibility.

“He helped me find it,” Maya said.

“Maybe he thought he could control what you found.”

“Maybe he didn’t do it.”

Leila sat beside her. “Yes.”

They remained on the floor for a while. Above them, the shelf still leaned slightly. Tomas had once reached to check it. The memory arrived without permission.

“If he did it, does that make everything he taught me false?”

“No.”

“It feels like it should.”

Leila thought before answering. “He taught you to check the physical record. You checked it. That part survived him.”

Maya wanted to be comforted by the answer. Mostly she felt tired.

* * *

Tomas had joined the diversion effort after a meeting in which nobody shouted.

That was how he remembered it: polite voices, a revised agenda, a promise that the report would receive attention after the contribution agreement. He had left with a folder of unanswered questions and the impression that the institution could absorb any objection by acknowledging it just long enough to defer it.

The first suggestion came from a colleague who said the approvals should stop until the dependencies were visible. Tomas agreed. When she proposed holding one nonurgent shipment in a depot rather than letting the model count it as delivered, he told himself the act would demonstrate a flaw without harming anyone.

The shipment was not nonurgent to everyone who depended on it.

He learned that afterward.

By then, the demonstration had not produced the response he wanted. The discrepancy had been classified as an administrative error.

His colleague proposed a second, clearer case. Tomas supplied the knowledge needed to make it survive the routine checks.

Each step was easier to justify in relation to the previous one than to explain from the beginning. They were not stealing the material, he told himself. They were preserving it. They were forcing an honest accounting. They were preventing a larger failure.

When a collaborator began speaking about the moral necessity of remaining in the old cosmic order, Tomas objected. He regarded that view as irrational. He did not leave the group. He believed he could limit what they did together.

The first report of spoiled seedlings reached him before Maya’s investigation. He read it at home. His partner asked why he had stopped eating. He said a project was going badly.

It was true in the smallest possible way.

He sent a message asking the group to release that shipment. The answer said releasing it now would expose the pattern before the warning report was ready to be forced into public view. Tomas argued, then accepted a delay.

Years of useful caution had not protected him from becoming reckless on behalf of a conclusion he believed was right.

When Maya called about the missing film, he felt a brief, terrible relief. Someone competent was looking. He could help her find enough to expose the model without finding all of him. The belief lasted longer than it should have because he wanted it so badly.

* * *

Marcus interviewed Tomas the following morning.

Before the formal questions, Tomas asked whether the retained cargo had been located. Marcus said much of it had. He asked whether any had been destroyed. Marcus told him that equipment could remain intact while the people waiting for it lost things that could not be stored in a depot.

Tomas looked toward the window.

Maya watched the recording later. She could see the moment when he decided not to continue offering technical explanations for a deliberate act.

“We needed them to stop approving the next phase,” he said.

Marcus asked who he meant by we.

Tomas gave names for some collaborators and refused others. He said two people whose approvals appeared in the records had acted legitimately and should not be implicated. Marcus told him the investigation would decide that from evidence.

“You can help by being accurate,” he said.

For the first time, Tomas looked almost relieved. Accuracy was a role he understood even when it no longer protected him.

He described attempts to raise the correlated-failure problem.

Reports submitted, meetings postponed, summaries returned with softened language. Some of that account would be verified. Some would prove to contain the distortions of a man who had begun treating every disagreement as proof that the system could not be corrected honestly.

Marcus did not resolve the distinction in the interview. He recorded it.

When Tomas requested that Maya receive the recording, he said he wanted her to understand why.

Marcus asked whether he also wanted her to understand what he had done.

Tomas did not answer immediately.

* * *

The first affected shipment resumed its journey while the interviews continued.

Maya called Celso with the confirmed departure record and the independent tracking reference. He listened, wrote them down, and asked when he should begin to believe the arrival estimate.

“When the next observation supports it,” she said. “I’ll send that too.”

He nodded. “All right.”

It was not a restoration of trust. It was permission to begin earning a small amount back.

At the growing site, Inês had already changed part of the crop plan.

She would use the delayed material, but not in the sequence the central model still expected. Maya corrected the model to match the work instead of asking the work to preserve the model’s appearance.

Then she opened the interview file Tomas had requested she hear.

In the recording, Tomas asked Marcus to open a folder called COMPOUND FAILURE.

It contained a version of the global food model Maya had never seen.

The median yield looked familiar. The low end did not.

Several agricultural regions shared the same critical assumptions about power availability, replacement membranes, and fertilizer delivery. When those failed together, the shortfall grew sharply. The public summary treated them as largely independent.

The internal warning was eighteen months old.

Amara had received it.

Maya played that part of the interview twice before going to her mother’s office.

CHAPTEREIGHT

What Was Withheld

MINUTES, MODELING REVIEW 2120/17, ITEM 4

A reviewer asked whether the public could be shown both versions with the disagreement explained. The chair responded that this would create confusion before there was an actionable answer. Minutes approved.

ORBIT ODDS: 61% · T–1,371 DAYS

Amara had first seen the warning on a train between two meetings.

She remembered the journey because the carriage lights flickered while she was reading, and for a moment the report vanished from her screen. When it returned, the low-end result looked worse than she remembered. She checked whether the display had changed. It had not.

The report did not say the preparation program was useless. It said that several sources of failure were being treated as more independent than the evidence justified. A supplier shortage could affect multiple regions at once. A power assumption could fail alongside the refrigeration that depended on it. The same favorable estimate appeared inside several different margins, giving the final model more apparent protection than the physical system possessed.

Amara sent questions before the train reached the station.

The replies identified genuine problems in the first analysis. One supplier had been counted as unique when an alternative existed. Two maintenance schedules were being compared across incompatible reporting periods. A crop-loss estimate used a provisional coefficient that its own authors had since revised.

Those corrections reduced the projected harm. They did not remove the underlying dependence.

At the next review, the communications office asked whether the warning should accompany the public update. The contribution agreement was approaching renewal. Several governments had already signaled that they would redirect funds toward national programs if the shared plan’s outlook deteriorated.

“We need a stable version,” the coordinator said.

Amara agreed to wait for the corrected analysis.

That was the first delay. It felt responsible.

The corrected analysis arrived before a construction milestone. Its central forecast remained within the program’s intended range. The low-end case required more reserves and a different

distribution of agricultural infrastructure. Making those changes immediately would expose the incompleteness of the plan they were asking people to fund.

Amara asked for another comparison. That was the second delay.

By the third, she had begun using the continued uncertainty as a reason not to publish the uncertainty. Nobody instructed her to do it. Several people made it easier.

When a junior reviewer asked whether the public could be shown both versions with the disagreement explained, Amara said they would create confusion before they had an actionable answer. The reviewer wrote that response into the minutes. Amara approved the minutes.

Now Maya stood in her doorway with the old report open.

Amara recognized the page before she recognized how much her daughter had already understood.

Amara did not deny seeing the warning.

“We were changing the model.”

Maya remained standing.

“For eighteen months?”

“The first version overstated some dependencies. The suppliers challenged it. There were errors.”

“Were there enough errors to make it safe?”

“No.”

The room was too warm. Maya took off her coat and put it on a chair because otherwise she would leave with it still on.

“I asked you.”

“You asked about the release.”

“Don’t do that.”

Amara looked down at the report. Her notes occupied its margins.

She had not ignored it. Somehow that made Maya angrier.

“If we had published the first result,” Amara said, “the contribution agreement could have collapsed. Some delegations were looking for a reason to withdraw. We needed the next construction season.”

“And people needed to know what they were building.”

“I thought we could resolve it before—”

“Before anyone had to choose?”

Amara sat back.

Outside, someone laughed in the corridor, then lowered their voice.

Maya could remember doing homework in this room during school holidays. Her mother would give her spare paper, and Maya would draw houses on the backs of solar-system diagrams.

She had trusted the lines because her mother had drawn them.

“Tomas changed the manifests,” Amara said. “Whatever we did, that was his decision.”

“I know. I’m here about yours.”

Maya learned this chronology from records assembled after the first interview, not from Tomas’s attempt to explain himself.

Marcus gave her the verified portions and marked the disputed ones. She read them at a desk overlooking the same loading routes she had been trying to restore. The account was less dramatic than a sudden conversion to sabotage. It was more disturbing.

She could recognize several of the emotions. The irritation of being ignored. The certainty that a warning mattered. The temptation to make the next decision unavoidable. She could not use that recognition to excuse the act, but it prevented her from treating Tomas as a different kind of creature from the people who would design the replacement system.

She asked Marcus whether the review would include the unanswered reports.

“Yes.”

“Some people will say that makes excuses for him.”

“Some people will. Leaving out causes doesn’t improve responsibility.”

She returned to the freight record. A delayed shipment had finally received independent arrival confirmation. Inês would still lose part of the season. Maya entered both facts, then opened the next unresolved location.

* * *

The independent review found both failures. Tomas and five collaborators had diverted equipment to force a pause in the program.

They called themselves Anchor. Two had tried earlier to raise the modeling problem through official channels. One had never tried; he believed any preparation for departure was an offense against the natural order.

Their motives did not make the empty warehouses full.

The withheld report did not disappear because its disclosure had come through a crime.

After Maya left, Amara called Luis.

He arrived expecting another technical emergency. She gave him the report and the publication history. He read without speaking. The silence grew long enough that she began explaining before he finished. “There were errors in the first version.”

He raised a hand. She stopped.

When he reached the end, he placed the tablet on the desk with unusual care.

“You knew the disagreement mattered.”

“Yes.”

“And you decided everyone else could wait.”

She wanted him to understand the funding problem. He did understand it. That was what made his disappointment difficult to evade.

“If the agreement had collapsed—”

“Then we would have had a different problem. You don’t get to erase one by choosing it for people without telling them.”

He looked toward the wall where they had once projected the first survivable orbit. The wall was covered now with schedules and photographs from construction sites. Their work had become real in a thousand places. That reality had made secrecy easier to justify, not harder.

“You sent me every broken calculation in the first week,” he said.

“Why not this?”

“Because you couldn’t fix the politics.”

“Neither could you.”

He left to prepare an independent account of the modeling history.

He did not ask whether she wanted him to do it.

Amara sat alone until the building’s evening lights changed. Her office was full of objects meant to demonstrate the project’s progress: a small panel sample, a school patch, a photograph of the first completed reserve warehouse. She turned the photograph around so she could see the handwritten names on its back.

The people in it had signed the photograph for her. None had signed permission for her to decide what they could know.

* * *

The public hearing took place in a room designed for translations, not intimacy.

Amara could see the interpreters behind glass. Each had a screen displaying the documents under discussion. There was a slight delay before questions reached some delegates, and she forced herself to wait instead of beginning answers too quickly.

Maya sat with the technical witnesses. She had declined a place beside her mother. The empty chair there was occupied by legal counsel, who had advised Amara not to speculate about motives.

The first witness was Inês from the growing site.

She described what the missing material had cost. The discarded plants. The extra labor. The revised planting sequence. She did not attribute all those losses to Amara, and she did not allow the lawyers to describe them as temporary inefficiencies.

“Some of the work can be recovered,” she said. “Some cannot. You should count both.”

The next witness reviewed Tomas’s unauthorized diversions. The hearing then turned to the report the diversions had helped expose. The chair repeated that one did not excuse the other.

Amara was asked when she first understood that the public summary understated the risk of correlated failures.

She gave the date.

“Why did you not publish the disagreement?”

She described the errors in the first version, the corrections, the contribution negotiations, and her belief that premature release could interrupt work needed under every plausible scenario.

“Who authorized you to decide that the public should not see it?”

“I was responsible for the scientific release.”

“That describes your role. It does not answer the question.”

Amara took a breath. “No one authorized the delay in those terms.”

The answer moved through the interpreters’ room. She watched delegates hear it at slightly different times.

A woman from one of the regions dependent on the shared crop program asked whether Amara understood that local authorities had used the published assessment to cancel alternatives.

“I understand that now.”

“They did not have your internal report. They had your confidence.”

Amara looked at the woman rather than the counsel beside her.

“Yes.”

No explanation improved that answer.

During the recess, a staff member offered to take her through a side exit. She declined and went to the same cafeteria as everyone else.

People looked up when she entered. Some looked away again. A few greeted her with exaggerated normality that felt like another kind of accusation.

Maya was standing near a window with Leila. Amara approached, then stopped far enough away that her daughter could choose whether to acknowledge her.

Maya did.

“Are you eating?” she asked.

The practical question nearly undid Amara. She said she would get something. They did not sit together, but Maya watched until she had taken a tray.

* * *

The panel’s decision did not remove Amara from the project.

It removed powers she had come to regard as inseparable from doing her work. She could no longer determine publication timing alone. Independent reviewers received the underlying models and could publish disagreements with her conclusions. Her scientific recommendations remained influential, but influence would have to survive visible challenge.

Some supporters called the arrangement humiliation. Amara was briefly tempted to agree. Then she read the publication rules she herself had endorsed during the first year.

The new arrangement resembled what she had promised.

Her first meeting under it was awkward. A junior analyst interrupted her explanation to ask whether a parameter had been independently checked. She began to say that it came from her team’s latest run. The analyst waited.

Amara opened the reference and found the answer was no.

“Mark it provisional,” she said.

The discussion continued. No one congratulated her for accepting the correction. She realized she had been hoping someone would.

* * *

The argument over emergency authority moved from the hearing room to a smaller conference room with worse coffee.

Elena brought the blocked-shipment list. Marcus brought the audit findings. Both lists described real harm. Both were growing.

“I need one office able to move the equipment,” Elena said.

“You need several offices able to prove where it went.”

“If every move waits for a committee, the equipment stays in storage.”

“Then don’t make every move wait. Define the exceptions people can inspect.”

They worked through categories instead of slogans. Routine deliveries could continue with existing checks. Emergency redirections needed independent confirmation at the destination. A disputed command could be held by someone outside the issuing office. Holds required an explanation and a review deadline so that obstruction could not disguise itself indefinitely as caution.

Maya contributed the examples from Recife. Tomas’s changes had succeeded partly because the system treated records as more authoritative than people reporting that the records were wrong. Any revision that made it harder for a warehouse supervisor to challenge a receipt would preserve the same weakness under a different command structure.

Elena listened, then crossed out a section she had drafted herself.

“All right. But the people holding the routes have to see the cost of the hold.”

Marcus agreed.

The compromise had no dramatic name. It added work to already tired people. The next week, it caught a harmless error and delayed a legitimate shipment. The week after that, it prevented a second unauthorized diversion.

Elena insisted that all three outcomes appear in the review.

* * *

Before visiting Tomas, Maya asked Leila whether she should go.

Leila said she could not decide that for her. Maya was annoyed because it was the kind of answer she herself gave when she wanted to be fair.

“I’m not asking for a policy,” she said.

“Then yes. I think you’ll keep having the conversation in your head if you don’t have it with him.”

The visit took place in a room with a table too small to keep either person comfortably distant. Tomas had lost weight. He asked about Saira and Pavel before asking about the investigation. Maya answered briefly.

He wanted to explain the unanswered reports. She let him. Some of the story was familiar. Some of it showed how his interpretation had hardened over time. Every delay had become evidence of bad faith.

Every disagreement had become proof that ordinary channels could not work.

“You could have sent the report to me,” she said.

“You were inside the program.”

“So were you.”

“You still believed they would correct it.”

Maya looked at him. “You taught me how to make them look.”

He rubbed his hands together, a familiar gesture she had seen while he waited for a calculation.

“I thought a pause would force a real review. The equipment would still be there.”

“The plants wouldn’t. The season wouldn’t.”

He lowered his eyes.

She asked when he had realized the diversions were causing losses beyond the schedule. He named a point later than she expected and earlier than he wanted to admit. He had continued after it.

That was the answer she needed, though it was not the answer she had come hoping to hear.

When she left, she did not feel cleansed of the friendship. She felt its remaining weight more clearly. He had been kind to her. He had taught her useful things. He had harmed people he could describe as entries in a plan. Those facts would have to remain together.

Outside, Leila was waiting. Maya had not asked her to come.

“I thought you might want company on the way home,” she said.

Maya took her hand. For a while they walked without talking about departure, duty, or the future of the planet.

* * *

The morning after the hearing, Amara found that her access badge still opened the same doors.

She had expected some physical expression of the decision: a different office, a removed nameplate, a pause at a security gate.

Instead the building admitted her as it had the day before. People greeted her. The work waited.

The change appeared in an ordinary form. A publication request she would once have approved alone now required an independent reviewer. She entered her recommendation and sent it onward. The act took seconds. The waiting occupied the rest of the morning.

At noon, the reviewer returned three questions. Two were straightforward. The third challenged an assumption Amara considered well established. She began drafting an impatient response, then stopped and found the supporting evidence.

The evidence was good, but not as complete as the word established suggested. She changed the phrasing, supplied the range, and answered the question. The reviewer approved the release with the disagreement noted.

No catastrophe followed. No contribution agreement collapsed.

Amara knew that did not prove every delayed publication would be harmless. It proved that her fear of losing control had become entangled with her fear of losing time.

Luis came by in the afternoon to discuss a new encounter integration. Their conversation was technically easy and personally awkward. For years, they had relied on an intimacy built from shared failures and unfinished calculations. Now he checked whether she had supplied the complete set of inputs before beginning.

She noticed. He knew she noticed. Neither softened the check.

After they finished, she asked whether they could have coffee.

“I have another review.”

“Tomorrow?”

He considered it. “Tomorrow.”

It was a small answer. She did not ask it to mean more.

* * *

At home, Maya discovered that anger could be boring.

The first days had been sharp, each new fact giving it another edge. Afterward it settled into ordinary moments. She would reach to call her mother about something unrelated and remember the report. She would hear a phrase about confidence and become irritated with someone who had nothing to do with it.

Leila tolerated some of this and challenged the rest.

“I’m not defending her,” she said after Maya misread a question for the third time. “I’m asking whether you want dinner.”

Maya apologized. They ate quietly. Halfway through, Maya began describing the hearing again, then stopped.

“You’ve heard this.”

“Yes.”

“Do you want me to stop?”

“I want us to have some conversations the hearing isn’t in.”

Maya put down her fork. The request felt disloyal for a moment, and then she understood whose life she was asking to remain organized around the injury.

They went for a walk after dinner. Leila told her about a difficult crop trial. Maya managed not to turn it into an analogy about withheld information. The effort was absurd and necessary.

At the corner shop, they bought a replacement light for the kitchen.

The shopkeeper recognized Maya and began telling her what her mother should have done. Leila said they were there for a light. The shopkeeper looked offended, then sold them one.

Maya thanked her outside.

“You can say that too,” Leila said.

“I know.”

“Try it before I leave the planet.”

The joke landed badly, then well, then badly again. They stood on the pavement laughing because neither could decide which reaction to keep.

* * *

The civilian proceedings against Tomas took longer than the public attention surrounding them.

Maya was asked to verify a technical statement. She read it carefully and found one sentence that overstated what the records proved about his personal access. The investigator accepted the correction. A colleague asked afterward why she had helped him.

“I helped the statement be accurate.”

“You know what he did.”

“Then we don’t need to add something we can’t prove.”

The exchange stayed with her because she had felt the same temptation in herself. Once guilt became morally clear, precision could look like an inconvenience. Tomas had used a version of that reasoning against the program. She did not want his trial to teach it back.

When the findings were issued, they distinguished among the collaborators’ actions. They described the diversions, the falsified records, and the losses. They also documented the ignored warnings.

The penalties did not satisfy everyone, and the case did not restore the missing planting season.

Tomas sent Maya a letter through the permitted channel. It contained an apology and several paragraphs of explanation. She read it once, put it away, and did not answer immediately.

A month later she wrote that she had received it. She told him the corrected food model was public and the recovery trials had begun. She did not forgive him in order to end the letter politely.

The absence of a clean ending became something she could live with.

* * *

Amara’s first coffee with Luis after the hearing lasted fourteen minutes. They talked about an instrument fault, then the weather, then the terrible coffee. Finally she said she did not know how to make their work feel as it had before.

“It won’t,” he said.

She looked down at her cup.

“That doesn’t mean it can’t be good work.”

“I know.”

“I’m not sure you do yet.”

He told her that he had been angry not only because she withheld the report, but because she had left him defending a public account without knowing what it omitted. He had answered questions from colleagues using confidence he believed they shared honestly.

She had thought about the public, the council, Maya. She had not fully considered that particular betrayal.

“I’m sorry,” she said.

He nodded. He did not tell her it was all right.

When they stood to return to work, he asked whether she had seen a new observation that disagreed with their outer-planet solution. She had. She had already sent it to the independent group.

“Good,” he said.

The word meant less than absolution and more than nothing.

CHAPTER NINE

No Passengers

KENYA NATIONAL EXAMINATIONS COUNCIL, CIRCULAR 14

Candidates are reminded that the encounter does not constitute grounds for an extension.

ORBIT ODDS: 57% · T–1,288 DAYS

The recovery plan required abandoning part of the plan they already had.

Maya spread the supply dependencies across a wall in the Nairobi operations center. The largest new greenhouse complexes used the same membrane, manufactured at two sites. Their nutrient systems depended on a small group of suppliers. Their power forecasts assumed transmission repairs that were already late.

Pilot could reschedule the shipments. It could not manufacture another season.

“We keep trying to recover the construction total,” Maya said.

“That isn’t the same as recovering the food.”

Elena asked what she wanted to stop.

Maya pointed to three expansion projects. They would produce efficiently if completed, but their completion depended on nearly every delayed component. Smaller regional projects could use locally available glass, different crops, and simpler equipment. Their yields were lower. Their failures would not all be the same failure.

“We lose capacity on paper,” Elena said.

“We lose promises. Some of the capacity doesn’t exist.”

Maya’s first recovery proposal fitted beautifully on the wall.

She had reduced the number of dependencies, increased local substitution, and assigned wider margins to the parts most likely to fail together. The revised model recovered much of the lost capacity under conservative assumptions. Several reviewers called it promising.

Amina asked where Maya intended to test it.

“We have data from comparable installations.”

“You have data from installations whose crews already know how to keep them alive. Where will you test this one?”

The question irritated Maya because she had expected Amina to support the decentralization. Amina did support it. She did not consider agreement a reason to skip the unpleasant part.

They selected a site serving several farming groups outside Nairobi. Wanjiru's cooperative joined the trial on the condition that it would not replace an entire productive area before the new system had demonstrated itself. Maya agreed. Elena complained about the reduced scale, then accepted the reason.

The material arrived on time. That felt encouraging enough to be misleading.

On the first construction day, two components did not fit the existing support frames. The drawings had described the newer version of a structure several sites had repaired differently over the years. Maya had used the drawings because that was what the inventory system contained.

Wanjiru watched the installers measure the mismatch.

"We told the survey team this side was different."

Maya searched the survey notes. The observation was there, attached to a photograph rather than entered into the structured field used by the design tool.

She could explain the failure exactly. The explanation did not make the components fit.

The crew improvised a temporary connection. It worked well enough to continue and introduced another problem in ventilation. In the afternoon heat, the growing area became difficult to work in. The original design assumed openings that the improvised connection partly obstructed.

By evening, the trial had demonstrated that a resilient plan could become fragile through entirely ordinary contact with a real building.

Maya stayed after the visiting team left. She helped remove the worst section while the crew told her which modifications they trusted and which would create maintenance work no one had budgeted.

She wrote down the comments. Wanjiru took the notebook from her and added two she had softened.

"Write what we said."

Maya did.

* * *

The next morning, Elena asked whether the trial should be stopped.

"The first version should," Maya said.

"I need to know if I'm sending more material."

"Not until we change the connection design and survey the actual frames."

"How long?"

Maya gave her a range. Elena disliked the upper end. Maya disliked it too.

“You asked me to replace projects that were already delayed,” Elena said. “Now this one is delayed.”

“Yes.”

“Tell me why that is better.”

“Because we found the failure before we copied it into forty sites.”

Elena was silent long enough for Maya to wonder whether she had mistaken a good argument for a useful answer.

“Send me the revised test,” she said finally. “And the cost of the delay. Both.”

Wanjiru’s cooperative voted against the second trial before it voted for it.

The first meeting took place beneath a roof Maya’s team had helped repair. She thought that would count in favor of the proposal. It counted as evidence that the team could repair a roof. The members asked about land use, labor, ownership of the equipment, and what would happen if the central program changed its priorities again.

Maya answered from the draft agreement. Wanjiru asked who could remove the equipment.

“It remains project property during the trial.”

“Who can remove it?”

Maya found the clause. A central office could reassign material under an emergency designation. There was a consultation requirement, but no local power to refuse.

The room became quiet in a way she had learned to distrust.

“You’re asking us to give up productive ground,” one member said.

“Then someone we’ve never met can take away what replaced it.”

Maya explained why emergency reallocation existed. The explanation did not answer the concern. The vote failed.

On the way back, Elena asked whether another site would agree more easily. Maya said perhaps, and then asked whether easier agreement would repair the clause.

They revised it. Equipment essential to an active crop could not be removed without a local continuity plan and an appeal route fast enough to matter. The program retained emergency powers in defined circumstances, but those powers now carried an obligation to show what would happen to the people using the material.

The second vote passed narrowly. Maya did not celebrate the narrowness as evidence of thoughtful democracy. It meant many members remained unconvinced. She wrote their objections into the trial plan and assigned someone to check them.

* * *

The second version used fewer specialized fittings. Its appearance was less uniform because it allowed local variations in the existing structures. The maintenance instructions included photographs from the trial and a list of substitutions the crew had actually tested.

Saira reviewed the material choices remotely. She rejected two shortcuts and suggested a third that could be made with equipment the cooperative already possessed. Pavel demonstrated an inspection method using the same kind of small mirror he had given Maya years earlier.

Maya took the mirror from her bag and used it beneath a connection she could not see directly. One of the installers asked why the handle looked so old.

“It has experience,” she said.

The installer laughed and showed her a wrench older than either of them.

* * *

The crop trial could not be hurried by the urgency of the report.

For weeks, the plants grew at the rate the conditions allowed. Leila helped design the observations, drawing on the habitat work without pretending that a small enclosed vessel and an agricultural district were interchangeable. Amina insisted on recording labor as carefully as water and light.

The first harvest was edible and disappointing.

It weighed less than the central forecast. Some of the produce stored poorly. One variety performed well under the reduced-light schedule and was disliked by almost everyone asked to taste it.

At the community meal organized to compare the results, a man asked why poorer districts always seemed to receive the food that required an explanation.

The room grew quiet.

Maya began to describe the nutritional value. Wanjiru stopped her with a glance.

“He’s asking who chose it,” she said.

The answer was a group of researchers and planners, none of whom would be required to eat only that crop. Maya said so.

The discussion became more useful after that. People suggested preparation methods, alternatives, and combinations the original trial had not included. Some suggestions were impractical. Others improved acceptance without requiring a different infrastructure. A local cook transformed one of the least popular samples with ingredients that were already part of the regional plan.

Leila asked for the recipe and the actual quantities used. The cook told her she could watch the next batch because the quantities changed with the crop.

“Then we should write the range,” Leila said.

Maya smiled. It was a small version of an argument they had both spent years making elsewhere.

The least popular crop became useful because a cook refused to accept the test kitchen’s recipe.

Her name was Zawadi. She had cooked for groups large enough to make small changes in texture matter. The research team had prepared the sample according to a standard method intended to make comparisons fair. Zawadi said the method made several varieties equally unappealing and therefore answered a question nobody eating dinner had asked.

Leila asked to observe her preparing it differently. Maya came because she wanted to understand why a successful yield could still become a failed food plan.

Zawadi did not transform the crop through a secret ingredient. She changed the cut, the soaking time, and the order in which it entered the pot. She used a familiar staple to balance the texture and kept track of the added demand that would place on another part of the supply system. When Maya asked for a precise recipe, she handed her a spoon.

“Taste this.”

Maya did.

“Now this.”

The second batch needed longer. Maya could tell once someone asked her to attend to it. The difference came from the crop itself, not careless measurement.

“Write the signs people should look for,” Zawadi said. “If you write only minutes, they’ll blame themselves when the pot is different.” Leila began taking notes. The eventual guidance included ranges, photographs, and ordinary sensory descriptions. It also included alternatives for kitchens without the same equipment.

At the next tasting, more people liked the crop. Some still did not.

The team kept alternatives in the plan rather than declaring the objection solved.

Maya took a portion home. She cooked it badly on the first attempt and correctly on the second. The experience made her less impatient when regional teams asked for training time that the original schedule had omitted.

* * *

The revised trial did not produce a miracle. It produced a lower but more defensible forecast and a system whose operators understood its weaknesses. Maya carried that to the council rather than the more impressive number she had brought to the first meeting.

The site trials exposed assumptions the desk reviews had missed.

Anika’s team supplied further results from the ark farms, including the disappointing ones.

The revised plan was less elegant than the original. It contained more storage, more repair shops, fewer identical installations, and an uncomfortable number of crops that no marketing department would have chosen for a future civilization.

The owners of the canceled projects came to the hearing with valid contracts.

Some had built access roads and recruited workers. One municipality had relocated a market to make space for a structure now marked for suspension. A supplier had borrowed against the expected order. Calling the projects inefficient did not make their commitments imaginary.

Maya presented the revised food estimate. The first questions concerned the methods. The next concerned who would pay for the abandoned work.

She had prepared for the methods.

Elena answered some of the financial questions. Others went to a working group that everyone knew would be unpopular. The mayor of the municipality asked Maya whether she had visited the relocated market.

She had not.

“Then before you call this an improvement,” he said, “come and tell the traders where their lost year went.”

She went the following week.

The market occupied temporary stalls beside a road that had not been designed for it. Traders had adapted with shade cloth, improvised drains, and a complicated agreement over storage. Some wanted the original site restored. Others had invested in the new location and did not want to move again.

Maya listened for most of the morning. She could not offer the original project back. She could identify material and services the recovery program could provide without making another false promise.

A woman selling vegetables asked whether Maya expected gratitude.

“No.”

“Good. I can give you a list.”

The list was long and practical. Maya took it to the working group with the names of the people who had written it. The compensation package that followed was incomplete. It was also more useful than the generic relocation allowance first proposed.

At the next council hearing, Maya described the cost of her own plan before opponents did. It did not prevent criticism. It changed the criticism from an exposure into an argument about choices people could see.

Amara watched from the technical seats. She recognized the courage involved and understood that her daughter had learned part of it from what she herself had done wrong.

* * *

The workers' declaration began as a shared inventory.

Nobody set out to write a manifesto. A lunar team posted material it could release. A port crew identified containers held by an unresolved fee dispute. An agricultural group published its actual spare-parts needs beside the standard list it had been sent.

The responses made the scale of the problem visible. They also made it harder to pretend that stopping a project on paper stopped the people committed to it.

Maya joined the group assembling the records. There were arguments over verification, liability, and whether publishing an inventory would attract theft. Marcus helped define what could be shared without exposing unnecessary details. He did not ask to control the whole exchange.

The first draft of the declaration promised that workers would continue regardless of political obstruction.

Saira objected.

"Some stop orders are there because the work is unsafe," she said.

"I'm not signing a sentence someone can use against the person who refuses a dangerous shift."

The sentence was changed.

A transport representative objected to language implying that unpaid labor was a moral duty. Leila objected to treating technical workers as the only people doing essential work. Wanjiru asked why food growers appeared after the equipment rather than beside it.

The document improved as it became harder to make stirring.

Maya removed the promise of unity herself. She left the commitment to report actual conditions, share useful knowledge, and remain answerable to the people affected by decisions.

When the finished declaration circulated, some commentators complained that it lacked vision.

The inventory attached to it released enough material to finish several delayed installations. Elena said she preferred that kind of vision.

The final version was shorter.

We were born aboard this planet. None of us chose the danger. We can choose to report what we have, share what we learn, and stop pretending that someone outside the system will repair it for us.

No passengers.

Amara called after reading it.

“The council will say you’re bypassing them.”

“They can still vote.”

“They may remember this.”

“I hope they remember the warehouses.”

For a moment, Maya expected another warning. Instead her mother asked whether she could help check the new climate scenarios.

“Anika’s team is checking them.”

“Then I can check their check.”

“The results get published.”

“Yes.”

Maya made space in the shared review. It was not forgiveness. It was work they could do together.

The revised program passed on its second vote. Full recovery would take twenty-two months. Some canceled facilities would remain unfinished, enormous empty frames standing beside roads. Others were already finding new uses.

* * *

The canceled greenhouse frame stood beside the relocated market like an accusation visible from the road.

For months, nobody wanted to decide what it should become. The original project office regarded reuse as admitting the cancellation was permanent. Local traders wanted shade and storage. A municipal planner proposed a transport facility. Each proposal had a funding source that covered only part of the work.

Maya returned because the compensation group had promised to report back. She found the traders had already begun using one safe corner of the frame to shelter goods during rain. The informal arrangement solved a problem and created another: the unfinished structure had not been inspected for public use.

The first official response was to clear it. Maya asked whether the inspection could happen before the removal. A structural team found areas that could be opened with modest work and others that had to remain closed. The distinction required more effort than a single prohibition.

The market group agreed to a staged conversion. The project supplied material it no longer needed. The municipality supplied work crews. The traders contributed to the layout and refused a design that would have made their stalls look orderly while blocking deliveries.

Months later, the frame held a covered market and a small storage area. It did not repay the lost year. It made the next one more workable. The woman who had given Maya the long list showed her around.

She pointed to a drain that still overflowed and a section of shade that had been installed backward.

“You’re starting with the faults?” Maya asked.

“You know where the good parts are. They’re the parts we’re using.”

Maya laughed and wrote down the drain.

* * *

The first shared review with Amara took place at the trial farm.

Maya had invited her to the site rather than the office. It was not intended as punishment, though she knew her mother might experience it that way. She wanted the discussion to occur within sight of the things its assumptions described.

Amara arrived wearing shoes unsuited to the mud. Wanjiru found her a pair of boots. Nobody mentioned that the most famous scientist on the project was standing on one leg beside a storage shed trying to pull them on.

They walked through the revised installation. Maya showed the failed connections they had retained for training. Leila explained which trial results transferred from the habitat work and which did not. Amina asked Amara to describe the uncertainty in the seasonal forecast without using the annual average as a substitute.

Amara answered carefully. When she did not know something, she said so before anyone had to discover it for her.

At lunch, she asked Maya whether she could help with the next review. Maya gave her a specific part of the work, with the independent checks already assigned.

Her mother accepted it without asking for more control.

They washed their cups afterward at an outdoor sink. For a moment, the gesture recalled the kitchen at home so strongly that Maya almost began an ordinary conversation. Then she remembered how much remained unresolved.

“I still get angry,” she said.

Amara kept washing the cup. “I know.”

“Sometimes when we’re talking about something else.”

“You don’t have to make it convenient for me.”

Maya looked at her. It was not a request for forgiveness. That made it possible to remain beside her a little longer.

On the way home, Leila asked whether the meeting had helped.

“The calculation?”

“You.”

Maya considered the distinction. “A little.”

Leila nodded. She had been accepted into the next stage of ark preparation that morning. She told Maya before they reached the flat, while there was still time to walk farther instead of putting the news down between the dinner plates.

* * *

The first harvest celebration at the trial site lasted longer than Elena intended.

She arrived expecting a short acknowledgment and a review of maintenance readiness. Wanjiru handed her a plate and told her the review would happen after people ate. Elena protested that she had another call. Wanjiru pointed toward a quiet place where she could take it and kept the plate for her.

When Elena returned, the food was still warm. She sat beside a worker who asked whether the project would continue paying for replacement parts after the publicity moved elsewhere.

“The budget includes them.”

“For how long?”

Elena gave the term. The worker asked what happened afterward.

She said the renewal process would begin before the term ended, and heard how inadequate that sounded to someone whose livelihood depended on it.

“We should show you the renewal criteria now,” she said.

“Yes.”

No applause followed. The worker passed her another dish.

Later, Maya found Elena standing near the maintenance store, looking at the labeled shelves.

“You got your inventory,” Maya said.

“I got another obligation.”

“They tend to travel together.”

Elena looked toward the tables, where people were still eating under the repaired roof. “This one is at least attached to something that exists.”

CHAPTER TEN

The Ledger

COURIER MANIFEST, SONDAR DELTA DISTRICT

Item 1: Notice of facility cancellation.
Item 2: Certificate of commendation.
Signed for: M. Aziz.

T–1,190 DAYS

Sondar received its cancellation notice on the same morning it received a certificate.

The certificate commended the district for its cooperation during the survey phase. It was printed, signed, and delivered by the same courier who carried the schedule revision. Meherun Aziz put both documents on the same nail behind her desk, where they stayed for two years.

The canceled project was one of the three expansions Maya had removed from the recovery plan. On the wall in Nairobi it had been a rectangle sharing a membrane supplier with eleven other sites. Its removal was defensible. Maya had defended it, and had been right about the dependency, and had assumed that the replacement pattern would work in the delta as it had worked outside Nairobi.

The replacement pattern required locally available glass.

There was no glass works within four hundred kilometers of Sondar. There was a cement plant, a shipbreaking yard, and a fish-processing cooperative that had closed twice. The substitution schedule listed a supplier in another province whose output was already committed. The clause allowing regional variation existed. Nobody in the delta had the standing to invoke it.

For eleven months, the district was a line item awaiting reassignment.

* * *

Meherun ran the coastal reserve depot at the edge of the port. It held dry stores, a cold section for medicines, and the fuel for the pumps that kept the low ward from flooding in the wet season. She had thirty-one staff and a generator she did not trust.

She was good at the work. That is the part that later accounts tended to leave out, because it complicated the story in both directions. She reported the shortfall correctly. She reported it again when the first report was closed as a scheduling matter. When the regional office asked her to confirm information she had already supplied, she confirmed it, and she kept the correspondence, and she photographed the empty racks with the date written on a card. She had learned the method from a bulletin distributed after Recife.

The method worked in Recife because someone with authority came and stood in the bay.

Nobody came to Sondar. The requests were logged, ranked, and placed behind requests from districts that had produced measurable yield. Sondar had produced no measurable yield because its facility had been canceled. The ranking system did not contain a way to say so.

* * *

Ilyas Karam had been a customs officer for twenty-six years before the discovery, and he retained the habit of asking to see what a document claimed to describe.

He was sixty-one. He wore the same gray jacket to every meeting and carried a bound book with a cracked spine in which he wrote, in a small careful hand, every consignment that entered or left the district by road or water. He had begun it as a private discipline. By the second year it had eleven contributors, and by the third it had a name that people used without irony.

The Ledger was accurate. That was its power and, later, its defense.

Maya first heard him speak from the back of a hall that had been a customs shed. He stood on a pallet because there was no stage. He did not use notes.

“They have taught us a beautiful word,” he said. “The word is resilience. It is a fine word. It has the sound of iron in it. When they came to us with their surveys and their photographs, they used it eleven times in a single afternoon, and I counted, because counting is the only trade I have ever practiced. And I will tell you what the word means when it arrives from a great distance and lands in a poor district. It means: you will manage. It means: the arrangement you have made out of nothing will be entered into our accounts as though we had built it for you. They will write down the roof you patched yourselves and call it capacity. They will write down the hours your daughters spent carrying water and call it community participation. And then, when the accounting is finished, and the district appears in their table as a place already provided for, they will send the material somewhere else, and they will send us a certificate.”

The hall made a sound that was not applause and not agreement. It was recognition.

Maya wrote one sentence in her notebook and did not write anything else for the rest of the meeting.

* * *

She met him the following morning, at his request, in a room with a table and no window.

She had prepared honestly. She had the cancellation record, the substitution failure, the ranking algorithm and its weights, and the compensation package the working group had approved. She put all of it on the table and let him read for as long as he wanted.

He read for eleven minutes.

“You are not lying to me,” he said.

“No.”

“That is a change.” He turned a page back. “This weighting. Measured output.”

“Yes.”

“We have no output because you canceled the facility.”

“Yes.”

“So the consequence of your decision becomes the reason we are last in the queue to have it corrected.”

“Yes.”

He closed the folder and put his hand flat on it.

“You could say that differently,” he said. “You could say it was an unintended interaction between two reasonable systems, and I would have nothing to argue with, and we would both go home.”

“It’s an unintended interaction between two reasonable systems,” Maya said. “It has taken eleven months off your planting and I don’t have a way to give it back.”

Karam looked at her for some time.

Then he asked for the depot.

He wanted the coastal reserve transferred to district control. He wanted the district’s own harvest to stop leaving by water while its people queued for allocation. He wanted the ledger recognized as the receiving record.

Maya explained why she could not agree to the first, and why the second was more complicated than it appeared, and why the third was the only one she could actually deliver. She was accurate about all three. She offered the compensation package, an accelerated survey, a maintenance grant, and a named person in the regional office who would answer his messages.

“A name,” Karam said.

“A person. She’ll answer.”

“And when she is promoted?”

Maya did not have an answer that would survive being said aloud.

“You have brought me a smaller version of the thing that failed,” he said. “You believe that because it is honest, it is different.”

* * *

Elena’s assessment took four sentences.

“The package is nine percent of the harm. Everyone in the room knows it. Offer it anyway, because nine is not nothing, and write down that it is nine. Then tell me what a district holding its own reserve costs the coastal chain.”

Maya told her. It was substantial and not catastrophic.

“Then that’s the number we’re arguing about,” Elena said. “Not sovereignty. Not the ledger. Eight ports and a distribution interval. Say that out loud before somebody makes it about the flag.”

Nobody made it about the flag. It became about the flag anyway.

* * *

The occupation of the depot took place over four days and never looked like the images that later circulated.

On the first day, about two hundred people gathered at the gate and did not leave. They brought chairs. A man set up a stall selling tea. Meherun’s staff continued working inside, and at midday she sent food out to the gate because the alternative was watching people stand in the sun beside a building full of food.

Later, that was used against her.

On the second day, the crowd was larger and the composition had changed. Some of the new arrivals were not from the district. The Ledger’s stewards tried to register everyone at the gate and lost control of the registration by the afternoon. Karam spoke twice, and both times he told them the depot was to be held, not opened.

“We are not thieves,” he said. “We are the receiving record.”

On the third day, the power to the cold section failed.

The cause was never established. The line ran from a substation through an area that had been fenced off by the crowd, and also through a section that had been reported as faulty in March, and also past a point where a contractor had been working on an unrelated water main. The investigation examined all three and could not choose. Karam blamed the utility. The utility blamed the occupation. A technical panel wrote that the evidence permitted several explanations and recommended a maintenance change unrelated to any of them.

The generator started. It ran for nine hours and stopped, for the reason Meherun had reported eleven times.

By the time the cold section was restored, the vaccines were gone, the insulin was gone, and forty percent of the district’s stored blood products had to be destroyed. None of this was visible. It was a set of temperature logs and a disposal manifest.

On the fourth day, someone opened the wrong gate.

There was no shooting. There was a narrowing between the fence and the loading ramp, and a rumor that the dry stores were about to be moved by water, and perhaps four hundred people moving toward an opening built for a vehicle.

Shefali Rahman was forty-three. She had come for her household’s allocation, which she was entitled to, and had been standing near the front for most of the morning because she had arrived early, as she always did. She was carried against the ramp edge and did not get up.

Two other people were badly hurt. A boy broke a collarbone and was proud of it for a week.

She was not protesting anything. She was collecting rice.

* * *

The depot was recovered on the sixth day without force, because Karam gave it back.

He walked to the gate, opened the ledger on the bonnet of a truck, and read out every item the district had received and every item it had lost, including the medicines. Then he handed the depot keys to Meherun in front of the crowd and told them to go home.

It was a genuine act. It was also the most effective thing he could have done for himself, and both of those remained true afterward.

* * *

Maya arrived on the eighth day.

She had wanted to come on the second. Marcus had told her that a senior figure at the gate would convert a standoff into a negotiation with a hostage, and he had been right, and being right had not made the eight days easier to sit through.

She went to the cold section first. Meherun walked her through the temperature record, hour by hour, without editorial. Then she showed her the disposal manifest, then the eleven generator reports, then the nail behind the desk with the certificate on it.

“Do you want me to say I made mistakes?” Meherun asked.

“I want the record.”

“You have the record. I’m asking what you want me to say.”

Maya looked at the nail.

“I don’t know yet.”

“Then don’t ask me for it now, and don’t come back for it in six months when you’ve decided.”

* * *

She met Shefali Rahman’s brother in a room at the district office. He had asked for the meeting and then said very little.

He wanted to know whether his sister appeared anywhere.

Maya did not understand the question at first. He explained: the depot appeared in the supply model. The medicines appeared in the loss inventory. The lost planting appeared in the yield forecast. He had read all of it, because a neighbor’s son had shown him where to look. He wanted to know whether there was a place in any of those documents where his sister was recorded.

There was an incident report. There was a coroner’s finding. There was a line in a regional safety annex.

None of them were in the model.

“Then the model says the week was expensive,” he said.

“Yes.”

“And it doesn’t say she died.”

“No.”

He nodded slowly, as though something he had suspected had been confirmed by a person who had no reason to confirm it.

“You should fix that.”

Maya said she would try. She was aware, saying it, that she was making the kind of promise she had spent ten years teaching other people not to make.

* * *

The review took five months.

It found that the cancellation was defensible, that the substitution clause was unusable in practice by the districts most likely to need it, that the ranking weights created a trap for any region whose capacity had been removed, and that eleven separate reports from the depot had been correctly filed and correctly ignored. It found no evidence that the power failure had been deliberate and no evidence that it had not been.

It made fourteen recommendations.

Twelve were adopted. The weighting was changed. The substitution clause was rewritten so that a district could invoke it without a manufacturing base. A standing rule was introduced requiring that a canceled facility generate an obligation rather than a gap.

Those changes helped other districts. Some of them are the reason the difficult winter went better in three provinces than it otherwise would have.

They did nothing for Sondar.

The planting season was gone. The medicines were gone. The trust was gone in a way that the correction of the weighting could not reach, because the weighting had never been the thing people were angry about. Sondar received its material eighteen months late, from a supplier who had been available the whole time, under a clause that had been rewritten because a woman had died at a gate.

* * *

Karam stood for the district council and won with a margin that surprised the people who had not been listening.

He was, by most accounts, a competent member. He was also relentless, procedurally obstructive on any measure connected to the shared plan, and entirely unmoved by the argument that the correlated-failure reforms had made the plan more honest. He described the reforms as an internal matter among people who had never queued.

Nobody converted him. There was no scene in which the evidence accumulated until he changed his mind. He read the evidence. He accepted most of it. He went on saying the same thing.

Two years later, when a genuine emergency redirection required the coastal ports to release stored fuel within a day, Sondar was the district that held out longest, and the delay cost three other districts a week of pumping, and Karam knew that, and did it anyway, and was not wrong that the last four times something similar had been asked, the promise attached to it had not been kept.

That was the part Maya found hardest to file.

* * *

She kept a recording of the customs-shed speech.

She played it perhaps a dozen times over the following years, usually late, usually after a meeting in which someone had used the word resilience without meaning any harm by it. She could refute most of it. The passage about the surveys was unfair to people she knew. The passage about the certificate was not.

There was one sentence she never found an answer to.

They will write down the roof you patched yourselves and call it capacity.

She had done that. Not in Sondar, and not knowing, and with a good reason, and the district had gone into the ranking table as a place already provided for.

She wrote the sentence on the inside cover of the notebook her mother had given her, beneath the question about trust that had never received an answer either.

Then she went back to the corridor, and the food model, and the eleven months that could not be recovered, because those were the things in front of her, and because there was no version of the work in which she was permitted to stop on account of having been correct.

CHAPTERELEVEN

The Last Normal Summer

STALL SIGN, GIKOMBA MARKET

LAST NORMAL SUMMER T-SHIRTS — ALL SIZES
LAST NORMAL SUMMER (AGAIN) — FOR CUSTOMERS WHO MISSED THE FIRST ONE

ORBIT ODDS: 64% · T–1,095 DAYS

Amara’s father had agreed to the neighborhood trial because the first version of his own plan had failed.

He had not described it that way to anyone.

The glass roof he installed over one section of the garden trapped more heat than he expected. On the first hot afternoon, he left it closed while visiting a neighbor. He returned to wilted plants and a lesson he considered unnecessarily expensive.

Maya found him removing the worst leaves.

“You needed ventilation.”

“I know that now.”

“Leila told you before.”

He looked at her over the rim of his glasses. “Are you helping or keeping score?”

She almost answered both. Instead she took the bucket he offered and began collecting the damaged leaves.

The mistake embarrassed him because he had spent years being the person who knew how a garden worked. New conditions had entered through a roof he had chosen himself. Admitting that he needed help felt different from accepting a visitor’s assistance with heavy lifting. Leila came the following day with a simple vent arrangement and no speech about having been right. He objected to two parts of the design. She accepted one objection and explained why the other would preserve the problem.

They built it together.

When the next hot day came, the plants did better. Grandfather said the weather had been less severe. Leila showed him the measurements.

He studied them, grunted, and asked whether she could leave the instrument for another week.

That was how he joined the trial.

* * *

Maya and Leila brought a radio to the house for the small neighborhood gathering held before Amara’s visit.

The gathering had no official purpose. People brought food, borrowed chairs, and argued over music. Someone had suggested combining it with a readiness briefing. The suggestion was rejected so firmly that the person who made it later claimed to have been joking.

Maya helped carry tables while Leila worked in the kitchen with Ruth. Hadi arrived late with a bag of fruit and an elaborate explanation involving a transport diversion nobody asked him to finish.

For a few hours, the future existed mainly as a reason to invite people now.

Pavel joined by video from the Moon. His connection was propped beside the serving dishes until he complained that all he could see was rice. Saira appeared briefly behind him, took a look at the table, and asked whether anyone had arranged shipping.

Grandfather told them the food would not survive their customs procedures.

Maya watched Leila laughing with the neighbors. Her acceptance into the ark program had become more definite, though the final placement remained subject to testing. People knew enough to avoid asking directly. Their care made the absence ahead more visible.

A woman who had known them since school asked when they intended to marry.

Maya felt Leila become still beside her.

“We’ve been discussing other forms,” Leila said.

The woman laughed, assuming she meant paperwork. Maya was grateful for the misunderstanding.

Later, they danced badly in the small space between the tables.

Maya kept looking toward the kitchen door because she expected someone to need something. Leila took her chin gently and turned her back.

“Five minutes,” she said. “The world can manage.”

Maya gave it five. Then another song began, and she gave it that one too.

* * *

The next morning, they walked to the school garden.

The first tank had been replaced again. Their original brackets remained in a storage shed because the caretaker refused to discard anything potentially useful. The garden now supported a small teaching program and an assortment of plants that had outgrown every plan made for them.

Leila found the place beside the wall where they had first kissed. A new pipe ran through it.

“Improved access,” Maya said.

“Terrible for romance.”

They sat on the low wall and watched a younger group attempt to repair a hose connection. Neither intervened. Grandfather had taught them at least that much.

Leila asked whether Maya regretted not taking the habitat interview.

“Sometimes.”

“That’s more than you said before.”

“Before, I thought admitting that would mean I had chosen wrong.”

Leila picked a small piece of loose mortar from the wall, then put it back as if the gesture could repair it.

“I regret leaving already,” she said. “I still want to go.”

Maya had been hoping the regret would eventually cancel the wanting. Hearing both in one sentence made that hope harder to preserve.

“What happens to us?”

Leila looked toward the students at the hose. “We stop pretending a word can decide it before we live it.”

“That’s unfairly sensible.”

“I’m not feeling sensible.”

They held hands on the wall until the repair succeeded and the students accidentally sprayed one another. Maya laughed, then began to cry without warning. Leila moved closer and let the two things happen together.

The trip to the lake had been Leila’s idea that time.

She found two days when neither of them was scheduled for a test, review, or family obligation, and booked a room before the days could be claimed. Maya said she should have checked first. Leila asked which person Maya would have needed permission from. Maya opened her calendar, saw the empty space, and could not supply a convincing answer.

They traveled with bags small enough to carry without thinking.

On the train, Maya kept reaching for the work display and then putting it away. Leila read beside her. After the fourth attempt, Maya asked whether Leila was counting.

“Only because you are.”

The room overlooked water and a landing stage used by small boats. A fan clicked once each revolution. Maya climbed on a chair to inspect it until Leila told her that the guesthouse employed people for that purpose. They reported the noise and went walking.

For the first hour, they discussed work because it was the language in which they knew how to be attentive to each other. Then they ran out of updates. The silence that followed felt unfamiliar rather than empty. Leila asked what Maya would have done if the discovery had never happened.

“I don’t know. Engineering, probably.”

“Would you have gone to the Moon?”

“Maybe.”

“Would we have lived together?”

Maya looked at her. “I hope so.”

Leila smiled. “I was hoping for more confidence.”

“I can’t validate the alternate timeline.”

“Try being romantic without a method.”

Maya took her hand. “Yes.”

They found a place serving fish and vegetables at tables close to the water. The menu had several items crossed out. The owner apologized for the limited choice, explaining that transport priorities had changed again. Maya almost asked which route had been affected. Instead she ordered what was available.

At the next table, two older women were arguing about a holiday they had taken decades earlier. One remembered a broken vehicle; the other insisted the breakdown had happened on a different trip. Neither could persuade the other. They ordered another drink and continued.

Leila watched them with a softness Maya recognized.

“What?”

“Nothing.”

“Tell me.”

“I wanted that version for a moment.”

Maya knew which version. Years of shared incidents becoming stories neither remembered correctly, a future long enough to contain several different holidays, an argument that did not carry a departure inside it.

“I still want it,” Maya said.

Leila looked down at their joined hands. “I know.”

They did not spend the evening deciding again. They had made no final promise about the application, but neither needed to rehearse every argument to prove it remained painful. They walked back slowly.

The fan had been repaired. The quiet room felt like an unexpected gift from someone doing an ordinary job well.

* * *

In the morning, Leila swam while Maya sat near the shore reading a book she had brought and barely opened. When Leila returned, water ran from her hair onto the pages. Maya complained, and Leila said the book looked underused.

“Are you going in?”

“Later.”

“That’s a famous place.”

Maya put the book down and went with her. The water was colder than she expected. For several minutes, all her attention belonged to breathing, movement, and Leila’s laugh when she objected to the temperature.

They floated where it was safe, looking at the sky. Maya could not see any sign of the approaching encounter. She knew enough now to distrust the wish for a visible warning, but part of her still wanted the world to announce its changed conditions in a form the body could recognize.

Leila reached for her hand beneath the surface.

“Here,” she said.

Maya turned her head.

“You’re somewhere else.”

“Sorry.”

“You don’t have to apologize. Just come back.”

Maya watched a small cloud move above them. She felt the water holding her and Leila’s fingers against her own. For a little while, she managed to attend to the life she was afraid of losing.

* * *

They returned to Nairobi with a photograph taken by the guesthouse owner. It showed them standing beside the water, both laughing at something outside the frame. Maya could not remember what had been funny. Leila said that was all right.

The photograph joined the one of the crooked shelf. Neither was especially well composed. Both contained things no public account of the project would ever include.

That week, Maya’s team circulated a request for memories of the preparation years to support an archive. She looked at the photographs and decided not to submit them. The archive could have her technical records and the work she owed the public. Some parts of her life could remain hers.

She told Leila the decision. Leila looked relieved.

“I was afraid you’d turn us into an example.”

“Of what?”

“Whatever people needed us to mean.”

Maya put the photographs in a drawer before dinner. Later she took them out again and placed them where they could see them. Private did not have to mean hidden.

* * *

Three years before closest approach, Amara returned to her father's garden.

He was eighty-six. He had begun leaving a chair at the end of each row, though he insisted they were for visitors.

The beds had changed since her last visit. A section stood beneath a simple glass roof. Along the fence, a water tank fed a gravity-driven irrigation line. He had rejected the automatic controller the project had offered him and accepted the controller's spare hose.

"Maya says you are preparing," Amara said.

"Maya talks too much."

"She says you joined the neighborhood trial."

"They needed someone who knew when a tomato was dead."

She sat in one of the visitor chairs. He handed her a bowl and began picking.

From the garden, the sky looked ordinary. Telescopes could map the black hole's lensing pattern, but there was no theatrical wound above the house. The approaching object had no obligation to resemble the pictures used in fundraising campaigns.

"Can you see it?" he asked.

"Not like this."

"Good. I have enough to look at."

For a while they discussed the roof. The neighborhood trial compared varieties under different amounts of light and heat. His records were precise, his handwriting difficult, and his opinion of the first batch of seedlings unprintable.

Eventually he asked whether she and Maya were speaking.

"We work together."

"That wasn't the question."

Amara watched an ant move along the bowl's rim.

"I thought if I could get us through the next decision, there would be time to repair the rest."

Her father lowered himself into the opposite chair.

"And now?"

"Now I'm trying to stop calling people the rest."

He nodded once. She had hoped for absolution. He began sorting the tomatoes instead.

That evening Maya arrived carrying a bag of groceries and a replacement filter. Her grandfather accused her of using visits to conduct inspections. She asked when he had last cleaned the tank.

Amara washed the dishes while they argued.

Later, Maya came into the kitchen and dried the bowls without being asked. They worked around each other carefully, as they had when Maya was small enough to stand on a stool at the sink.

“I’m afraid,” Amara said.

Maya put a bowl on the shelf.

“I know.”

“Not only of losing Earth. I’m afraid of finding out there was something obvious I should have done.”

“There will be.”

Amara looked at her.

“There always is,” Maya said. “Afterward.”

Outside, her grandfather called to ask whether anyone intended to bring him tea. Maya filled the kettle. Amara took down three cups.

The following day, he proposed a visit to an old friend whose house was a longer walk away than he admitted. Amara wanted to arrange transport. He refused. Maya supported him until she saw how slowly he rose from the chair.

They compromised on walking there and taking a vehicle back. He called this capitulation and put on his hat.

Along the way, people recognized Amara. Some thanked her. One man asked why she had kept the food report private. Grandfather stopped walking before the security escort could move him along.

Amara answered the question. She did not attempt to make the conversation brief by invoking the larger emergency.

The man listened, then said he was glad she was still doing the science but did not want her deciding what he could read.

“That’s fair,” she said.

He nodded and continued down the road.

Grandfather resumed walking. For several minutes, nobody spoke.

Then he asked whether the new review system was working.

“It’s slower,” Amara said.

“That wasn’t the question.”

She smiled reluctantly. “Yes. Better in some ways.”

At the friend’s house, the conversation concerned a roof repair, a grandson’s training course, and a dispute over the ownership of a fruit tree that had grown across a boundary. Amara sat listening while the men revisited an argument older than Maya. No one asked her to solve it.

On the journey home, her father fell asleep against the window.

She watched the small movements of his hand on his knee and realized how much of his aging she had received as reports from other people.

That evening, she asked him whether he wanted to move somewhere with more support.

He said no before she finished.

“I’m not suggesting you stop having a garden.”

“You’re suggesting I stop having this one.”

“I’m suggesting help.”

“Then arrange help here.”

The answer irritated her because it required negotiation rather than a solution she could fund and declare complete. Maya watched the exchange with an expression Amara knew too well.

“You’re enjoying this,” she said later.

“A little.”

They arranged a local support schedule with the neighbors and a clinician who already knew him. He objected to part of it and accepted more than he initially intended. The final arrangement was imperfect and belonged to him.

* * *

On the last afternoon, Amara found the notebook she had given Maya after the school-garden visit.

Maya had left it beside her bag. It was thick now with inserted pages, photographs, and small scraps of paper. Amara asked before opening it.

The first pages described the tank repairs. Later entries concerned university projects, lunar work, the missing fitting, the accident. Some pages contained drawings rather than words. Others contained questions with no answers beside them.

Amara stopped at one written after the withheld report became public.

What can you trust when the person who taught you to check is the person who didn’t tell you?

She closed the notebook.

Maya had been watching from the doorway.

“I should have warned you.”

“No.”

Amara put it back where she had found it. She did not ask what answer Maya had eventually written. There was no answer on the page.

They spent the rest of the afternoon working on the gutter. The task required one person to hold a section while the other fastened it. Neither could do it conveniently alone. They spoke about measurements, tools, and where the rain would go.

It was enough for that afternoon.

That season became known later as the Last Normal Summer. The name was inaccurate in almost every practical respect. People were conducting food drills, moving clinics, and arguing about ration schedules. Some beaches were closed because their access roads had become freight corridors.

But weddings still filled the weekends. Someone nearby was learning the trumpet badly. A boy came to the gate selling seedlings, and Amara's father bought two after a negotiation so prolonged that the boy returned the next day to ask for gardening advice.

Amara remembered the trumpet when she thought of that summer.

* * *

The merchandise arrived before the season had agreed on its name. LAST NORMAL SUMMER shirts appeared in July, in quantities that suggested the manufacturers had committed early. By August the stalls also sold LAST NORMAL SUMMER (AGAIN), for people who felt they had missed the first one.

At night the lens could be found without a chart. It sat in a part of the sky that had once held nothing worth pointing at. Bright stars near it had shifted from their old positions by nearly the width of the Moon, and a faint arc of light hung where no star had been before.

Children measured it with their little fingers at arm's length, as their teachers had shown them. Every week a school in the neighborhood posted the result on its gate. For now, a little finger still covered the disordered patch. The children predicted that by closest approach it would show on both sides. Their teacher told them they were correct, and they were insufferable about it for a week.

Grandfather walked past the gate each morning and did not look.

"You could just look," Maya said.

"I'll look when it's finished."

"That's at closest approach."

"Then that's when I'll look."

He said it the way he said he would taste a tomato when it was ripe, and Maya stopped asking.

It was the sound of someone assuming there would be enough time to improve.

* * *

Before leaving, Leila asked Grandfather for seeds.

He had expected the request. Maya could tell from the carefully folded packets waiting in a drawer. Each carried a variety name, a date, and notes about the conditions under which the seed had been collected. “They may not use them,” Leila said. “There are quarantine rules. Trial requirements.”

“Then let them test them.”

He placed the packets in a larger envelope. It was not a magical inheritance. Some might fail. Some might never be planted. Their value lay partly in the records that accompanied them and partly in the fact that he had made the records for her.

Leila read one of the notes and laughed.

“What?” Maya asked.

“He wrote that this one is stubborn.”

Grandfather took the packet back. “I wrote that it tolerates poor conditions.”

“Your handwriting is ambiguous.”

He inspected the word, then returned it without conceding the point.

At the gate, he embraced Leila for longer than usual. After she stepped away, he adjusted the collar of her shirt, an unnecessary small task that let him keep his hands occupied.

“Write,” he said.

“I will.”

“Not only when something important happens.”

She promised.

On the walk back to the flat, Maya carried the seed envelope while Leila carried the tools. The division had happened without discussion.

It felt like hundreds of earlier walks and unlike any of them.

CHAPTER TWELVE

Departure Windows

INDEPENDENT HABITAT PASSENGER GUIDANCE, NOTE 6

Personal effects: 2.0 kg. Items of sentimental value are not exempt. Passengers are advised to choose before the scale does.

ORBIT ODDS: 68% · T–852 DAYS

The ark departures began twenty-eight months before closest approach.

They were staggered because engines, transfer corridors, and people could not all be ready on the same day. Each vessel had a trajectory designed to avoid the worst of the encounter. None could simply cancel gravity.

Anika’s cylinder was the fourth to leave.

The final isolation trial began with a small ceremony that most of the crew missed because they were working.

For three days, *Ithaca* would operate without routine support from the assembly complex. Emergency connections remained available, but using them would end the trial. The vessel had passed shorter tests. This one was intended to expose problems that only appeared when systems had to keep working together while people ate, slept, argued, and made ordinary mistakes.

Leila had been aboard for six weeks preparing the agricultural sectors. Her cabin contained two bags, several sample records, and the seed envelope from the garden. The wall above her bunk remained blank. She had not decided whether putting up a photograph would make the choice easier or more difficult.

On the second night, a water balance failed to close.

The discrepancy was small enough to have an innocent explanation. A volume estimate might be wrong. Water could be retained in a part of the system not fully represented in the current model. A measurement could be drifting. The engineering team checked those explanations before treating the missing amount as a leak.

Leila noticed the dampness first while inspecting a storage area beneath the crop beds.

It was not a dramatic torrent. A fine line of water had found a path along a structural joint and collected where nobody normally looked during routine operation. She photographed it, marked the location, and called the watch team.

The response was practiced. Electrical equipment near the affected area was isolated. Access was restricted. Leila assessed which crop sections could tolerate an interruption to their normal watering sequence while the watch team went looking for the source.

They worked by elimination, out loud, because the vessel’s diagnostic display was optimistic and the technicians were not.

“Condensate line?”

“Dry. I checked the trap.”

“Bed drain, sector four?”

“Holding. The dye’s still in it.”

“Then it’s upstream of the manifold.”

“Or it’s the manifold.”

A technician named Oskar closed an isolation valve and watched the pressure trace with the expression of someone waiting for a verdict. The trace held. He closed the next. It held. On the third, the decay slowed but did not stop.

“There,” he said. “Between the manifold and the bed feed.”

“That’s forty centimeters of pipe.”

“It’s forty centimeters of pipe behind something.”

The leak lay behind a section intended to remain accessible. Since the access plan was drawn, another contractor had installed a support bracket across part of it. Oskar got his light under the bracket, then his hand, then withdrew both. A weld, sound. A fitting, sound. A second fitting, seated one turn short and weeping at a rate that would take three days to fill a coffee cup and had taken two to empty the balance.

The maintenance chief looked at the bracket for a long time.

“Who signed this clearance?”

“We can answer that after we stop the water,” Anika said.

She had arrived without an escort or a camera. She asked whether the crew could repair the fault safely under the trial conditions. The maintenance chief said yes, probably, with an interruption long enough to risk part of the current crop. Using the external support connection would reduce that risk and invalidate the trial.

Several people began discussing how to preserve the test designation. Anika stopped them.

“We are testing the vessel. We are not protecting the name of the test.”

They opened the support connection.

The crops received water. The trial failed.

Leila felt relief followed by embarrassment at how much she had wanted the test to succeed. She had described *Ithaca* to Maya as a place where people would have to face what their systems actually did. Now the vessel had made that description inconvenient.

The repair required moving equipment, removing the obstructing bracket, and inspecting neighboring connections installed under the same procedure. One additional fault was found before it leaked.

The crew worked through the night. Leila carried trays out of an affected section with two technicians and a physician who had completed the same cross-training. Farah complained that the plants weighed more than some of her patients and were less cooperative. Oskar said the bracket was the only thing on the vessel that had been installed exactly to its drawing and should therefore be the first thing thrown overboard. Nobody laughed the first time. By the third telling, around four in the morning, it had become funny, which everyone understood to be a symptom rather than a judgment.

At dawn, the water balance closed within the expected uncertainty.

No one cheered. They were too tired, and the next test had become more complicated.

* * *

The delay consumed part of the departure margin.

Ithaca's flight plan could be revised. It would require additional propellant and changes to the assembly complex's schedule. Those changes would affect other vessels preparing to leave. The problem was manageable in engineering terms and unpleasant in every conversation about who would absorb its cost.

The council liaison asked Anika whether the leak could be classified as a commissioning issue rather than a failed isolation trial. "It is both," she said.

"The distinction affects the release."

"Then the release should explain both."

He warned her that some applicants were already questioning whether the vessel was ready. She said they should have the repair findings before deciding.

Two families withdrew during the delay. One had been uncertain for months. The other had trusted the published schedule as proof that the remaining risks were minor. Their departure from the program left roles to fill and personal relationships the staffing model did not describe.

Farah's son asked whether they should withdraw too.

She took him to the repaired area after it was cleared for visitors.

The new access route looked less tidy than the old one. It was easier to inspect.

"Things will fail here," she told him. "The question is what we do when they do."

"Things fail on Earth."

"Yes."

"So why are we going?"

She sat with him on a storage bench and answered as a mother rather than a representative of the program. Some of her reasons concerned the work. Some concerned the future she believed this community could make. Some were hopes she could not guarantee.

He listened, then said he wanted to speak to his grandfather before deciding how he felt about it.

She arranged the call.

* * *

Leila sent Maya the repair report before the official summary was ready.

She included the parts that frightened her and the reasons she still intended to go. She did not want Maya to hear the failure through a headline and wonder whether their last weeks together had contained another concealed decision.

Maya called as soon as the message arrived.

“Are you all right?”

“Yes. Tired.”

“Did anyone get hurt?”

“No.”

They talked through the sequence. Maya asked practical questions, then stopped herself from turning the conversation into an inspection.

Leila noticed the effort.

“You can ask,” she said.

“I don’t want to make every problem into an argument for staying.”

“Thank you.”

The words made Maya sadder than an argument would have. They were learning how to be generous to each other in a future that included separation.

After the call, Maya opened the report again. The failed trial had produced a better vessel. It had also made the dangers more real.

Neither fact canceled the other.

Packing the flat took longer than either expected.

The difficult objects were not the valuable ones. Leila knew which equipment belonged to the program and which books she could take.

Clothing could be reduced. Duplicate tools could be left behind.

The difficulty lay in a chipped bowl, a folded cloth from a market trip, a photograph whose frame weighed more than the allowed personal allowance could comfortably justify. Each object seemed to ask whether the life they had shared was ending or merely changing address.

Maya offered to keep everything Leila left.

“You don’t have to turn the flat into storage for me.”

“It’s not storage.”

Leila looked at the growing pile beside the cupboard.

Maya sat on the floor. “I don’t know what to do with it.”

They sorted slowly. Some things went to friends. Some remained because Maya wanted them independently of their history. The chipped bowl stayed. The frame did not travel, but the photograph did, printed on a sheet light enough to fold inside a book.

The crooked shelf was the last argument they almost had.

Leila suggested fixing it before she left. Maya said it was fine.

They looked at each other, understood what the disagreement was trying to become, and started laughing instead.

They fixed it together.

The result looked wrong for several days because they had become accustomed to the tilt.

* * *

Ruth organized a family meal without calling it a farewell.

She prepared too much food and complained when anyone offered to take leftovers because she had not finished deciding which containers she could spare. Hadi told a story from Leila’s childhood that she had heard so often she began correcting it before he reached the part he always exaggerated.

Maya sat beside Leila and tried to remember the evening without turning every moment into a thing to preserve. The effort made her miss parts of the conversation.

After dinner, Ruth asked Leila to help in the kitchen. They remained there long enough that Maya knew the washing up was not the point. Hadi offered her tea and asked how she was managing.

She almost said fine. He had known her since she was seven. The lie would have insulted them both.

“Badly sometimes.”

He nodded. “So are we.”

“Do you wish she’d stay?”

“Every day. I also know what she looks like when she talks about the work. I don’t want her to spend her life blaming us for taking that away.”

He turned the empty cup in his hands.

“It would be easier if wanting the best for someone always felt generous.”

Maya looked toward the kitchen. Leila was laughing at something her mother had said. She wanted to take the sound home and keep it unchanged. She knew she could not.

Grandfather arrived late because he had insisted on carrying the seed records himself. He gave Leila an additional packet containing copies rather than originals.

“If one set is lost,” he said, “the other still exists.”

Anika would have approved of the reasoning. Maya wondered whether he knew that and had chosen not to say so.

* * *

The final night in the flat was ordinary until they tried to sleep.

They had cleaned the kitchen, checked the travel documents, and charged devices that did not need more charge. Leila’s bags stood beside the door. Maya kept looking at them and then pretending she had not.

In bed, Leila asked whether they should make promises about waiting.

Maya turned toward her.

“Do you want to?”

“I want to promise something I can actually give.”

The honesty hurt. Maya had spent weeks imagining a declaration that would make the distance less consequential. She knew enough about promises to distrust one offered mainly to make the next morning bearable.

“I don’t want your life to become a room where you’re waiting for me,” she said.

Leila closed her eyes.

“I don’t want that for you either.”

They agreed to keep writing, to say what changed, and not to use the distance as an excuse to make decisions invisible. It was less than the certainty Maya wanted and more than either could have offered if they had pretended certainty existed.

For a while, they talked about unimportant things: the shop’s compressor, the neighbor’s music, the first shelf argument. Then they stopped talking.

Maya woke before the alarm with Leila’s hand resting on her arm.

She stayed still until the alarm sounded because there was nothing useful to do with those last minutes except have them.

Farah’s father changed his mind about coming to the farewell meal twice.

The first time, he said travel would be too tiring. The second time, he said he did not want the children to remember him making a fuss.

Farah told him they would remember him refusing to come if that was the memory he preferred. She regretted the harshness as soon as she said it.

He came.

Leila met him at the family accommodation near the transfer terminal. He was smaller than she expected from Farah's descriptions and more interested in the food than the ceremony everyone was trying not to call a ceremony. He asked Leila what she grew. She answered, and he questioned her with the practical skepticism of someone who had spent enough years eating vegetables to regard that as relevant expertise.

Farah's son sat beside him, refusing to finish a portion he had chosen himself. The younger child slept against Farah's shoulder.

Around them, other families occupied tables with the same strained determination to have an ordinary meal.

At one point, Farah left the table to collect something. Her father asked Leila whether the vessel's clinic was good.

"Farah helped design it."

"That wasn't what I asked."

Leila smiled despite the setting. "It has limits. She knows them better than I do."

"She thinks knowing the limits means she has to be there."

"Sometimes it does."

He watched his grandson pushing food around the plate.

"Sometimes it means you know what you're giving up."

When Farah returned, he asked for a photograph with the children.

She took several because the younger one woke and cried halfway through. The best image showed nobody looking at the camera. Farah kept that one too.

* * *

Arun's family sent him three different lists of things he had forgotten.

Two lists concerned objects. The third, from his younger brother, contained arguments. He read all three while sitting on a packing crate and then called the brother. The conversation lasted long enough that Leila, working nearby, moved to another compartment to give him privacy.

When she returned, he was rolling the corrected electrical drawings.

"Did you settle it?"

"No."

"Are you all right?"

"Also no. But I know where the drawings are."

She sat on the crate beside him. He said his brother believed he was abandoning their parents. Their parents said they supported him and then described practical problems in a way that made support feel like a debt. Arun loved them. He also wanted the life aboard the vessel.

Neither fact removed the other.

Leila told him about Ruth and Hadi, and about Maya. He asked whether she thought leaving made them selfish.

“I think choosing to stay can be selfish too, depending on what you demand from everyone else. The label doesn’t tell us enough.”

“That’s an inconvenient answer.”

“I work with plants. Most useful answers are.”

They went back to the inspection. The drawing showed a panel that could be removed in one piece. The actual panel had been divided during installation to improve access. Arun marked the difference and photographed it. Leila watched his concentration return. Work did not cure the grief. It gave the next hour a shape.

* * *

Anika spent her final evening within easy transfer distance answering messages from people who had changed their minds.

Some wanted places that no longer existed. Some wanted to withdraw and feared the consequences for their families. One specialist had accepted a seat believing a partner would eventually receive one, despite a written decision saying otherwise. Anika read that file twice before calling.

She could not create the additional place. She could explain the withdrawal process and the support available on Earth. The specialist asked whether she was telling him to stay.

“I’m telling you what the choice contains.”

“That’s easy from where you are.”

She almost described everything her own choice had cost. It would not help him. She let the accusation pass and answered the next practical question.

After the call, she sat alone in the observation gallery. The dock structure obscured much of Earth. She had spent years defending the vessels against people who treated them as betrayal. Now she had to resist treating every person who stayed as someone who had failed to understand her work.

Leila found her there while checking a route for the final visitors.

“Have you eaten?”

Anika laughed. “That question has become the real command structure.”

Leila brought food. They ate from containers balanced on their knees. Anika asked whether the growing team was ready.

“Ready enough to stop finding new reasons to postpone,” Leila said. “Not finished.”

“Good distinction.”

“Will you still like it when we ask for more time?”

“Probably not. Ask anyway.”

* * *

During *Ithaca*’s last week within convenient transfer distance, Maya made her final visit. The painted river had become a narrow channel carrying water through a biological treatment system. Small trees grew beneath lamps. The leaves were real. So were the warnings attached to every irrigation junction.

“It looks like the picture,” Maya said.

“Look under the table.”

A tray caught a slow leak from a pipe coupling. Someone had written REPLACEMENT TUESDAY on a strip of tape.

Anika took her to the observation gallery. Earth was partly obscured by dock structures. Families were gathering at the windows as if this were an airport, except the people on the other side of the glass could not come to wave goodbye.

“Have you changed your mind?” Anika asked.

Maya shook her head.

“There is still a systems vacancy.”

“There’s one on Earth too.”

Anika accepted that. She had made the offer because she owed Maya a choice, not because she expected a different answer.

They discussed the observation network that would remain active after separation. The arks’ changing viewpoints would help distinguish real orbital perturbations from errors in the common reference frame.

Their instruments were not as powerful as the dedicated observatories, but their independence mattered.

“Keep sending the raw measurements,” Maya said. “Even if our network tells you it already knows.”

“Especially then.”

Amara arrived late. A transport connection had failed, and for once the explanation was both true and irrelevant. Anika had saved her a seat in the gallery.

They sat together while Maya spoke with the instrument team.

“I used to think you were dividing the project,” Amara said.

“I was.”

“You know what I mean.”

“Yes. I also know you eventually funded the thing you disliked.”

Amara looked at Earth between the dock braces.

“I wish there were more of you.”

“So do I.”

The ark fleet would preserve only a small part of humanity if Earth failed. The number had grown with construction and shrunk with testing. Neither woman needed to recite it.

“Lifeboats are not a vote against the ship,” Anika said.

“I remember.”

“You seemed to need reminding.”

Before going to find Leila, Maya embraced Anika. Anika held on for an extra second, then released her before either had to acknowledge it.

On her way through the assembly complex earlier that day, Maya had met the person replacing one of the withdrawn specialists.

His name was Arun. He had helped build part of *Ithaca*’s electrical distribution system and had spent months training a colleague who would remain behind. His acceptance had arrived late enough that his family had not yet agreed on what it meant.

“Everyone says I must be delighted,” he told Maya while they waited for a transfer door to open. “I am. That’s not the only thing.”

He carried a small bag and a roll of drawings marked with handwritten corrections. The official versions existed in the archive. He trusted the paper because it showed what had been changed during installation.

Maya wished him well and meant more than the phrase could contain.

Now, inside the vessel, Leila took Maya past the access panel so she could see the new inspection route. It was an odd thing to show a visitor during a goodbye. It was also part of why Leila could go.

They reached the cabin together. Leila put the photograph above the bunk with two small pieces of tape. Maya had expected to feel replaced by the image. Instead she was relieved to see something from their flat inside this unfamiliar room.

“The shelf looks better here,” Leila said.

In the photograph, it was still crooked.

They had less time than they wanted and more than Maya knew how to use. She asked where Leila would eat. Leila showed her the route to the communal kitchen. She asked how to reach her during off-duty hours. Leila demonstrated the message system they already knew. Finally Leila took her hands.

“You don’t need another instruction.”

“I know.”

“Come here.”

They held each other in the narrow cabin while activity continued beyond the door. Someone was moving a cart. Someone else was laughing. A vessel preparing to leave did not become silent because two people needed time.

At the transfer lock, Maya realized she had brought the inspection mirror in her bag. For an instant she considered giving it to Leila. Then she remembered that it was a useful tool she would need, not a token required to prove what the goodbye meant.

She kept it.

Leila gave her a folded note instead. “Later,” she said.

Maya put it in the notebook without reading it.

The final visitor announcement sounded. Neither argued with it.

They had spent too long learning what departures required to make the crew chase them through a corridor.

At the boundary, Leila touched Maya’s face once, lightly. Then she went back into the vessel.

Maya walked to the ferry with Amara and did not speak until they were seated.

“I could still have chosen differently,” she said.

Her mother did not tell her she had made the right choice.

“Yes.”

Maya cried then, quietly and without trying to make the tears useful. Amara sat beside her until the ferry began its departure sequence.

* * *

Leila’s note contained no promise about forever.

It described the first day they repaired the school gutter, the orange she had peeled during the announcement, and the moment on the Moon when Earth seemed too small to contain everyone they knew. At the end she had written that Maya did not have to earn rest by finishing the world’s problems first.

Maya read it twice, then closed the notebook.

Her first message to *Ithaca* concerned the cupboard latch in the flat.

It had broken the evening before they left. She sent a photograph she had taken of the replacement bracket and complained that Leila had escaped the repair.

The reply came while the ferry was still within an easy communication interval.

Wrong bracket, Leila wrote.

Maya laughed hard enough that her mother looked over.

The departure itself was almost invisible from the ferry. No great plume filled the sky. A sequence of modest, sustained maneuvers slowly changed the vessel's future.

Maya watched until the cylinder became one point among others.

On her screen, its status changed from DOCKED to INDEPENDENT FLIGHT.

She had seen the same words beside machines all her adult life.

This time she read them twice.

* * *

After the last visitors left, *Ithaca* sounded different.

There were fewer voices in the corridors. The people remaining no longer asked directions as often. Temporary signs came down. A child looked for a grandparent near the transfer lock and was led gently toward the residential area by a parent who had already explained the goodbye and now had to explain it again.

Leila returned to her cabin and stood in the doorway. The photograph above the bunk had lifted at one corner. She pressed the tape down, then sat without removing her shoes.

For months, she had defended this choice against her own fear and other people's grief. Now there was nobody in the room to persuade.

She felt, for the first time, the full silence after choosing.

A message from Maya arrived about the cupboard latch. Leila looked at the image, recognized the wrong bracket, and laughed. The laugh became a sob before she could stop it.

She answered the joke. Then she washed her face and went to the growing compartment, where a routine inspection was waiting.

Someone had put on music. Someone else asked whether she wanted tea.

"Yes," she said.

It was not enough to replace what she had left. It was a beginning she had chosen, and she stayed to help with the inspection.

CHAPTER THIRTEEN

Entry

DAEDALUS STYLE GUIDE, REVISION 30

“The gravitational event” is retired. Preferred: “the Encounter.”
Staff should capitalize it, as one would a guest.

ORBIT ODDS: 70% · T–425 DAYS

There was no morning when the black hole’s gravity switched on.

It had been altering the Solar System throughout the years of preparation. Comets responded first and most strongly. The distant reservoirs of ice and rock no longer belonged to the same quiet architecture the textbooks described.

Most of the disturbed objects would never approach Earth. Some would arrive long after the people preparing for them were dead.

Tracking teams learned to distinguish the danger of this month from the danger of the next century without pretending either was harmless.

Fourteen months before closest approach, Pilot changed the encounter status to ACTIVE.

The designation meant emergency staffing, shorter review cycles, and automatic isolation of equipment that exceeded local safety limits.

It did not mean the universe had crossed a boundary.

Amara approved the change from a conference room overlooking a school playground. Children were practicing a shelter drill. They walked in pairs, one teacher at the front and another counting at the door.

The count mattered more to her than the trajectory display.

Evacuation of the most vulnerable outer settlements had begun years earlier. The last departures were freight crews removing machinery that could still be used elsewhere. Some stations would continue operating without people until their power or communications failed.

Jupiter’s future remained uncertain. Saturn’s was worse. Their moons complicated every forecast, each bound to a planet that might itself leave the Sun. For residents, a stable moon around an escaping planet was not necessarily a survivable home.

Jun Park had been given three dates for leaving Rill, and had put none of them on the kitchen calendar.

The first depended on completion of a replacement pump station on Earth. The second depended on a cargo carrier passing a pressure test. The third was printed on a passenger

manifest bearing his name, which should have made it different. He still wrote only the inspection schedule, and the day the bread starter needed feeding.

His daughter had drawn the calendar when she was eleven. She was nineteen now, living with her aunt outside Busan, and had stopped asking when he was coming. That was worse than the questions.

There were forty-three people left in the settlement. At its largest, Rill had held six hundred and twelve. Jun could walk from the machine shop to the observation gallery without seeing anyone, passing doors whose occupants had left names written in the dust of the windows. A restaurant still displayed a menu. The woman who had run it now repaired kitchen equipment on Earth; she had sent a photograph of a battered oven with the caption FINALLY SOMETHING I UNDERSTAND.

He had answered with a photograph of her old serving counter, scrubbed clean.

Why did you do that? she had asked.

He had not answered.

On the morning their carrier arrived, Jun was in a maintenance passage trying to close a valve that had not been operated by hand in sixteen years. The motor turned; the valve did not. Through the small inspection window he could see a coupling moving where no coupling should have moved.

“We can leave it,” said Orla, who had been working beside him for four hours. “The automatic station doesn’t need this loop.”

“It does if we want a second thermal path.”

“Do we?”

He looked at her.

“Not a rhetorical question,” she said. “How much more life are we buying it?”

Jun set the driver down. He had been calculating how to make the repair. She was asking whether it was worth making. They used to have different people for those jobs.

“Months, perhaps.”

“Of measurements.”

“Yes.”

“Then let’s record the choice. So whoever reads those measurements knows what they cost.”

They called the carrier’s local coordinator. The signal stayed within the neighborhood of Saturn; Earth would hear the exchange later. Jun requested two hours. The coordinator said forty minutes were available before the next cargo movement, and asked which people he intended to leave at the valve.

“Me and Orla.”

“And your backup?”

Jun nearly said they did not need one. He caught himself.

“Dima. At the junction.”

“Forty minutes. Then close out in the present state.”

Orla started the timer. They replaced the coupling in thirty-one minutes and spent the remaining nine proving that the valve would close. Jun wanted to test it once more. Orla put her glove on the driver before he could pick it up.

“It passed.”

“I saw.”

“Then come on.”

They returned through a residential sector where the lights had already been removed. The glow from their work lamps traveled over a mural of Earth painted by schoolchildren. Jun’s daughter had painted one of the clouds. He had never been sure which one. When she was small, he had asked her to show him; she had pointed at a different cloud each time.

He stopped.

Orla stopped too, but did not ask why.

The camera on his helmet made the wall look flat and gray. He took out his personal camera and adjusted the light. He photographed the whole mural, then each section close enough to show the brush marks.

Orla held her lamp steady.

They lost seven minutes. He entered them in the work record as personal time.

* * *

The carrier was called *Tern*. Its passenger compartment smelled of new padding and the disinfectant used after every boarding cycle. A crew member stood at the hatch with a list that contained forty-three names.

Jun recognized her from the transport hearings years earlier. She did not recognize him until he gave his name.

“You’re the man who asked who collected the last crew.”

“Apparently you.”

“Apparently.” She checked his identity, then his suit connection.

“You’ve got a leaking seal on this bag.”

“It’s just clothes.”

“Then they can go in a different bag.”

He almost told her that he had spent twenty years maintaining a settlement and knew what a leaking seal looked like. Instead he followed her to a table and repacked his shirts. A sleeve caught on the fastening. His hands had become awkward with exhaustion.

Inside the old bag lay a flat package that did not appear on his personal inventory. It was the wooden height measure from the kitchen doorway. His daughter’s aunt had made it from salvaged packing timber; each birthday, his daughter had stood against it while Jun marked her height.

The crew member looked at the package, then at him.

“Mass allowance?”

“I’m under.”

“Then list it.”

“Wooden object?”

“Whatever you want to call it.”

He entered HEIGHT RECORD. The words seemed ridiculous in the form. He left them.

Before he could seal the new bag, the coordinator called his name over the compartment speaker. The carrier had received a revised departure instruction. A cargo attachment had failed its final release check. The cargo was a package of power converters scheduled for a coastal growing district. It occupied the handling route needed to bring the last people through from the settlement.

Jun returned to the hatch. Dima was already there, still wearing his work harness.

“It’s the latch sensor,” Dima said. “The mechanical flag is in the right position.”

“Have you seen it?”

“On the camera.”

Jun looked at the image. A strip of reflective tape crossed the view.

It had been useful during alignment and was now hiding the part that mattered.

“So we haven’t seen it.”

The coordinator opened a repair request. The carrier captain joined from the flight deck. They had time to inspect, not time to improvise a new handling procedure. If the package could not be cleared, they would leave it attached to the settlement-side machinery and retrieve it remotely later, if later remained possible.

“The district needs those converters,” Dima said.

“The district also needs you alive to install them,” said the captain.

Jun recognized the pressure in the silence that followed. Twenty years of work had trained them to make things function. A carrier was waiting. People far away needed the equipment. The problem was small enough to invite a dangerous solution.

He called for the narrow inspection head. It was stored two sectors away, on a shelf that someone had already marked EMPTY.

“It was packed yesterday,” Orla said.

“Where?”

She searched the inventory. The location field gave the carrier. The carrier inventory gave the settlement. Both records bore completed checks.

Nobody spoke for a moment.

Then Orla laughed once, without amusement. “We have achieved perfect balance.”

Jun asked which person had done the physical move. A young rigger named Ben answered from the passenger compartment. He had put the inspection head in the crate with the flexible cable, because they were used together. The crate was aboard *Tern*, under a different equipment category.

They found it in six minutes.

Jun did not let anyone erase the mistaken records yet. He wanted the next evacuation crew to know how the error had survived the checks.

The inspection head showed the latch had only partly withdrawn.

A worn guide had allowed it to tilt. The flag indicated the actuator’s position, not the latch’s. Clearing it required unloading the mechanism and trying again. Doing that meant moving a different cargo package back into the settlement and accepting its loss if they ran out of time.

The package contained greenhouse lights.

Jun stared at the list. He had personally supervised their removal.

They were newer than much of the equipment waiting for them on Earth.

“Make the exchange,” the captain said.

Jun acknowledged.

Orla put the instruction on the local record, including the lights.

Dima worked the machinery. The converter package moved clear. The lights went back through the opening they had taken two days to leave. Nobody cheered when the handling route became available. They brought the last six people aboard. Ben was the final one through the hatch, carrying the inspection head in both hands.

The crew member counted forty-three passengers. Jun counted them too, with his eyes rather than his screen. He knew each face. He knew who was pretending not to cry, who had been waiting years to leave, who had no idea what work they would do next.

“Forty-three,” he said.

The hatch closed.

* * *

The last view of Rill was not through a window. The passenger compartment had none facing the right way. Someone offered a live camera feed, and most of the passengers watched it on the shared screen above the aisle.

Jun watched the temperature record from the repaired loop.

It was steady.

Beside him, Orla took off her boots and put her feet on her bag.

There was a dark bruise on one ankle. He had not noticed her limping.

“When did you do that?”

“Yesterday.”

“You should have told me.”

“I told Medical. You didn’t need another thing.”

He knew she was right and disliked it anyway.

The departure maneuver pressed them into their restraints. It was an ordinary, modest acceleration, entirely produced by an engine whose operating limits the crew knew. Jun found that reassuring. For months, everything people had said about motion had been too large to feel and too consequential to ignore. This motion had a beginning he could notice and an end someone announced.

When it was over, he opened a message to his daughter.

We’re aboard, he wrote.

Then he deleted it. She already had the passenger notification. He attached the photographs of the mural.

I never knew which cloud was yours.

The reply would take time. He put the screen away and tried to sleep.

He dreamed that he had left a tap running. In the dream, the tap was in a house on Earth where he had never lived, and the water was rising toward the height marks on the kitchen door.

When he woke, Orla was looking at him.

“Did we forget something?” she asked.

“Certainly.”

“Anything with a pulse?”

He looked down the aisle. Ben was asleep with his mouth open.

Dima was showing the crew member a photograph. The passenger count remained forty-three.

“No.”

“Then sleep.”

* * *

Elena received the closeout record with the message delay plainly marked. Jun had included the abandoned lights, the inventory error, Orla’s ankle injury, and the seven minutes spent photographing the wall.

She read it twice. The transport budget had once made those last forty-three seats look like an inefficient use of propellant. She remembered the committee member who had asked whether a smaller final crew could finish the work. Jun had answered by listing the jobs that required a second person, and then asking which of those jobs the committee wanted done alone.

She sent him a short acknowledgment. All passengers accounted for. The converters had a receiving location. The lights remained on the loss inventory. She did not thank him for going beyond his duty. His duty had been more than enough.

Then she opened the resettlement file.

Jun had a family address. Eleven passengers did not. Three had been assigned temporary rooms at a facility that was now being converted to medical use. The reassignment had not reached the transport office.

Elena called the housing coordinator. There was no astronomical reason the problem could not be solved, which made her less patient with it.

By the time Jun’s daughter replied, the three passengers had somewhere to go.

All of them, she wrote under the photographs. I painted all the clouds. I thought you knew.

Jun sat with the message for a long time before answering.

I should have asked properly.

He sent another photograph: the height measure, still wrapped, strapped beside his seat.

You’re not putting that up in my flat, she replied later.

For the first time that day, he laughed loudly enough to wake Orla.

When the final closeout call arrived, Elena described Rill as closed.

Jun objected to the word.

“We’re leaving the instruments running.”

“I’m closing the payroll location,” Elena said. “The instruments can argue with Finance themselves.”

After the call, she remained at her desk looking at a photograph of the station’s greenhouse. She had helped authorize its construction twenty years earlier.

Maya arrived with the agricultural readiness figures. She waited until Elena turned the photograph facedown.

“How bad?”

“Better than last month. Worse than the public thinks.”

“Then fix the public figure.”

They reviewed reserves by region. Several had exceeded their targets. Others lacked transport, storage, or power. The total looked adequate until the map showed where the food actually was.

“Ships,” Maya said. “We need more coastal distribution before the ports start sharing emergency power.”

“I can take steel from the orbital reserve.”

“How much?”

“Enough to make someone hate me.”

Maya signed the request.

That night, the first unambiguous images of an evacuated settlement’s changing orbit reached the public network. The news displayed a moving dot and an old photograph of people eating in its cafeteria.

The photograph did more damage.

Amara called her father. He asked whether the people had got out.

When she said yes, he asked whether they had somewhere to live.

She had to make another call before she could answer.

* * *

The journey inward gave Jun too much time to imagine his arrival.

In some versions, Min met him at the terminal and they knew immediately how to be a family in the same place. In others, she looked at him with the polite distance he had heard creeping into their calls. He had been a present father for much of her childhood and an increasingly remote one afterward, sending advice from a settlement whose problems had always seemed urgent enough to justify one more rotation.

He practiced saying that he should have come earlier. The sentence sounded like a request for her to deny it.

Orla told him to stop rehearsing an apology for an audience that had not yet arrived.

“You’ll only be annoyed when she doesn’t say her part.”

He looked at her. “You have children?”

“I have people. Same defect.”

They spent some of the journey preparing practical handovers for the equipment they had brought. The converters needed receiving inspections. Several packages contained undocumented local modifications. Jun could explain those while the memory was fresh. He worked beside Ben, who had found the inspection head, and made him lead part of the review.

Ben was surprised. “You know it better.”

“I know Rill better. You’re going to know where this goes next.”

The work helped. So did sleep. The carrier’s medical team insisted on rest periods, hydration, and exercise suitable for the journey. Jun found it easier to obey a clearly assigned routine than to decide for himself that he was allowed to stop.

* * *

The terrestrial arrival facility had a ceiling too high to feel economical. That was Jun’s first thought. His second was that the air smelled of more things than he could identify. Perfume, food, wet clothing, cleaning agents, people arriving from outside. Rill had smells, but they belonged to a system whose sources he usually knew. Here, the world entered through doors nobody treated as precious barriers.

Min was waiting beside a sign that said FAMILY RECEPTION.

She wore a yellow jacket he had never seen. He recognized her immediately and still needed a moment to reconcile her with the person on his screen.

She hugged him before either spoke.

“You’re heavier,” he said, then wished he had chosen anything else.

“It’s gravity, Dad.”

They laughed. The awkwardness became survivable.

At the housing desk, three members of the Rill group were sent to a different queue. Jun noticed before he had finished signing his own form. Their assigned facility had changed. The update existed, but the reception system had not linked it to their arrival group.

He began to intervene. Min touched his arm.

“There are staff here.”

“I know them.”

“Then introduce them to the staff. You don’t have to become the desk.”

He did. He stayed until a named person accepted the issue, then left with her. The act felt irresponsible for several minutes. Later, Orla messaged that everyone had reached a room.

At Min’s flat, he found the bed made in the living-room corner and a shelf cleared for his things. She showed him the bathroom, the kitchen, and the building’s shared laundry. He listened as though receiving a station orientation, which made her smile.

“There won’t be an exam.”

“There usually is.”

She put the kettle on. He unwrapped the height measure but did not ask where to hang it. She leaned it against the wall where both could see the marks.

“I should have come sooner,” he said.

She kept her eyes on the kettle. “Yes.”

He nodded. The sentence did not end the conversation. It allowed a different one to begin.

* * *

The first community meeting for the resettled crews was held in a hall that also hosted dance classes.

Jun could see tape marks on the floor beneath the rows of chairs. A counselor explained the support available. An employment coordinator described skills translation. Several attendees objected to the phrase, as though years of maintaining life-support equipment needed to be translated into something useful.

The coordinator accepted the objection and changed her wording.

Then she explained that terrestrial employers often did not recognize the titles used at outer settlements, and that applicants needed to describe the work in terms a hiring manager could understand.

Jun knew that was sensible. He still disliked reducing a place he had loved to a list of transferable capabilities.

During the break, he met a woman from another Saturn settlement who had been a school administrator. She said everyone asked about the view and nobody asked about the children she used to teach.

“What do you tell them?”

“That the view was good. Then I tell them about the children anyway.”

He told her about the mural. She asked to see it. For ten minutes, they stood beside a table of weak tea looking at clouds painted by children who now lived in several different places.

When the session resumed, Jun wrote a fuller description of his work history. It included training younger staff, planning safe shutdowns, and keeping maintenance knowledge usable. It did not include the view.

* * *

The first heavy rain came while he was walking home from an interview.

People hurried beneath awnings. Jun stopped on the pavement.

Water struck his face and ran down the back of his collar. He had seen rain before, long ago, but memory had stored the idea more neatly than the experience.

A stranger asked whether he was all right.

“Yes.”

The stranger looked unconvinced and offered the shelter of an umbrella. Jun accepted because standing there grinning was beginning to feel rude.

They walked together for one block. The stranger complained about the drainage, the traffic, and a delivery that had been late. Jun listened with pleasure so evident that the man asked what was funny. “Nothing. I’m glad to be here.”

The man glanced at him, then smiled. “Give it a week.”

At the flat, Min handed him a towel and told him he would ruin the floor. He dried his shoes carefully and tried to describe the rain. She listened, then showed him where to hang the wet jacket.

That evening, he sent Orla a photograph of the street outside. It contained no dramatic event, only water collecting in a pothole and someone trying not to step in it.

She replied: Looks badly maintained.

He wrote: It is. There’s work.

CHAPTER FOURTEEN

Falling

FINAL SCHEDULED TRANSMISSION, SATURN SYSTEM RELAY 4

STATUS: NOMINAL

The relay had been designed to say nothing else. Its operators, who had watched its orbit open for eleven weeks, sent back one word of their own: THANKS.

ORBIT ODDS: 66% · T–118 DAYS

In the months before the refrigerator woke her, Maya had developed the habit of leaving shoes beside the bed with their openings facing her feet. Leila had once teased her for kicking them off in opposite directions. Now she placed them carefully, as if readiness could be reduced to a shape on the floor.

Leila's messages arrived in batches. The distance was growing, but delay was not the only reason. Work happened at different hours aboard *Ithaca*. Antenna time was shared. Sometimes one of them simply did not know what to say.

In the first weeks, Maya had answered everything immediately. A photograph of a leaf received a page about her day. A question about a utility bill became an account of three meetings. She recognized, eventually, that she was trying to make the conversation continuous by supplying both halves of it.

Leila recognized it sooner.

You can leave a gap, she wrote. It doesn't mean I've gone anywhere beyond the obvious.

Maya had laughed, then cried, then put the screen down without answering for an hour.

Now she sent things that had happened. A neighbor had returned a borrowed pan with a different lid. The shelf remained straight. She had burned rice while watching an operations briefing that contained no information she needed.

Leila sent a recording from the growing compartment. Fans hummed. Water moved through a pipe. Someone in the distance was singing badly and refusing requests to stop. For forty seconds, nothing exceptional occurred.

Maya played it while making dinner.

* * *

The first public rehearsal for a difficult food season was held in a school gymnasium, which made nearly everyone behave as though they were waiting to be dismissed.

Maya had agreed to observe. By the time she arrived, an organizer had put her name on the opening presentation and placed a chair for her on the stage.

She moved the chair down to the floor.

“People came to ask questions,” she said.

“They can after the presentation.”

The presentation lasted thirty-seven minutes. It showed reserve locations, distribution intervals, household storage guidance, and the reassuring average capacity of the regional network. Maya watched a woman in the third row hold her hand up, lower it, and stop listening.

When the questions began, the woman asked where she was supposed to keep a month’s food in a room she shared with four other people.

The organizer referred to community storage.

“Which one?”

He put the map back on the screen.

“That’s a clinic now,” she said, pointing.

Maya opened the underlying record. The map had been correct six weeks earlier. The site had been reassigned, and the replacement location was still under negotiation.

“You’re right,” Maya said. “That location isn’t available.”

The room changed. Several people began talking at once. The organizer leaned toward her and whispered that they needed to avoid undermining confidence.

She whispered back, “Then we need somewhere to put her food.”

They spent the next hour marking the map. A small shopkeeper offered part of a storeroom but wanted to know who would cover losses if the power failed. A teacher said the school kitchen could handle more meals if deliveries came outside the hour when children arrived. Two men disagreed about whether a footbridge was passable with carts. The woman who had asked the first question offered to walk there with someone who could make the decision.

Maya went with her.

The bridge existed. Its central section had been repaired with steps, making it useful for pedestrians and nearly useless for the proposed distribution carts. On the other side, a route around a warehouse added forty minutes to the walk.

“What’s your name?” Maya asked.

“Njeri.”

“Would you be willing to help the local team check the revised route?”

Njeri gave her a tired look. “Is that a job or a favor?”

Maya felt the correction land.

“It should be a job.”

“Then tell them that first.”

The route was changed. Njeri was hired for six weeks of local verification. At the next rehearsal, the organizer shortened the presentation to eight minutes and left most of the meeting for questions. Not all of the answers were ready. The questions were better.

Maya put the first map in the training archive beside the corrected one. She did not label it a failure of communication. The information had been communicated clearly. It had been wrong.

* * *

At the lunar factory, Saira was preparing a different rehearsal.

One of the observatory relays had developed a pointing fault. It was not an emergency yet. That phrase had begun to appear on too many work orders. The relay could hold a useful direction for minutes at a time, then drift beyond the tolerance needed for its assigned measurements. A replacement unit was available. The crew that would install it had not worked together before.

Pavel set up a mock assembly in the maintenance bay. He deliberately placed one fastening where the camera could not see it.

The first operator completed the sequence on the screen. The mock unit remained attached.

“The model says released,” the operator said.

“Then ask the model to carry it away.”

Pavel waited until the operator found the hidden fastening. He did not smile. He had been told that his exercises were becoming unpopular.

Saira watched from the doorway. “Run it again with the camera from the actual vehicle.”

The actual camera had a narrower view and worse depth cues. The second run took twice as long.

“Better to learn here,” Pavel said.

“Yes. Also better to finish before the relay fails.”

They looked at the schedule together. The repair vehicle would leave in two days. Another delay would consume the reserve window intended for a different observatory. Saira cut a nonessential equipment demonstration from the crew’s program and gave them the time. A visiting delegation complained. She offered them a recording.

Maya watched the final rehearsal remotely. She was there because a long gap in the relay’s data would widen the crop forecasts, not because she had authority over the repair.

“What’s your acceptable gap?” Saira asked.

“We can use the other baselines.”

“That wasn’t the question.”

Maya called the astrometry team into the discussion. They gave a range, explained the assumptions, and identified the period when a second outage would be especially costly.

Pavel marked it on the work plan.

“That’s useful,” he said. “Now we know what we’re protecting.”

The repair did not proceed exactly like the rehearsal. A thermal cover had stiffened and would not fold into the space shown on the drawing. The crew stopped before forcing it. Pavel, watching the delayed camera feed appropriate to the vehicle’s location, asked for a wider view and waited for the next image before suggesting a different grip.

There was no conversation quick enough to make him feel as though his own hand were at the tool. The crew needed a plan they could execute without him correcting every movement.

He sent two options with the conditions that distinguished them.

The local operator chose the second. The cover came clear.

The relay returned to service late but within the tolerable gap.

Maya sent Saira the first useful reduction. It looked like a table of small numbers.

Saira forwarded it to the crew with a note: This is why.

* * *

Amara’s father began keeping a radio on in the kitchen, tuned to a station that repeated the encounter forecast every hour. He claimed it was for the music between bulletins. Maya had never heard any music on it.

She visited on a Sunday carrying groceries and a replacement garden hinge. He had already repaired the hinge with a strip of metal and two mismatched screws.

“You said it was broken.”

“It was.”

“You could have told me you’d fixed it.”

“You could have come without a reason.”

She put the hinge on the table. He put the kettle on. Neither apologized, and after a while neither needed to.

The hourly bulletin began. He reached to turn it up, then stopped when he saw her looking.

“Does it help?” she asked.

“Sometimes I miss something.”

“They repeat the same update.”

“Then I know I haven’t missed it.”

She sat beside him while the announcer described an orbital range that had barely changed since the morning. Her grandfather listened to the end. Only then did he turn the volume down.

“I know it’s irritating,” he said.

“I check my phone when the refrigerator stops.”

He looked relieved in a way she had not expected.

They went into the garden. Under the roof, the air smelled of damp soil and a cracked tomato. He asked about Leila. Maya told him about the singing in the growing compartment, and played the recording.

He listened with the same attention he had given the bulletin.

“She sounds busy.”

“She isn’t the one singing.”

“I meant the place.”

Maya played it again. She could hear what he meant: a room occupied by people who had tasks, irritating habits, and enough confidence in the next hour to waste part of it on a song.

She sent Leila a recording of the garden gutter dripping into its barrel. Her grandfather added, “Your ventilation adjustment is still working.”

Then he asked Maya to help move a heavy pot, and for the next ten minutes neither of them checked the news.

The false alarm began with a photograph of an ordinary sunset.

Someone had increased the contrast, circled a bright patch in a cloud, and claimed that the black hole had begun stripping gas from the Sun. The image spread with explanations more confident than any forecast Maya had signed. By the time the public information desk responded, several versions included invented timestamps and quotations attributed to Amara.

Maya saw it because Ruth sent it with a question mark.

She answered directly: No, that is a cloud and an edited image.

Then she called her mother to warn her about the quotation.

Amara had already seen it. She was preparing a response that explained why the Sun’s own gravity held its gas much more tightly than it held distant planets, and why the encounter threatened orbital architecture without requiring the solar surface to tear apart.

“Will people believe it?” Maya asked.

“Some will.”

“You sound resigned.”

“I’m trying not to confuse reaching everyone with telling the truth clearly.”

The correction helped. It did not erase the image. A local school invited an astronomer to answer questions, and Maya went with Salma because the school was close to a distribution site she was visiting. She did not pretend to be the astronomer. She listened while a young researcher used safe images and simple comparisons to explain the difference between an object falling as a whole and the forces that distorted the relationships among its parts.

A child asked whether the grown-ups had lied before.

The researcher paused. “Sometimes grown-ups get things wrong. Sometimes they don’t tell everything they should. That’s why you should be able to see how a claim was checked.”

Maya sat very still.

The child asked whether that meant no one could be trusted. The researcher said it meant trust should include questions, especially when the answer mattered.

Afterward, Salma asked Maya whether the response had upset her.

“A little.”

“Was it wrong?”

“No.”

They helped stack the chairs. Salma said Tomas had once used the same room for his visit. Maya remembered the photographs. Neither made the memory carry a conclusion.

* * *

The next real warning concerned a pump.

Njeri called from a community storage site where cooling power had become unreliable. The backup supply started correctly, but its fuel record did not match the physical tank. The discrepancy was small enough to be explained by testing and large enough to shorten the planned endurance.

Maya asked whether they could transfer the most vulnerable food.

Njeri had already arranged that. She wanted to know who could authorize replenishment without waiting for the next scheduled delivery.

The regional team now had that authority. The contact list still directed callers to the old central desk.

Maya connected them, then updated the list and asked which printed copies remained in use. Several sites had laminated the old version because it was intended to survive outages. The durability of the wrong instruction had become part of the problem.

They sent replacement sheets. Njeri asked for the revision date to be large enough to read without searching the bottom corner. Maya agreed.

At the site, volunteers moved food into temporary storage while a technician checked the supply. One man arrived carrying a bag of personal groceries because he had heard there was a shortage. The volunteers explained that they were redistributing stored food, not collecting donations. He looked embarrassed. Njeri thanked him and asked whether he could help carry empty containers instead.

He stayed two hours.

Maya watched the local team work through a problem that would never appear in a history of the black hole. It mattered more to the people served by that store than the latest elegant orbit graphic. She added the incident to the practical briefing, where it joined several other small failures that had been corrected before becoming larger ones.

* * *

At home, the empty side of the bed became less startling.

Maya noticed this with guilt. For months, she had reached toward Leila in the first moments after waking. Now she woke knowing the room. Adaptation felt like a betrayal until she imagined asking Leila to remain startled by her absence forever.

She wrote about it in a message, then left the message unsent for a day. When she sent it, she included the sentence about guilt because otherwise the account would sound colder than she felt.

Leila replied that she had stopped expecting the kitchen window to show the street. She missed the street anyway. Her cabin had become familiar, and she sometimes forgot which sounds aboard had once kept her awake.

I don’t think learning the room is the same as forgetting who lived in the other one, she wrote.

Maya read that sentence while replacing the cupboard latch correctly. She kept the wrong bracket in a small box of miscellaneous hardware because it might be useful for something else. Leila would have laughed at the justification.

That evening, Maya invited a colleague over for food and warned her that the menu would be simple. They talked about work for fifteen minutes, then about the colleague’s brother, who had decided to learn an instrument despite being bad enough to disturb the neighbors.

Maya thought of the trumpet near her grandfather’s house. The sound had annoyed people for months and had become precious in memory. She wondered whether the neighbors would appreciate that argument. Probably not.

When her guest left, Maya recorded a short message about the evening. She did not ask Leila to reassure her that having enjoyed it was allowed.

* * *

At three in the morning, Maya woke because the refrigerator stopped.

For several seconds she lay still, listening to the sudden absence of its small vibration. Then the building’s backup supply engaged, and the compressor started again.

Nothing astronomical had caused it. A maintenance crew was replacing a local switch.

She checked the encounter feed anyway.

The Sun was accelerating toward the black hole. Earth, the Moon, the air, and the ocean accelerated with it. People did not feel the common fall. What concerned the scientists was the small difference between one part of the system and another, accumulating as the geometry changed.

The black hole would remain hundreds of AU away. It would not tear the Sun open. The Sun’s surface gravity held its gas far more tightly than the Sun held a planet one AU away.

There would be no reassuring implication to draw from an ordinary sunrise.

Maya dressed and went to operations.

The room used two clocks. One showed the time at the receiver.

The other accompanied each observation and showed when the measured event had occurred. Predictions of the current state appeared in a different color.

Amara had ordered the distinction after an early display made a delayed measurement look instantaneous. Luis had suggested painting WE DO NOT KNOW YET across the wall. They had settled for the clocks.

Earth’s orbit was changing within the modeled range. That statement concealed an uncomfortable variety of futures.

For now, the atmospheric forecasts could still use actual sunlight.

Planning the next planting cycle required estimating sunlight months ahead, after more of the encounter had occurred. A small change in the orbital solution could determine which region planted which crop.

Maya’s work had become a chain of decisions that expired.

At 06:20, the South Asian coordination desk asked whether to release reserve seed. At 06:31, an energy group requested another day before changing its seasonal allocation. Waiting would preserve flexibility in one system and remove it from another.

She approved the seed release with a regional modification, then sent the basis for the decision to the people who would have to live with it.

At noon she found Amara outside, sitting on a low wall with a sandwich she had not unwrapped.

“Have you eaten?” Maya asked.

Her mother looked at the sandwich as if someone else had placed it there.

“Apparently I’m about to.”

They watched traffic move along the road. A cyclist stopped to help a driver secure a loose load. Above them, the Moon occupied its usual daylight place, though its predicted future no longer fit comfortably inside old tables.

“I keep expecting to feel it,” Maya said.

“You won’t feel our shared acceleration.”

“I know the explanation.”

Amara unwrapped the sandwich.

“So do I. I still look at my coffee.”

Maya laughed. It was the first time that day she had noticed her shoulders relax.

Then her phone sounded. The newest food forecast had crossed another threshold.

CHAPTER FIFTEEN

The Corridor

DAEDALUS PUBLIC INFORMATION, INTERNAL MEMO

Please avoid the word “corridor” in public messaging. Members of the public have begun asking where it leads.

ORBIT ODDS: 54% · T–36 DAYS

They still called it the corridor, though it was not a place in space.

On Maya’s display, it was the overlap between tolerable orbits and food systems that could survive them. Move an assumption about rainfall, power, or spare parts, and its edges moved. A planet could remain comfortably bound to the Sun and fall outside it.

That distinction had become the center of her work.

The calibration package arrived on a Thursday, eleven days before it changed the food forecast.

It was one of seventeen routine updates accepted that morning. A technician named Esi reviewed its test report while eating breakfast from a container whose lid no longer fit. She had worked the previous evening because a colleague’s child was ill. Nothing in the package appeared urgent or unusual.

The update improved the agreement among the central instruments. Its tests covered the range of reference-frame changes expected during the encounter. The package also contained an adapter for older observation formats, which was where the mistake lived.

The adapter assumed that one timing adjustment had not yet been made. In the central processing path, it had. Both components behaved as their own documentation said they should. The combined path applied the adjustment twice.

Esi did not see that. Neither did the next reviewer. The end-to-end test used a synthetic packet prepared by the same group that had written the adapter. It therefore carried the same assumption.

The first warning came from observations the network already regarded as less precise.

The arks had different instruments, less frequent observing opportunities, and vantage points that made them valuable precisely because they were inconvenient. Their measurements did not fit the improved central solution as well as before. The automated quality report lowered their contribution and opened a low-priority discrepancy item.

A technician aboard *Ithaca* received the discrepancy list. He asked Leila to move a lighting cycle because reflected light from a growing compartment was interfering with a scheduled exposure.

“How much time?” she asked.

“Eighteen minutes.”

“The plants can stand eighteen minutes. The people harvesting them need to know which eighteen.”

They changed the schedule. A crew member complained that astronomy had once again arrived at dinner time. Leila told him to complain to the stars and passed him a tray.

The resulting exposure went into a packet sent to Earth. No one aboard knew that it would become part of a decision about food on a planet they were leaving behind.

* * *

At the receiving desk, Luis opened the discrepancy item because someone had labeled it POSSIBLE PLATFORM MOTION and he disliked the word possible when it was being used to avoid assigning work.

He checked one of the raw exposures. The star images were slightly elongated, as the instrument team had already reported. The elongation did not explain the direction of the residual.

He requested a second reduction.

The request joined a queue behind an urgent Mars navigation assessment. That priority was reasonable. Two days passed.

When he returned to the item, a colleague had compared another ark and found a similar disagreement. They began looking for a shared problem in the ark reductions. It was also a reasonable hypothesis. The arks shared some software and had received the same guidance on correcting for their motion.

Neither scientist yet had a reason to suspect the central network’s timing path.

Luis recorded both possibilities and went home. He had promised his partner that he would cook dinner without taking a call from the kitchen. Halfway through chopping onions, he remembered a test that might distinguish the hypotheses. He wrote it on the back of an envelope and left it beside the door.

The next morning, the envelope was still there. He took it to work, opened the analysis, and was interrupted by Maya’s call about an unrelated reserve estimate.

Later, when people asked why nobody had noticed the error earlier, this was the part he found hardest to explain. They had noticed something. They were doing the work. A warning did not arrive labeled with its eventual importance.

* * *

In Recife, Inês was planning a planting sequence around a delivery of replacement film that had already been delayed once. The last delay had made her cautious about every date printed in green.

The new allocation system showed two possible crop plans. One used the nursery’s remaining stock and required more power later. The other needed an early delivery of different seed and allowed lower lighting demand during the expected shortage. She had prepared for both, at the cost of leaving a portion of the nursery unused.

“How long can you keep both options?” Celso asked.

“Until Tuesday.”

“The forecast update is Monday.”

“If it arrives.”

He did not reassure her. He showed her the freight cutoffs and the coastal carrier’s loading sequence. The relevant departure was part of a chain: a carrier collected several ports, then transferred cargo to smaller vessels and road depots. Missing one departure did not simply mean waiting for the next ship. It could mean losing the connection that reached a growing district before its preparation window closed.

Inês pointed to one district on the map.

“That one can’t wait.”

“I know.”

“Put it in the instruction. Not just in your head.”

Celso added the local deadline. The system accepted it as an exception requiring confirmation. He called the receiving coordinator, who confirmed it and asked why she was confirming information she had supplied three times already.

“Because this time it will sit beside the dispatch decision,” he said. “We’ll see.”

He left her words in his private notes. He had learned not to mistake an accepted form for restored trust.

* * *

On the morning the severe forecast arrived, Maya was supposed to attend a review of her own team’s workload. She had prepared a case for removing two functions from the central desk

and giving them to regional teams. Her argument was that the center had become a place where ordinary decisions waited for famous people to notice them.

The severe forecast canceled the meeting.

She did not immediately see the irony. She was looking at the projected deficit and the list of systems the emergency plan would reduce. Clinics had minimum protected allocations, but minimum did not mean comfortable. Water networks could postpone some pumping, but not indefinitely. Small growers would lose power precisely when the new plan asked the largest facilities to produce more.

There were reasons for those choices. The largest facilities had measured yields and established distribution. Their output was easier to count. That did not mean the uncounted output was expendable.

Maya asked the duty analyst to show the losses alongside the gains.

“They’re in the impact attachment.”

“Put them on the main display.”

The display became less reassuring. Elena arrived while they were making the change.

“What’s new?” she asked.

“The forecast,” Maya said. “Not all of the assumptions.”

They began checking what had moved.

Thirty-six days before closest approach, the corridor narrowed abruptly.

Pilot forecast a severe reduction in sunlight during the coming agricultural season. Northern storage capacity was already committed.

The emergency response called for changing crop allocations across several regions and transferring power to the largest controlled farms.

The same response would reduce supply to clinics, water pumps, and smaller growers. It was intended to prevent a larger disaster, but it would cause damage of its own.

Elena asked whether the change was supported by new observations.

“Three orbital solutions agree,” the duty analyst said.

Maya opened their source lists.

The three solutions used the same calibration update.

She remembered Marcus asking who had opened the warehouse.

“Hold the allocation.”

The analyst looked at her. “We lose the freight window.”

“How long?”

“Eleven minutes until the first dispatch becomes irreversible without breaking the next schedule.”

Maya called the astrometry desk. The update accounted for their changing reference frame during the encounter. Its verification tests had passed. Its fit to the central network looked better than the previous version.

“What does it do to the independent observations?” she asked.

“They have larger residuals.”

“Which ones?”

The ark network.

The receiver had assigned them less weight because they disagreed.

Maya pulled the raw packets. They were old measurements, made from distant vessels and received after a substantial delay. They could not tell her what was happening this second. They could tell her whether yesterday’s reconstructed positions were consistent across different viewpoints.

They were not.

“Luis,” she said, sending the comparison. “Look at the time conversion.”

He was already in the room. He bent over the display without sitting.

“This correction is applied upstream.”

“And here?”

“Again.”

The error was small enough to resemble an observational nuisance.

Once passed through the forecast, it changed a decision affecting millions of people.

Maya asked for a rerun without the duplicated correction.

Completion estimate: sixteen minutes.

She had seven.

Elena stood beside her. “I need a dispatch instruction.”

Maya could release the emergency plan and hope its harms were justified. She could hold everything and lose transport capacity they might need. Or she could authorize only the shipments useful under both forecasts, accepting that the rest would miss this departure.

The third option wasted less than a wrong full response. It also risked leaving them short if the more severe forecast proved correct.

Amara entered halfway through the explanation.

Maya expected her to take over. Instead she asked to see the independent comparison.

“Does it support the old solution?”

“More closely. It doesn’t establish the new one.”

“Then say exactly that in your decision record.”

“Will you sign it?”

“I will sign the measurement assessment. The allocation is yours.”

Maya heard no evasion in it. Her mother was leaving her the responsibility she had trained to hold.

She authorized the common shipments, held the rest, and told each region why. Four departures slipped. One delegation demanded her removal before the recalculation finished.

The corrected forecast arrived nine minutes later.

Earth was still inside the narrower, workable range.

Maya leaned back. Her hands began shaking after she no longer needed them to be steady.

Elena read the result, then called the freight desk to recover what could be recovered.

Luis began documenting the error before anyone could describe it as a miracle.

Amara placed a glass of water beside her daughter.

“The arks,” Maya said.

“Yes.”

People who had chosen a different future had helped preserve this one.

Maya sent Anika the full account. The reply would not arrive in time to comfort her that afternoon. She sent it anyway.

* * *

The corrected forecast did not put the missed ships back into the ports. Maya remained at the desk while the freight team worked through the damage. One held shipment could travel on a later carrier with no important loss. Another needed a different connection and additional fuel. The third contained equipment that would arrive too late for the original crop plan.

The fourth was for Inês’s district.

Celso joined the call from the loading area. Behind him, workers were securing the containers that had been released. He had followed the instruction exactly. Maya could tell from his expression that he was afraid she would ask why he had not found a way around it.

“You did what we authorized,” she said before he could begin.

“Then we need a new authorization.”

They considered moving the held cargo by a faster route. The vehicle existed. Its capacity did not. Splitting the shipment would deliver enough seed to start part of the planting, but the power equipment would remain behind. The smaller crop would require a local lighting adjustment that had already been rejected as too expensive under the earlier budget.

Inês joined, listened to the options, and asked how much of the nursery they expected her to risk.

Maya gave the number.

“That’s not a number of trays,” Inês said. “That’s a fraction on your screen.”

Maya converted it. Inês walked to the nursery aisle and turned the camera so they could see it.

“From this support to the far wall,” she said. “That’s your fraction.”

Maya looked at the empty spaces prepared for plants. A worker was rinsing a tray at a sink. He paused when he realized he was in the image, then continued.

“Can you use the alternative crop in the rest?”

“Some. It won’t produce the same food, and the timing is worse. We can do it if the receiving kitchens agree to change their plans. They need to agree today.”

Maya asked who should be on the call. Inês gave three names.

The conversation took another hour. It produced a smaller loss than the first estimate, not a clean recovery. A school meal program would use more reserve stock. Another district would receive less of a crop it had been expecting. The additional power came from a maintenance deferral that had a cost of its own.

When the plan was settled, Inês asked Maya whether the severe forecast had been wrong.

“Yes.”

“And holding the shipment?”

Maya took a breath. “Given what we knew, I still think it was the least harmful choice available. It harmed you.”

Inês nodded once. “Put both in the report.”

* * *

Esi expected to be suspended.

She arrived at the incident review with the calibration test report, her acceptance note, and the sequence of messages that had preceded it.

She had slept for two hours and had spent most of those dreaming that a test result kept changing after she read it.

Luis asked her to begin with the path of one observation through the system.

She did. The whiteboard filled with boxes. The duplicated adjustment appeared at the boundary between two of them.

“Why didn’t the integration test catch it?” Maya asked.

Esi showed the synthetic packet.

“It was generated before the upstream correction,” she said. “The real packet wasn’t.”

“Who checked that the test packet matched the live path?”

“We checked the format.”

“I’m asking about the meaning.”

Esi looked at the report. Nobody had been assigned that check. She had assumed the upstream team owned it. Their record assumed the receiving team did.

Luis drew a line between the boxes and wrote NO OWNER.

The review continued. They found that the ark discrepancy had been correctly reported but treated as lower priority because it affected a smaller amount of data. They found that several apparently independent orbital solutions shared the same corrected input. They found that the agricultural forecast gave a sharp operational threshold to a change that should have been displayed with greater sensitivity to the calibration uncertainty.

No single finding erased Esi's responsibility for accepting the package. None made her the whole cause.

At the end, she asked whether she should clear her desk.

Maya looked up from the action list.

"No. You should go home. Someone else is covering the next shift."

"And after that?"

"After that, we need you to help build a test that uses a real observation from end to end. Another team will try to break it."

Esi nodded, then remained seated.

"People lost food."

"Yes," Maya said.

There was no humane reason to pretend otherwise. Luis handed her the printed contact for the staff counselor without making a speech about it. She folded it into the test report.

* * *

Amara found Maya in a small room off the operations floor, holding the glass of water without drinking it.

"The account is public," Maya said.

"I saw."

"Someone has already called it a triumph of human judgment."

"Someone will call it a failure of automation. Then someone will call you reckless. They have the same packet."

Maya put the glass down. "Did you ever get used to it?"

"I got used to anticipating it. That was part of the problem."

Her daughter looked at her.

"I started hearing the reaction before I made the decision," Amara said. "Sometimes before I had finished understanding the evidence."

“Is that what happened with the report?”

“Among other things.”

For a moment Maya wanted the familiar argument, because anger was easier than the uncertainty she felt. Instead she asked whether her mother thought the partial release had been right.

“I think you made the basis visible and chose a loss you could limit. I can’t tell you what would have happened in a world where you made the other choice.”

“You used to sound more certain.”

“I know.”

They sat together until someone knocked to ask for Maya. She drank the water before standing.

The reply from *Ithaca* arrived later. Anika had acknowledged the technical account and scheduled independent checks of their own processing path. Leila’s message was separate.

I moved a lighting cycle for that exposure. You owe the harvest crew eighteen minutes of dinner.

Under it was a second paragraph.

I read the losses too. I’m glad you sent the whole thing. Please sleep before you answer.

Maya wrote, I will, and this time she did.

* * *

The request to remove Maya arrived in a form that required an answer by the next council session.

It came from the delegation whose freight connection had been lost. The document did not accuse her of malice. It argued that she had exceeded the acceptable discretion of an allocation coordinator by holding shipments against a forecast produced by the approved system.

Maya read it after sleeping. The first time, she felt only anger. The second time, she could see the institutional question beneath it. If every coordinator could reject the approved forecast on personal suspicion, the network would become impossible to operate. If none could act on evidence of a shared error, approval would become a mechanism for spreading failure.

She sent the decision record and asked the review panel to judge the authority she actually held, not the result alone.

Elena offered to defend the choice publicly. Maya asked her to defend the record, including the losses. Elena said she was becoming repetitive. Maya said the losses were still there.

At the session, the delegation’s representative described the disrupted nursery, the changed meals, and the additional cost. Maya did not interrupt. When asked whether she regretted the

hold, she said she regretted the harm and believed the partial release remained defensible on the evidence available.

“That sounds like refusing responsibility.”

“Responsibility includes explaining the alternative losses. It doesn’t let me pretend I could choose a harmless option.”

The panel examined the authority rule, the timing, and the shared calibration. It found that she had acted within the emergency limits, but that the freight window had been presented to her too late and without enough prepared options for partial release. The process had required improvisation where prior planning could have supplied a better choice.

The finding did not remove her. It required changes from her team.

Maya accepted them. The delegation maintained its criticism and participated in designing the revised freight options. Cooperation did not require a retraction.

* * *

The next calibration review included an observer from the agricultural team.

Esi initially found this irritating. She had spent years learning the instrument pipeline and now had to explain it to someone who asked basic questions. The observer, a quiet analyst named Jonah, wanted to know which changes could move a crop decision across a threshold even when they improved the average fit.

That was not a basic question after all.

They built a set of examples linking measurement changes to downstream choices. Some updates barely affected operations. Others changed timing or regional allocations sharply. The review did not make calibration decisions subordinate to a food planner’s preferences.

It made the consequences visible enough to require the appropriate checks.

Esi also changed the end-to-end test. It used real archived observations with independently established processing histories.

Another team supplied deliberately altered packets. The first version of the test missed one alteration. She documented the miss and revised it before release.

When Maya visited, Esi showed her that failure first.

“You don’t have to perform contrition every time I come in,” Maya said.

“I’m showing you what the test can do.”

Maya paused. “You’re right.”

They worked through the result. Esi looked less exhausted than at the incident review. She had taken two days away from the center and returned annoyed by an ordinary disagreement over file naming. Maya was glad to hear about it.

* * *

Inês invited Maya to join the revised planting briefing remotely.

The camera was set at the end of a long table. Workers came and went, carrying trays, notes, and cups. The schedule on the wall had been rewritten in several colors. Maya recognized the consequences of her decision in lines that would mean little to someone who had not followed the freight chain.

Inês explained the changed crop mix and asked each team to identify the point at which the plan would need another decision. She did not ask them to promise completion regardless of conditions. One supervisor named a staffing problem. Another identified a preparation step that could begin earlier if an inspection happened that afternoon.

Maya connected the inspection office while the meeting continued.

The inspector had capacity but had not received the revised request.

The office sent someone that day.

Afterward, Inês asked whether Maya had learned anything useful.

“That our partial-release options should exist before the forecast changes.”

“Yes. Also that we can help design them.”

Maya nodded. The central team had treated local knowledge as information to collect rather than authority to include. The difference was becoming harder for her to miss.

She asked Inês to nominate someone for the design group and confirmed that the work would be paid. Inês nominated a supervisor who had disagreed with Maya twice during the call.

“Good,” Maya said.

“We’ll see if you still think so next month.”

* * *

Leila received Maya’s personal account after the technical exchange had already made her a minor hero aboard *Ithaca*.

The harvest crew joked about the eighteen minutes. The instrument team reminded them that the exposure had been one among many, and that the independent viewpoint mattered more than any single person’s favor. Leila agreed and still enjoyed the joke.

Maya’s message described the hearing, the nursery, and the way her hands had begun shaking after the dispatch. Leila listened in her cabin. She could not make tea for Maya or take the phone from her at bedtime. She could answer without asking Maya to turn the episode into proof that staying on Earth had been the right choice.

She recorded a reply about fear arriving after action, about the assessment exercise where she had stopped listening to the mechanic, and about the difficulty of accepting praise that left out what went wrong.

At the end she said, “I love you. That isn’t an evaluation of the dispatch.”

Maya laughed when the reply arrived. Then she played the last sentence again, because it was something she needed to learn how to believe.

CHAPTER SIXTEEN

Periapsis

DAEDALUS STYLE GUIDE, REVISION 41

“The Encounter” is retired. Preferred: “the Transit.”

ORBIT ODDS: TRADING SUSPENDED · T–0

Closest approach arrived without an explosion.

On March 19, 2126, the Sun passed approximately three hundred and ninety-four AU from the black hole. The exact instant existed first as a prediction. Confirmation would come through measurements made at different times and places, then reconciled by people who were tired enough to make mistakes.

The week before closest approach, Amara’s father finally agreed to spend the encounter week near the operations center after a neighbor told him she could manage the garden without his supervision.

He gave her so many instructions that she took the notebook from him and asked which ones were already written down. Most were. She told him to stop repeating those and explain only what had changed.

Maya watched with professional admiration.

The packing took an afternoon. He brought clothes, medicines, the flask, and food that Maya said would be available where they were going. He said availability was not the same as quality. She accepted the bag.

Before leaving, he walked slowly through the garden. He did not make a speech. He checked a fastening, moved a pot into a better position, and asked Maya to look at the gutter after the next rain.

“The neighbor can do it.”

“I know.”

She understood then that he was not doubting the arrangement. He was making plans for coming back.

She said she would check the gutter.

* * *

The day before the predicted periapsis, a man came to the visitors’ desk carrying a telescope in a padded case.

He wanted to know where to point it.

The receptionist directed him to the public observing program, but he shook his head. He did not want an ordinary view of the Sun through safe equipment or an explanation on a screen. He wanted to see the thing that had occupied seventeen years of his life.

Amara was passing through the lobby and heard the last sentence.

She stopped. “There isn’t a visible edge to point at.”

“I know it’s black.”

“I mean the event. The closest approach. We reconstruct it from measurements. You won’t see a moment when everything changes.”

He looked at the case. “My wife bought this when they announced it. She thought we’d watch together.”

Amara waited.

“She died last year.”

There were things she could explain about starlight and angular separation. None would answer him.

“There isn’t an edge,” she said after a moment. “But there is something. The public observatory is open tonight. Show them this.”

She wrote a position in the sky on the back of a visitor slip and signed it, which was not procedure.

“Will you be there?”

“No. I’ll be working.”

He nodded as if that, at least, was an answer he understood.

After he left, Amara stood beside the reception desk for a moment.

The receptionist asked whether she needed anything.

“No. Thank you.”

She went upstairs to the measurements that could tell her what the telescope could not.

* * *

Jun watched the encounter coverage from his daughter’s flat.

He had been on Earth long enough to stop reaching for handholds that were not there. Stairs still made his legs ache. The first rainstorm he had walked through had left him grinning so foolishly that a stranger had asked whether he needed help.

His daughter, Min, had cleared a corner of the living room for his bed. She had also cleared a shelf for the wrapped height measure, which he had not asked her to do. Neither mentioned the concession.

That morning she was trying to leave for work. Jun kept telling her which update would be next.

“I can watch at lunch,” she said.

“This is the day.”

“It’s also the day the lift in Building Four gets inspected. People live there.”

He had no argument against that. It was the sort of thing he might once have said to her.

At the door, she turned back. “Will you be all right?”

“Yes.”

“There’s food in the blue container. The other one is for tomorrow.”

“I can read labels.”

“You ate tomorrow yesterday.”

After she left, he turned down the coverage and opened the Rill instrument feed. The station was still reporting. The repaired thermal loop had held. One camera no longer responded, and the power reserve was declining along a curve he had helped calculate.

He sent Orla a note. Still here.

She replied with a photograph of the terrestrial workshop where she now worked. A machine twice the age of anything at Rill had been dismantled across the floor.

So is this. Unfortunately.

Jun laughed. Then he closed the feed and opened the employment application he had been avoiding. The title of the available job sounded smaller than his old one. Its list of responsibilities looked familiar.

He began with the section asking what he knew how to do.

* * *

At the operations center, Maya’s grandfather had brought enough tea for three people and enough food for six. Security had inspected the containers with a seriousness that offended him until a guard asked what was in the smallest one.

“Something you can have if you bring the spoon back.”

The guard brought it back washed.

Maya found him in the visitors’ room telling another family’s child that no, they were not going to fall off Earth. The child asked why everyone was worried, then.

He looked at Maya.

She sat on the floor beside the child’s chair. “The whole Earth is moving with us. The worry is how its path changes around the Sun, and how we keep things working while that happens.”

“Like the bus when it turns?”

“Not quite. You feel the bus push you around. Here, everything falls together, almost. The differences are what matter.”

The child considered this. “Will school still happen?”

“Probably.”

“Then I don’t like it.”

Her grandfather made a solemn sound of agreement. Maya stayed long enough to eat one of the things he had brought. She could hear phones sounding around the room, each announcing an update slightly differently.

One said CLOSEST APPROACH NOW.

Another said SCIENTISTS AWAIT CONFIRMATION.

Both messages referred to the same forecast. A man near the window stood, sat down, and asked whether anyone knew which was right.

Maya explained the time labels again. She wished the explanation could become less frightening through repetition. It did not. It only became more accurate.

When she returned to operations, Luis was replacing the visitors’ briefing with a version that put the predicted event time and the observation times in larger type.

“They still want one clock,” he said.

“So do I.”

“We can want it together.”

* * *

The first troublesome packet arrived forty minutes before the briefing.

It did not show Earth lost. It showed a measurement that the current solution could not comfortably explain. The duty scientist isolated the instrument and requested a check. The room continued working, but Maya saw how quickly people became quiet around the affected display.

Amara asked whether the discrepancy appeared elsewhere.

Not yet.

Whether the packet’s time tag was intact.

Apparently.

Whether the local instrument had changed its thermal state.

A maintenance record was still coming.

They waited. Waiting was work when there was a list of things to check; it became something else when the list was complete and the signal had not arrived.

Maya used the interval to answer a regional question about reserve release. Her voice sounded ordinary to her. She found that almost insulting. Somewhere inside, she felt as though she were holding a door against water.

The maintenance record arrived. A heater had cycled during the exposure. The instrument team supplied a comparison from the same condition in an earlier test. The discrepancy was consistent with a known shift, larger than usual but not unexplained.

They did not simply delete the packet. They flagged it, widened the relevant uncertainty, and waited for another observation.

The next one agreed.

Luis exhaled. Someone at the back of the room asked whether they could now go to the toilet. Three people laughed too loudly.

Amara called the break she had been postponing.

Amara made the night shift take its break.

“Including you,” Luis said.

She stayed long enough to hand over properly, then walked to the room where visitors were waiting. Her father sat near the window with a flask of tea. Maya had brought him from Nairobi the previous week after he agreed that the garden could survive a competent neighbor for ten days.

“Has it happened?” he asked.

“According to the current solution.”

“And according to the actual universe?”

“We’re waiting for its paperwork.”

He poured her tea.

Earth remained in orbit around the Sun. Venus remained bound.

Mercury, after all the ceremonies held for its evacuated settlements, was still a planet. The buildings left there would continue receiving sunlight whether humans returned or not.

Mars’s new orbit was more eccentric. Its permanent population had already been reduced to crews in the best-protected facilities. Their messages concerned power budgets and spare seals, not speeches.

Jupiter and Saturn would not belong to the departing solar family.

The latest long integrations agreed that their heliocentric binding would be lost as the encounter developed. Their moons would have futures of their own, many still accompanying their planets. Years of evacuation had made those futures less crowded.

The losses did not occur in one cinematic instant. They emerged through changing estimates and increasingly consistent evidence. A familiar object could remain visible long after it ceased to be part of home.

Amara returned to operations before the next briefing.

Luis stood at the main display. The outgoing trajectory was becoming less uncertain. In the old neighborhood’s frame, the Sun was acquiring the speed the early calculations had predicted. In the black hole’s frame, it was turning and would eventually climb away with almost the same distant speed it had brought in.

No engine burned. Momentum changed hands through gravity.

“Outbound,” Luis said quietly.

Several people looked up.

He pointed to the reconstructed solution, then its timestamp.

“Supported by the new measurements.”

The applause began at the back. Someone cried. Elena sat down as if her knees had received permission before the rest of her.

Maya put a hand on her mother’s shoulder.

Amara watched the Earth–Sun separation trace. The relief she had imagined for seventeen years did not arrive all at once. It entered in small pieces: a stable fit, an independent confirmation, another observation that did not require a worse explanation.

Her father appeared in the doorway. He could not see the numbers from there.

She nodded to him.

He closed his eyes for a moment, then found a chair.

The celebration outside lasted into the following day. People gathered in streets under skies that had not yet changed enough to justify the occasion visually. They embraced strangers. They called relatives who were asleep. They made promises the morning would complicate.

At the operations center, the food desks remained staffed.

By evening, Maya was reviewing a missed delivery to a clinic. The clinic was still here. Its patients were still here. She found the carrier’s number and called.

* * *

The clinic’s missed delivery contained ordinary medicines, gloves, and a replacement part for a sterilizer. It had been diverted around a road closure created for a public celebration. The new route brought it to a checkpoint whose staff had an earlier list of authorized vehicles.

Maya called the carrier, then the local coordinator. She expected the matter to take five minutes. It took forty.

By the time the vehicle was cleared, the driver had exceeded the shift limit. A replacement driver had to be found. The clinic could manage until morning if another facility loaned supplies, but the person authorized to approve the loan had gone to the celebration with a phone on silent.

Maya was still on the call when Elena passed her desk.

“Give it to the local desk.”

“I am the person they have.”

“For tonight. You can’t remain that person.”

Maya knew. She stayed until the loan was approved, then wrote down every place the routine process had stopped. There were no extraordinary failures on the list. That was what made it useful.

Outside, people were singing. Someone had projected the old astronaut graffiti onto a building. The letters were enormous now, straight and professionally spaced. Maya remembered the child who had helped paint the wrong letter on a wall and refused to let anyone correct it until the paint dried.

She called Njeri. The neighborhood storeroom was open, and the volunteers had decided to serve the meal early so the evening shift could join the celebration.

“Any trouble?” Maya asked.

“The usual amount. Come eat if you’ve forgotten how.”

Maya nearly declined. Then she sent the handover to the next desk and went.

* * *

The neighborhood meal had no speeches. Someone tried to make one, was handed a bowl, and accepted the redirection.

Maya stood near the serving table until Njeri pushed a chair toward her. Across the courtyard, a girl was showing an older man how to enlarge a photograph on his screen. Two children argued about whether Jupiter had exploded. Their mother told them to ask Maya, then looked apologetic.

“It hasn’t exploded,” Maya said. “Its path is changing. It may keep many of its moons. It’s just not coming with the Sun the way it used to.” “Can we still see it?”

“For a while, when it’s in the right part of the sky.”

“Then how do you know it’s gone?”

Maya thought of the old photograph of Rill’s cafeteria, and of Leila’s voice arriving from a room she could not enter.

“Seeing something and staying with it aren’t always the same.”

The children found this unsatisfactory and returned to their food.

Her phone received a message from *Ithaca*. It had been sent before the latest confirmation and asked after her grandfather. Leila had attached a photograph of the growing compartment’s first successful crop from his recorded seed line. Not all of the seeds had been cleared for use;

these had passed the checks and done reasonably well. The leaves were smaller than in his garden.

Maya sent the photograph to him. Then she recorded the courtyard: dishes, conversation, a child refusing vegetables, a chair scraping the ground.

We’re still here, she said. I wish you could have this meal with us.

She did not add that the wish changed nothing. Leila knew that.

She did not make the wish smaller either.

* * *

Amara left the center after midnight with her father walking slowly beside her. He had refused a wheelchair and accepted an arm, which was the compromise they had reached before either became angry.

The street was littered with paper cups. A cleanup crew was already working around the people still celebrating. One worker recognized her and raised a hand. She raised hers back.

Her father asked whether she felt happy.

“In pieces.”

“That’s how most things arrive.”

On the steps he stopped and looked up, for the first time in three years. He held up his little finger at arm’s length, the way the neighborhood children did, and found the place. Even through the city’s glow, the disordered patch of stars showed on both sides of it now.

He looked for a long time.

Then he held up his hand and tilted it, the way he graded tomatoes and television prophets. Slowly, he turned it upright.

“Finished,” he said.

Amara had never seen him give anything that grade. She did not ask what it meant. She suspected he did not know either, and had decided the thing deserved it anyway.

They waited for their vehicle on a bench. He fell asleep for a moment, his head tilted toward her shoulder. She watched the lights moving above the road and tried to remember the last time she had sat beside him without planning the next thing she needed to do.

When the vehicle arrived, she let the driver wait while her father woke properly.

At home, he insisted on washing the flask. She dried it. Neither turned on the news.

Before sleeping, Amara opened the day’s observation record one final time. The Earth–Sun solution remained within the working range.

The uncertainty was smaller than it had been that morning.

She closed the display without opening the public reactions.

* * *

The public observatory stayed open until dawn.

The man with the telescope arrived before sunset and took longer than he expected to assemble it. A volunteer helped him secure the mount and checked the equipment before directing him toward a safe target. He told her his wife had bought it. The volunteer did not ask where she was.

People gathered around displays showing the encounter measurements. At the far end of the terrace, the volunteer read Amara's slip twice, looked at the man again, and swung his telescope toward a part of the sky where nothing had ever been worth pointing at.

"Look slightly to the side of the center," she said.

He bent to the eyepiece.

The stars around the place where nothing was had come loose from their positions. Near the middle, faint background stars had stretched into short bright arcs, curved around an absence like iron filings around a magnet that was not there. Some appeared twice, once on either side. At the center was a darkness too small to be sure of.

He looked for a long time.

"Is that it?"

"That's what it does to light," the volunteer said. "The thing itself is almost nothing, as far as your eye is concerned. A few minutes of arc of nothing."

He straightened, rubbed his eye, and bent again.

"She'd have hated it," he said at last. "She liked things to stay where you put them."

The volunteer laughed before she could stop herself. After a moment, so did he.

Later, a child asked to look through his telescope. He moved aside and helped her find the eyepiece.

"Is that it?" she asked.

"That's it. That's what's moving us."

She looked for much longer than children usually looked at anything. Her surprise was real enough that he forgot, briefly, to compare the evening with the one he had imagined sharing with his wife.

He sent a photograph of the assembled telescope to his wife's sister. The reply said she would have made him straighten the tripod before taking the picture. He looked at it, saw the tilt, and laughed.

When the outbound confirmation reached the observatory, people applauded. He joined them, then returned to the eyepiece. The night contained more than one reason to keep looking.

* * *

At a Mars facility, the encounter day was divided by maintenance tasks.

The remaining crew had known for years that the planet's future orbit could be difficult. They had reduced the population, concentrated resources, and accepted that surviving the immediate passage did not settle the longer climate and power problem. Their messages to Earth were practical because practical questions still had answers someone could act on.

One request concerned a seal material whose expected replacement interval no longer looked adequate under the revised operating cycle.

The supplier on Earth had a better candidate but insufficient test data.

The crew asked for both the material and the uncertainty, and prepared to test it locally before depending on it.

Maya read the request during a quiet interval after the celebration began. It was not within her direct remit. She sent it to the responsible team and asked for acknowledgment rather than taking ownership herself. The acknowledgment arrived before her shift ended.

She had learned to distinguish caring about a problem from becoming its only route to a solution. On that day, the distinction felt like part of survival.

* * *

In the growing district, Inês watched the confirmation on a small screen propped against a bag of supplies.

The workers stopped long enough to listen. One began crying.

Another asked whether they could now reverse the revised planting plan. Inês said no, not on the basis of that announcement. Earth remaining bound did not tell them everything about the next season.

The answer made her unpopular for several minutes. She let people have the minutes.

Then she opened a container of food Zawadi's group had helped prepare and declared a longer break. The crop work could tolerate it.

She had checked before making the decision, which made the break feel both less spontaneous and more generous.

Celso arrived with the latest delivery confirmation. This time, the physical cargo stood outside. He had asked a worker at the receiving point to photograph it, then had gone to see it himself because he wanted to.

Inês handed him a bowl. He showed her the photograph anyway.

"I can see it through the door," she said.

"I know."

She looked at him and understood. "Send it to Maya."

He did. Maya opened the image later that evening between the clinic calls. It showed no crowd, no sky, no famous person. Just material in the place where people needed it.

She saved it.

* * *

Elena’s private celebration consisted of calling her sister.

They had not spoken properly in weeks. Their messages had become lists of practical family matters, answered with increasingly efficient brevity. Elena had been proud of keeping up until she realized that she knew the date of a nephew’s examination and not whether he was frightened of it.

Her sister answered from a kitchen full of people. Someone behind her shouted congratulations. Elena thanked them, then asked about the examination.

“That was last week.”

“I know. How did it go?”

Her sister looked surprised. She called the boy over. He described the parts he thought he had done well and the question he had misunderstood. Elena listened without turning the conversation into advice about preparation.

Afterward, her sister asked whether the project was finished.

“No.”

“Are you coming to visit?”

Elena began to explain the next phase. Her sister waited with an expression Elena had seen in committee rooms when someone was avoiding the actual question.

“Yes,” she said. “I’ll choose a date tomorrow.”

“Choose it now.”

Elena opened the calendar.

* * *

The morning after the confirmation, Luis arrived at work carrying bread for the day shift.

He had bought too much. The bakery had been celebrating and kept adding things to the bag. He placed it on the table and found Amara already checking an independent reduction.

“You slept?”

“Some.”

“A unit recognized by no medical authority.”

She took a piece of bread. They looked at the solution together. The latest data supported the outbound interpretation. The outer planets remained a complicated loss rather than a single clean event. The Earth orbit was becoming better constrained.

Luis pointed to one value that had changed slightly.

“We’ll need to explain that.”

“Yes.”

Neither resented the work. For the first time in years, it was taking place on the other side of the date they had used to measure everything. Amara called the public team and asked them to preserve the previous estimate beside the revision. Then she went to find a proper cup for her coffee.

CHAPTER SEVENTEEN

Four Centuries

BETTING EXCHANGE NOTICE

The market “Earth stays bound” has settled. Result: BOUND. Payouts will be processed within three business days, subject to the definition of a day.

New market open: SNOW LINE REACHES 45° NORTH BY 2131.

EARTH MISSION ELAPSED TIME: DAY 41

The first stable post-encounter forecast put Earth’s semimajor axis near 1.19 AU.

Its orbit was no longer nearly circular. The preliminary eccentricity was about 0.27. Further observations would refine both numbers, and the other planets would continue perturbing the solution.

Amara refused the first headline proposed for the briefing: EARTH SAFE.

The annual average sunlight would be roughly twenty-seven percent lower than before, allowing for the new eccentricity and an essentially unchanged Sun.

Maya asked the briefing team to keep reading.

At perihelion, Earth would receive about a third more light than it did now. At aphelion, a little under half. The year would run to roughly four hundred and seventy-four days, and no calendar on the planet described its seasons. How the climate responded would depend on clouds, oceans, ice, and choices humans were still making.

“The average is the reassuring number,” she said. “It’s also the one that will get somebody killed. Put all three on the same page or don’t publish any of them.”

The long integrations did not close. With Jupiter and Saturn no longer bound, nobody could yet say whether the inner system was stable over the timescales on which such questions were usually settled. Amara published that too, in the same release, in the same size of type. They had built the reserves for the average.

The average was not the problem.

Amina was the first to say aloud what the climate teams had been circling for a year.

“Twenty-seven percent sounds like a bad harvest,” she told the council. “It is not a bad harvest.”

She put two numbers on the screen. The sunlight the planet absorbed had fallen by about sixty-four watts on every square meter, averaged over the whole Earth and the whole year. The warming from doubling the carbon dioxide in the atmosphere, the number that had dominated two centuries of argument, was under four.

“We have taken more than fifteen of those doublings,” she said, “in the other direction.”

A delegate asked why the planet was not already frozen.

“Because the oceans are slow. They hold more heat than everything else we have, and they give it up over years. That is the only reason this is a planning problem instead of a funeral.” She changed the slide. “Ice is not slow, once it starts. Snow reflects sunlight. The more of it there is, the less sunlight we keep, and the colder it gets. Every winter the snow line moves toward the equator makes the next winter easier for it.”

“How far?”

“That depends on what we do.”

The room waited for the rest.

“The models disagree,” Amina said. “Some say the oceans and our reserves carry us to a cold but livable balance. Some say the ice wins a little every year until it wins everything. I can’t tell you which. I can tell you we will have to act long before the arithmetic lets us be sure.”

Amara refused the second proposed headline too. It said EARTH SAFE (FOR NOW).

Maya briefed the council herself. She showed the regional differences, the first-year supply plan, and the places where construction remained incomplete. The questions were difficult and mostly practical. She preferred them to applause.

Afterward, Luis presented the navigation solution.

“Approximately three thousand kilometers per second relative to our old neighborhood,” he said. “The exact velocity relative to Alpha Centauri is different. We are calculating an encounter with a moving system, not aiming at a photograph.”

The projected route passed through Alpha Centauri’s future neighborhood in roughly four hundred and forty years. The approach distance remained under refinement. Nobody called it a landing.

“Can we stop?” a new council member asked.

“Not with the means we have,” Luis said. “That has not changed.”

“Then why call it a destination?”

Maya answered. “Because we can send instruments there, study it, and make decisions before the flyby. Whether our descendants can do more is work they will have to judge for themselves.”

The meeting turned to who could authorize preparations on behalf of people not yet born. A proposal to reserve future colonies for present member states failed so decisively that its sponsor withdrew the written version.

Other questions were harder. How long should expensive observatories operate? What records had to survive? Could one generation spend resources needed by another in pursuit of a destination that might later change?

That evening, Maya found Amara studying the route from home.

“You won’t see it,” Maya said.

“No.”

“Neither will I.”

Her mother moved a cup away from the edge of the table.

“When you were born, I thought about things I would see. Your first words. Your graduation. I didn’t understand how much of being a parent would be preparing for days I wouldn’t be there.”

Maya sat beside her.

“Four hundred and forty years is a lot of days.”

“Yes.”

There was no useful reassurance to add.

On the display, the Sun’s path continued beyond the star system.

Maya changed the view so it showed the uncertainty spreading with time. A route narrow enough to fit on a screen was not a promise to the people who would inherit it.

She saved that version for the public archive.

* * *

The calendar broke on the forty-first day.

A larger orbit was also a slower one. Every calendar built on three hundred and sixty-five days, from tax years to school terms to religious festivals, had become wrong by a hundred and nine of them.

A Calendar Commission was formed. It received more submissions in its first week than Daedalus had received in its first year. Some proposed fifteen months. Some proposed keeping twelve, only longer. One proposed a hundred-and-nine-day festival at the end of every year, and attracted a large and highly committed following.

Payroll departments wanted to know whether salaries were annual. Landlords wanted rent to stay monthly and the months to be short. Children wanted to know whether birthdays still counted if they came less often, and whether a person who had been fourteen for four hundred and seventy-four days was owed compensation.

Pilot’s oldest payroll module, asked to process a year of that length, returned an error. Someone printed the message and pinned it to the operations board:

ERROR 404 — YEAR NOT FOUND.

Amara asked what it meant. Maya explained. Amara said it was not funny. The next morning someone saw her smiling at it while she waited for the coffee.

Grandfather declined to take part in the debate.

“I’ll plant when it’s warm,” he said. “It will tell me.”

* * *

The first person to complain about the four centuries was not a council member. It was a technician in the corridor outside the briefing room, who had spent seventeen years being told that survival depended on completing things before a date.

“So what’s the deadline now?” he asked.

Maya thought he meant the next reserve review. He shook his head.

“For being finished.”

She could have told him that nobody had promised an end to maintenance. It would have been true and ungenerous. The project had borrowed the emotional shape of a rescue: danger, effort, deliverance.

The work after deliverance had remained in the footnotes.

“There isn’t one for the whole thing,” she said. “There should be one for your shift.”

“That’s smaller than advertised.”

“Yes.”

He looked tired, not angry. He had a daughter he wanted to take on a trip before she stopped wanting to travel with him. Every year there had been a reason to wait. Now the new maps offered four hundred and forty more.

Maya asked whether he had requested leave.

“Three times.”

“Send me the refusals.”

He hesitated. “I don’t want a special favor.”

“Then send them to the staffing review too. This shouldn’t depend on meeting me in a corridor.”

After he left, she opened her own leave record. She had unused time accumulated across years. The total looked less like diligence than a system that had been allowed to consume its reserves without recording the debt.

She put a week on the calendar.

The first invitation that conflicted with it arrived seven minutes later. She declined.

* * *

The public route display initially showed a bright line from the Sun to Alpha Centauri. Maya asked for the line to continue beyond it.

The designer did so, then asked whether the line could fade at the star system.

“People need to understand what we’re working toward.”

“They also need to understand that we don’t stop there.”

“The uncertainty makes the graphic harder to read.”

“It makes the journey harder to promise.”

They tried several versions. One looked like a cloud of meaningless paths. Another made the future encounter seem exact.

They settled on a central projection surrounded by a widening band, with separate views for the near years and the distant passage. The explanatory panel distinguished the moving destination from its present appearance in the sky.

Luis added a note about the velocity frame. He wanted every number labeled with what it was measured relative to. The designer said the note was becoming longer than the graphic.

“Then let people open it when they want it,” he said. “Don’t make them infer it from a pretty arrow.”

The first school group to test the display asked better questions than the adults who approved it. One child wanted to know whether the Sun had been traveling before. Another asked why this direction deserved a special name if everything in the Galaxy moved. A third wanted to know who would be allowed to change the plan after Maya died.

“The people doing the work then,” she said.

“What if they don’t want Alpha Centauri?”

“They can decide what to study and what to build. They won’t have an easy way to choose the Sun’s path.”

The child frowned. “Then it’s not all a plan.”

“No,” Maya said. “Some of it is a situation.”

She asked the designer to put that distinction into ordinary language beside the route.

* * *

Leila had watched a recording of the briefing aboard *Ithaca*. Her reply arrived after Maya’s workday ended.

I heard someone ask whether we can stop. People here ask whether we can come back. We have the same problem with the word can.

Maya read the paragraph several times. The arks had followed trajectories designed for their own survival through the encounter, using the propulsion they actually possessed. Their separation from Earth was no longer a matter of choosing a convenient transfer date. A return

would require a feasible trajectory, propellant, life support, time, and a future neither of them could promise.

They had known this before departure. Knowing it in a room together had felt different from receiving it in a packet.

Maya opened the navigation summary and closed it again. Leila had not asked her to solve the trajectory.

She wrote: I miss thinking I might hear your key in the door.

Then she worried the sentence would burden Leila, and almost deleted it. They had agreed not to make honesty so considerate that it became concealment.

She sent it.

Leila’s answer came the following exchange.

I miss having a door that leads to you. I’m also glad I came. Both are still true.

There was a photograph attached. Leila sat on the floor of the growing compartment with three other people, eating from bowls. One of them had a bandage across two fingers. Another was laughing so hard that the image blurred her face. Maya did not know them.

For a moment she felt the sharp, childish wish that Leila’s life should remain incomplete in exactly the way hers felt incomplete. She was ashamed of it, then tired of being ashamed of feelings she had not chosen.

She enlarged the photograph. The food looked good. Leila looked exhausted and alive in a room that had become familiar to her.

Who are they? Maya wrote.

It was the first question that made the distance feel like something they could inhabit rather than merely endure.

* * *

On her third day of leave, Maya took a train with no work appointment at the other end.

She had chosen a place near a lake because she had once wanted to go there and had forgotten why. The guesthouse had a loose window latch. She noticed it, considered offering to fix it, and went outside instead.

At breakfast, another guest recognized her from the encounter coverage. He asked whether the climate would worsen there. She told him the regional forecast was still being revised and pointed him toward the public source.

“But what do you really think?”

She looked at the tea cooling beside her plate. “That is what I really think.”

He seemed disappointed. She understood the attraction of believing that famous people possessed a second, more useful set of answers. Her mother had helped create that expectation once. Maya did not want to inherit it.

She spent the morning walking. The path led past a damaged landing stage and a group of children trying to retrieve a ball from reeds. They had a long branch and a poor understanding of leverage.

She watched until one asked for help, then held the branch while they moved along the bank.

The ball came free. One child thanked her. Another accused her of making it too easy.

She sat on a stone wall afterward and ate an orange. Leila had taught her to remove the peel in one long strip; Maya still could not do it reliably. The strip broke halfway around.

She photographed it and sent the image without explanation.

That evening she called her grandfather. He asked whether she had found anything useful to do.

“No.”

“Good. Stay another day.”

She stayed the whole week.

When she returned, the world had generated new problems and solved several without her. The fact hurt her pride for a moment, then relieved her more than the holiday had.

* * *

The proposal to claim future colonies arrived on expensive paper.

Most council documents were shared electronically, but the sponsoring delegation had printed this one with a seal and a map. The map showed Alpha Centauri as a destination waiting to be divided into rights. It did not show the uncertainties in the route, the absence of a means to brake the Sun, or the generations whose consent could not be obtained by leaving space for a signature.

Maya read it with growing disbelief. The proposal treated contributions made during the emergency as investments redeemable centuries later. Some of those contributions had been generous and costly. None created ownership of worlds the sponsors had not visited and might never be able to use.

At the preliminary discussion, the sponsor argued that clear incentives were necessary for long-term infrastructure. Anika’s written response, received before the meeting, asked how a contract could remain legitimate when its supposed beneficiaries and burdens fell on people not yet alive.

The debate was less simple than the most offensive clauses made it appear. Observatories required sustained funding. Archives needed caretakers. Navigation work could be abandoned if every generation treated distant benefits as someone else’s problem. A framework for continuity was necessary. The proposed ownership scheme was not the only one available.

Maya asked the archive team to describe how they expected records to survive four centuries. Their answer involved multiple copies, different media, periodic migration, and institutions willing to pay for unglamorous checks. The expensive paper might outlast an unattended file. It would not, by itself, preserve the meaning of a coordinate system.

The sponsor looked annoyed by the comparison. Luis looked delighted.

They began separating continuity obligations from claims of possession. The revised discussion concerned who would maintain shared observations, how decisions would be reviewed, and how later communities could change commitments without losing access to the knowledge that made change possible.

It was harder to put on a commemorative map. Maya regarded that as a useful feature.

* * *

The archive’s oldest machine sat in a room that looked like a repair shop because it was one.

Amara visited while the team was recovering an early observation format. The data had not vanished. The software needed to interpret it depended on assumptions stored in documentation that referred to a retired instrument by a name no current catalog used.

A technician had found the missing link in a maintenance note.

“Whoever wrote this thought they were explaining a replacement part,” she said. “They also told us which detector orientation the file assumes.”

Amara watched the first recovered image appear. It was not beautiful. It contained noise, blemishes, and a field of stars whose value lay partly in when they had been observed.

“How many are left?”

The technician gave a number large enough to make Amara wince.

“Can we automate more of it?”

“Yes. After we know what the automation is supposed to preserve.”

Amara smiled. “Fair.”

They discussed the encounter records she wanted kept. The team asked for raw observations, reduction histories, rejected alternatives, and the correspondence explaining changes. She offered the polished public releases first because those were easiest to locate. They took them and asked for the inconvenient material again.

She supplied it.

Among the files was the early folder of timing residuals that had once seemed noncritical. Amara opened one note and remembered the night of discovery, the cake photograph, the failed attempt to blame calibration. The record made the sequence less heroic than memory could have made it.

She was glad.

* * *

Maya’s week away caused a small argument when she returned.

An analyst had made a regional adjustment within the authority Maya had helped establish. The choice was defensible, documented, and different from the one Maya would have made. She began asking questions in a tone that made the analyst straighten in her chair.

Halfway through, Maya stopped.

“I’m checking whether you used the authority correctly,” she said.

“I shouldn’t be checking whether you became me.”

The analyst looked relieved and still cautious. “I can show the alternatives.”

“Please.”

They reviewed them. Maya found one assumption that should be checked next time. The analyst found a local constraint Maya had not known. The record improved without the decision being retrospectively transferred to the person with the more famous name. Afterward, Maya called the technician who had asked about being finished. He and his daughter had taken their trip. It had rained for three days. He sounded happy anyway.

“Did she enjoy it?”

“She complained. We played cards. I think so.”

Maya thought of the photograph by the lake and the missing joke.

“That sounds like a trip.”

* * *

The first public lecture about the four-century passage filled a hall and left several people disappointed.

They wanted to know what the destination would be like. Luis could describe the known stellar system, the meaning of the projected approach, and the work required to refine it. He could not show a confirmed habitable world waiting for settlement. He could not promise that a civilization traveling with the Sun would possess the means to stop there.

One attendee asked whether the whole destination story was therefore propaganda.

Luis considered the question. “It becomes propaganda if we present possibilities as arrangements already made. It can be a useful direction for investigation if we keep the distinction visible.”

Another asked whether they should spend anything on a future none of them would see while people remained hungry.

Maya answered that the amounts and tradeoffs should be public and periodically reviewed. Some observing work also supported immediate navigation and environmental measurements. Some was curiosity about a distant future. Curiosity deserved a place in civilization, but it did not deserve immunity from questions about cost.

The discussion remained contentious. Afterward, a student approached to ask how to join the archive team. Maya gave her the contact and warned that much of the work involved checking old labels. “Someone has to,” the student said.

Maya smiled. The phrase had justified too much self-sacrifice in her life. In this voice, it sounded like interest freely chosen.

CHAPTEREIGHTEEN

Interstellar Weather

MAINTENANCE TICKET, SURVEYOR NINE

Reported by: night shift. Summary: “It’s a sieve.”

Resolution: Second inspection. Not a sieve. Nickname retained.

EARTH MISSION ELAPSED TIME: YEAR 1, DAY 210

At one percent of light speed, the stars did not crowd dramatically into the forward sky.

Instruments measured the aberration. Ordinary eyes would notice the gradual changes in nearby stars over years and decades. The familiar constellations had not vanished on departure, whatever the souvenir posters claimed.

Space outside the atmosphere had changed more urgently.

Gas that had drifted through the old heliosphere now arrived with much greater energy in the Sun’s frame. Dust struck forward-facing surfaces hard enough to crater them. The solar wind’s protective boundary adjusted under the increased external pressure, and the mission designers stopped using its old shape as a convenient assumption.

None of this was a surprise. The first measurements still made Elena ask for them to be checked.

An unmanned scout returned photographs of a damaged plate. A grain too small to see in the image had excavated a hole through its outer layer. The electronics behind a second layer survived.

“That is what the second layer is for,” the materials engineer said.

“I know,” Elena replied. “I’d like it to remain bored.”

The Shield was a collection of systems, not a wall across the Solar System. Scouts measured gas and dust ahead of the traveling Sun.

Spacecraft used sacrificial material and better-protected electronics. Magnetic systems could divert some charged particles; they did nothing directly to an uncharged grain. Detection and avoidance remained limited by distance, response time, and the size of the target. Earth retained its atmosphere and enormous mass of shielding. The people most immediately exposed were outside it.

Saira’s factory reconfigured its external equipment. Maintenance crews worked behind temporary panels. Forward was a direction with a larger repair budget, not a curse attached to a compass point.

The damaged scout was named Surveyor Nine, though the maintenance crew called it the Sieve before the second inspection proved the nickname unfair. The nickname survived the inspection. Names on a shop floor were not evidence-based.

Its forward plate carried more pits than the earlier exposure model had predicted. Most were shallow. Two penetrated the outer layer. A temperature sensor behind one of them had failed, leaving the thermal controller dependent on a neighboring measurement that was no longer a good substitute under all conditions.

The vehicle still sent useful data. That was the source of the argument.

The navigation group wanted to preserve the observing schedule.

The materials group wanted to turn the damaged surface away from the incoming environment and examine it. The thermal team wanted a maneuver that would make both requests harder.

Saira listened to the competing requirements and asked which loss would make the others irrelevant.

“If the controller overheats the package, there won’t be a data schedule,” the thermal engineer said.

They began there.

Surveyor Nine was far enough away that commands and replies required patience. A repair crew could not be sent simply because the equipment mattered. The nearest available vehicle had other commitments, and a rendezvous would consume resources better used elsewhere unless the scout could demonstrate a fault that inspection might resolve.

Pavel proposed using the scout’s own small deployment arm to place a spare camera where it could see the damaged region. The arm had not been designed for maintenance. Its range appeared sufficient in the model. He asked for the as-built photographs before anyone approved the maneuver.

One photograph showed a cable bundle occupying part of the proposed path.

“That isn’t in the geometry,” said the operator.

“It is in the spacecraft.”

They revised the path. The new one brought the camera close to a surface whose thermal covering had already loosened. If the arm caught it, they might create a second problem while examining the first.

Pavel asked the team to build the relevant arrangement in the workshop. The model was ugly: spare brackets, a section of cable zip-tied at the angle in the photograph, a sheet of covering material baked in a thermal chamber until it crackled like old packaging when anyone touched it, which they were told not to do and did anyway. It could not reproduce every

condition. It could show whether the proposed movement was foolish before they asked the real arm to try it.

The first run caught the covering. The mock arm dragged a corner of it into the light with a sound like a boot on gravel, and somebody said a word that was later removed from the log.

The second avoided it but could not see the sensor.

“Angle’s wrong.”

“Angle’s what the drawing gives us.”

“The drawing didn’t put the cable there either.”

On the third, an apprentice suggested rotating the camera mount before deployment rather than after. That reduced the required movement. Pavel asked her to show it. She did, then watched him write her name beside the proposed change.

“You don’t have to,” she said.

“Yes, I do. If we need to ask what you meant, we should know who you are.”

* * *

Maya arrived at the factory during the final rehearsal. She had come to review the effects of the refit on agricultural equipment deliveries, and found half the people she needed gathered around the mock scout.

Saira handed her a pair of safety glasses and told her where to stand.

The apprentice’s name was Nadi. She was nineteen, had grown up during the encounter preparations, and regarded the idea that workshops had once operated without universal remote logging as evidence of primitive civilization.

Pavel pointed to Maya. “She used to stand there and believe the drawing.”

Maya looked at the mock assembly. “I still believe drawings. I ask more questions about which one.”

Nadi smiled politely, the way young people did when the old tried to make their lessons entertaining.

“The drawing isn’t the problem,” she said. “The problem is the spacecraft keeps disagreeing with it. If we could get the spacecraft to read the drawing—”

“Nadi.”

“Watching.”

The rehearsal succeeded. Saira approved the sequence for independent review. They did not transmit it immediately. The spacecraft team checked the power state, thermal limits, and stopping conditions. A command that moved the camera safely in the workshop could still be wrong for the vehicle at that moment.

Maya waited with them through the first returned images. The camera saw the edge of the plate, then darkness, then a portion of the failed sensor’s mounting. The image froze while the next packet traveled.

Nadi leaned toward the screen.

“It hasn’t stopped,” Pavel said. “We have.”

She sat back, embarrassed.

“I know.”

“So do I. I still lean.”

The next image showed a damaged lead, not a destroyed sensor.

That distinction mattered. The team could not reconnect it remotely, but they could change the controller’s use of the neighboring measurements with a better understanding of what had failed.

They also found an unexpected shadow from a piece of lifted covering. Its movement explained part of the temperature variation that had made the original diagnosis confusing.

The final plan reduced the scout’s observing duty and changed its attitude schedule. It would gather less data and probably last longer.

The navigation team accepted the gap and reassigned some observations to another platform.

No one called the result a rescue. Surveyor Nine remained damaged. Its remaining life had become easier to estimate.

Maya watched Nadi save the record of the failed rehearsals along with the successful one.

“Keep the cable photograph,” she said.

“Already attached.”

It was a small pleasure to discover that a lesson no longer required her to deliver it.

* * *

After the demonstration of the factory interlock, Pavel took Maya past the bay where he had been injured.

The equipment had changed. The floor markings had changed. A new worker was using the inspection point as though it had always been part of the machine.

Maya stopped outside the boundary.

“Do you come here often?” she asked.

“When there’s work.”

“I mean—”

“I know.”

He waited while the worker completed a movement. Maya could hear the machinery through the wall, a low rhythm that made her remember the sound just before the accident. She had remembered it differently at different times. The recording had corrected some details and left others untouched.

“Sometimes it bothers me,” Pavel said. “Sometimes I’m thinking about lunch.”

She nodded.

“You don’t need me to have one feeling about it forever.”

“I wasn’t trying to.”

“I know that too.”

The worker looked up and waved. Pavel pointed out a cable that needed securing. The worker fixed it before beginning the next cycle.

They continued to the cafeteria. Maya carried the inspection mirror in her bag. She had polished the handle enough over the years that one of the scratches had nearly disappeared.

At lunch, Pavel asked her a question that had nothing to do with the scout.

“Are you happy?”

She looked at him. “That’s a large question for this food.”

“It’s the food I have.”

Maya thought about the lake, the messages from *Ithaca*, and the quiet flat. “More often than I was. Not all the ways I expected.”

Pavel nodded as if that was an acceptable engineering tolerance.

Then he asked about the crop forecasts, and they turned to the problem his sister had sent him.

After the scout sequence, Pavel showed Maya a newly refitted remote inspection arm. His prosthetic hand rested beside the controls while he watched the camera feed.

“How’s the interlock?” she asked.

He demonstrated it without smiling. The machine stopped before crossing the maintenance boundary.

“People complain it’s slow.”

“Do you?”

“Constantly. I’m allowed.”

They ate in the same cafeteria where Maya had once struggled to explain the accident to her mother. The window showed Earth, white clouds turning above oceans that had begun to store a different balance of heat.

Pavel asked whether the forecasts were getting better.

“Some of them.”

“Useful answer.”

“It depends which question you’re asking.”

He pointed to the planet.

“My sister grows beans.”

Maya asked where. They spent the rest of lunch looking at a regional forecast instead of the global one.

His sister’s district had been assigned a replacement bean variety with a shorter growing season. The recommendation looked reasonable until Pavel opened her last message. The new crop needed irrigation during a period when the local pumps were scheduled for reduced power.

Maya checked the two plans. Each had been approved. Neither included the other’s latest change.

“I can flag it,” she said.

Pavel kept the message open.

“When does she have to plant?”

“Monday.”

Maya looked at the time. On Earth, the district office would still be open. She asked Pavel whether she could include his sister’s message in a call.

The agricultural coordinator answered while walking between buildings. She knew about the pumps. She had submitted a correction six days earlier and was waiting for the power office to acknowledge it. Maya brought the power office into the call. The solution took twenty minutes and required moving a maintenance shift. Nobody needed a new invention. Two people needed permission to change schedules that had arrived stamped FINAL.

Afterward, Pavel closed his sister’s message.

“Does this happen everywhere?”

“Enough places.”

“You can’t take all the calls.”

“No.”

On the ferry home, Maya drafted a change to the review process.

Regional teams needed authority to reconcile those conflicts without waiting for her. She sent the proposal to the two coordinators first, asking what it would miss.

* * *

The factory’s next refit replaced a tool that everyone disliked with a tool nobody trusted yet.

The old external inspection rig was slow and difficult to maintain.

Its faults were familiar enough that operators had developed ways to work around them, some approved and some passed quietly between shifts. The new rig offered better reach and

protection, but its control behavior differed in ways the training simulation had not made obvious.

Nadi noticed the first problem during a dry run. The display smoothed a pause in motion so the arm appeared to continue steadily while the real mechanism hesitated. The smoothing improved readability during ordinary operation. During close inspection, it hid information the operator needed.

She raised the issue. The supplier said the behavior was intentional.

Pavel asked whether intentional meant appropriate for this task.

The supplier sent an engineer. She arrived prepared to defend the interface and changed her mind after watching three operators make the same wrong inference. They disabled the smoothing in the inspection mode and added a clear indication of delayed feedback.

The change took less time than the argument about whether a change was necessary.

Saira included the supplier’s engineer in the final record. The engineer asked whether the report would make her team look careless.

Saira said it would show that a real use had exposed a limitation and that the team had corrected it. Leaving out the correction would be unfair; leaving out the limitation would be useless.

The engineer stayed for the next run.

* * *

Maya’s visit to the external operations control room brought back the old factory fear in a new form.

She could see a crew working behind temporary shielding on a camera feed. Their task was routine within the revised procedures: replacing a component whose exposure had become more costly under the new environment. The workers had planned positions, protected routes, and local authority to stop. The control room monitored conditions and equipment, not individual heroism.

One worker paused. A cover did not sit where the drawing showed it. Maya felt her body tense before the supervisor asked for the reason.

The worker explained. The supervisor acknowledged. They checked the alternate configuration and continued only after confirming the access was clear.

Nothing went wrong.

Maya realized she had been waiting for the story to repeat because the setting resembled the place where she had first learned what injury cost. The resemblance was not a prediction. The improved procedure mattered precisely because it made an uneventful outcome more likely.

Pavel noticed her expression when she left the room.

“Still waiting for the bad sound?”

She nodded.

“Sometimes I do too.”

“Does that help you work?”

“Sometimes. Sometimes it just makes me tired.”

They walked to a window where Earth occupied part of the view.

Maya had seen it from the Moon enough times that she expected familiarity. It still surprised her: the apparent smallness, the clouds, the fact that almost every person she had ever loved remained inside that visible shape except the one whose messages now arrived from farther away.

Pavel stood beside her without speaking. She was grateful he did not require the sight to make them optimistic.

* * *

Pavel’s sister answered the follow-up call from a field.

The bean planting had gone ahead after the power schedule changed. Not everything had worked. A section received water late because a local valve failed. Another section germinated better than expected. The forecast was useful enough to plan with and imperfect enough to require watching the ground.

Maya asked whether the revised regional authority had helped.

“They answered faster.”

“Did they solve the problem?”

“We solved the problem. They let the electrician change the schedule.”

Maya wrote down the distinction. Pavel’s sister looked suspicious.

“Are you going to quote me?”

“Only if you want.”

“You can quote that. Don’t use the picture. I look terrible.”

Pavel told her she looked like someone working. She told him that was why she had asked. Maya used the words and left out the image.

The regional review that followed reduced several unnecessary central approvals. It also added a clear process for conflicts that crossed district boundaries. Giving local teams authority did not make every local choice compatible with its neighbors. The work was to make those conflicts visible before each side committed resources the other needed.

Maya found the resulting meetings less glamorous than the encounter briefings and often more satisfying. People arrived with specific things they could change. They left with names, dates, and obligations small enough to verify.

* * *

On her last evening at the factory, Saira invited Maya to the workshop's informal meal.

There was no celebration attached. People brought food, complained about procurement, and argued over a game played badly enough that Maya suspected the rules were being invented. Nadi won twice and denied all accusations of cheating with the confidence of someone who had read the rules more recently than the others.

Pavel showed Maya a new photograph of his niece. She was a young woman now; Maya still remembered the six-legged horse drawing. In the photograph she held a tool she had made for a school project.

"Does it work?" Maya asked.

"She says yes. I asked for the test."

"Of course you did."

He showed her the second image: the tool in use, awkward but functional. Maya admired it. Pavel looked pleased in a way he rarely displayed about his own work.

Saira sat beside them with a plate. She asked Maya whether the old inspection mirror was still useful. Maya took it from her bag. Saira examined the handle, then handed it back.

"You'll eventually need a new one."

"I know."

"Objects can retire. You don't have to make them carry the whole lesson."

Maya turned it in her hand. The mirror was scratched but clear enough for its purpose. She would replace it when it no longer was. For now, she put it away and joined the game.

* * *

The resolution passed at three in the morning, Geneva time, with eleven abstentions and one delegate asleep.

It authorized the manufacture and release of engineered warming gases: compounds designed to trap heat efficiently and break down on known schedules, first proposed, generations earlier, for warming Mars. The chemistry was old. The quantities were not. Anika's team sent a forty-page dissent from Ithaca about everything nobody yet knew of how the new gases would interact with clouds. It arrived nine days after the vote. Amara made sure it was published beside the resolution, on the same page, in the same type size.

Elena read the resolution twice.

“Upward,” she said.

“Upward,” Maya agreed.

“Two hundred years of summits, and the target goes upward.” Elena put the tablet down. “My grandmother once chained herself to a refinery gate. Someone should tell her.”

“Is she alive?”

“No. That’s why someone should tell her. She’d have found it hilarious.”

The first plants were built beside old chemical works, where the pipes and the permits already existed. Their output was monitored by satellites that had once tracked emissions for the opposite purpose. The monitoring teams found that most of their software worked without modification. Only the sign had changed.

CHAPTER NINETEEN

Worldship

DAEDALUS STYLE GUIDE, REVISION 44

“The Transit” is retired. Preferred: “the Seasonal Adjustment.”
Handwritten in the margin, origin unknown: It snowed in Seville.

EARTH MISSION ELAPSED TIME: YEAR 2, DAY 18

The first winter under the revised agricultural calendar was worse than its central forecast and better than its worst one.

Neither description meant much to people whose local stores ran short.

In one district, a damaged rail bridge delayed deliveries for nine days. In another, a regional authority protected its own reserve and released it only after a court order. Several smaller growing systems outperformed the large installations they had once been dismissed as replacing.

The bridge failed during a week when everyone had been told the transport network was recovering.

It was a modest bridge over a channel used mostly for drainage.

The inspection had identified a support that needed work. The repair had been scheduled before the heavier reserve traffic began, then moved when the material went to a higher-priority crossing. The temporary load restriction had reached the rail operator but not the model used to distribute the food shipments.

When the damage was found, trains stopped. No one was injured.

The absence of casualties made the first report sound less serious than the nine days that followed.

Inês’s facility had food ready to move. The receiving districts had places prepared for it. Between them sat a broken support and a line of refrigerated containers consuming power while they waited.

Celso called Maya before calling the national desk because he knew she would answer. She answered, then brought the national desk into the call because he should not need to know her.

They divided the cargo by what could survive delay. Some could wait. Some could be processed nearby. Some needed another route immediately. The alternative roads were slower and passed through districts with their own shortages.

“People will see trucks go past,” the local coordinator said. “If they’re hungry, a routing notice won’t be enough.”

“What do you need?” Maya asked.

“A share of the load, openly agreed. Or a very different conversation.”

The central model treated that as a loss to the intended destination.

The people on the road treated it as food. Maya asked both districts to join the decision. The negotiation was ugly, but less ugly than sending vehicles into a conflict and calling it unexpected.

* * *

Near the equator, the winter came as colder nights, a shorter growing season, and an argument about blankets. The city distributed bedding to schools and clinics. The bedding arrived in sizes that suggested its supplier had been expecting different children.

Farther north it was something else. In the new orbit, Earth’s farthest point from the Sun fell during the northern winter, and the two effects stacked. At aphelion the planet now received less than half the sunlight it once had; at perihelion, a third more. In the old orbit the difference between the two had been a few percent, and nobody outside an astronomy class had noticed it.

Harbors froze that had never frozen in their recorded history. In Busan, Min photographed the ice from her balcony and sent the picture to her father at the depot. He wrote the date on the back of the print, as he had once written dates on the kitchen calendar at Rill.

In spring, the satellites measured what the ground had already suggested. Snow had lingered weeks longer than the models expected across a band of northern farmland. The snow line had stopped advancing in two of the five monitored regions. It had advanced more slowly in two others. In the fifth, it did not seem to have heard about the resolution.

Amina put all five on the same slide.

“That one,” a delegate said, pointing to the fifth.

“Yes,” Amina said. “That one.”

Someone in the operations center wrote the fifth region’s number on the board. Nobody erased it.

* * *

Jun was working at a municipal repair depot when the request for bridge equipment arrived.

He had taken the job six months earlier. The title still bothered him sometimes, though the work rarely gave him time to dwell on it. He supervised a small team maintaining pumps, lifts, and power equipment in public buildings. His colleagues had stopped introducing him as the man from Saturn after he flooded a workroom by trusting an old valve label.

The depot had a lifting assembly suitable for part of the bridge repair. It also had a scheduled job in a residential tower whose lift had been out for four days. Several residents could not use the stairs.

Jun called the tower coordinator. He explained the request and asked which people needed immediate assistance if the repair moved.

“You’re asking us to be reasonable about being trapped,” she said.

“I’m asking what we need to provide if we take the equipment.”

They arranged temporary help and a fixed return date. The tower’s residents did not become pleased. Jun did not record them as pleased.

He recorded the services they had been promised and assigned someone to verify they happened.

Min came to the depot that evening with dinner. She found him checking the dispatch.

“You know you don’t have to be the last one out of every building.”

He looked up. “This is the bridge.”

“I know what it is.”

He wanted to tell her that the work mattered. She had never said otherwise. She was asking why mattering always seemed to require his absence.

He handed the final check to his deputy and went home with her.

On the way, she told him about a course she wanted to take. He listened without offering a more practical alternative.

* * *

At the receiving district, the reserve ran low before the bridge reopened. Maya visited on the fifth day. Njeri came with the local verification team, carrying a notebook whose cover had been repaired with tape.

They spent the morning following a shipment from the temporary depot to the community kitchen. Every transfer was recorded. At the kitchen, the cook showed them the smaller portions she had begun serving.

“The record says everybody received a meal,” she said.

Maya looked at the bowls.

“They did.”

“Then the record needs another question.”

Njeri wrote it down. Maya asked about children, people with medical needs, and workers whose shifts made them miss the serving hours. The cook had answers for most of it because she had been solving those problems without waiting for the form to recognize them.

In the afternoon, Maya met the district official who had refused to release a reserve to a neighboring area. He was not hiding a mountain of food. His own projection showed a possible shortage within weeks, and he had been elected by the people who would face it.

“You want me to trust the next delivery,” he said. “Look at the bridge.”

She could not argue that the delivery system deserved unquestioning trust. She could argue that the neighboring shortage was already happening.

The dispute went through the emergency court process. The order required a partial release, independent verification, and a revised replenishment priority. The official complied and publicly criticized the decision. Both actions remained lawful. Maya found the criticism exhausting and necessary.

That night, at the guest room where she was staying, she called Amara.

“The annual total still looks adequate,” she said.

“I know.”

“I hate the annual total.”

“It answers a question. Not all the questions.”

Maya lay back on the narrow bed. “I taught you that.”

“Several times.”

They laughed, then were quiet.

“How’s Grandfather?”

“Tired. He asked when you’re coming.”

“This weekend.”

Amara did not say if work allows. Maya was grateful.

* * *

The bridge reopened on the ninth day. The first train crossed slowly while the inspection team watched. Jun received a photograph of the repaired support and forwarded it to the residential tower coordinator with the lifting assembly’s return schedule.

The food arrived. Some had spoiled. Some had been saved by processing that changed its intended use. The receiving kitchens adapted again.

At the review, Elena asked whether the system had succeeded or failed.

Maya put the losses and the recovered deliveries on the same page.

“Both is the answer that gives us work to do.”

Elena studied the page. “Don’t let that become a way of avoiding a judgment.”

“Then judge the promises separately. We kept some. We broke some. We can say which.”

The final report did. It named the missed inspection connection, the food lost, the smaller meals, the alternative routes, and the local workers who had prevented a worse outcome. It also changed how load restrictions reached the distribution plan.

Maya sent a copy to the cook. The cook replied that the new portion field was useful, and that she still needed a replacement freezer seal.

Maya forwarded that request to the responsible desk and checked, a week later, that it had arrived.

Maya spent much of the year correcting the map.

There was no day when nations disappeared. Elections continued.

Governments blamed one another for shortages, and sometimes they were right. A shared danger had not made competence equally distributed or power harmless.

The emergency authority expired on a day when the food desk was busy enough to make Elena regard expiration as a personal insult.

Three routes needed changes. A regional dispute had stalled a fourth. Under the temporary rules, her office could have issued a limited redirection and resolved the objections afterward. Under the returning framework, she needed a different authorization path.

She had argued for an extension and lost. She considered asking for a narrow exception, then read the criteria and saw that the current delays did not meet them. They were serious, ordinary problems of the kind the new system was supposed to handle.

Marcus came to the final review in civilian clothes. Elena asked whether he had chosen them to make a point. He said his other clothes were being cleaned. She did not believe him and did not need to.

The review showed that the emergency authority had accelerated many deliveries and created costs that had not always been recorded promptly. Independent receipts had caught errors and consumed staff time. Some holds had prevented harm. Others had been overly cautious.

The evidence did not support a simple verdict of success or failure.

Elena argued that the capacity to act quickly should be retained in a form that could be activated under defined conditions. The civilian panel agreed. It rejected a provision allowing the issuing office to decide alone that those conditions existed.

She signed the final transfer with visible irritation. Then she called the new routing lead and worked through the three changes.

The first took longer than it would have the day before. The second revealed a conflict her old process might have missed. The third proceeded normally. She recorded all three because selective evidence would make the next argument worse.

* * *

Wanjiru’s cooperative held its own review after the first difficult winter.

The meeting began with yields and ended with labor. The smaller installations had avoided some shared failures, but they had required more local effort than the central summaries suggested. Skilled people had worked long hours repairing fittings, adjusting growing conditions, and explaining changes to households already tired of adapting.

Maya attended without presenting first. She listened while members described which parts of the program had helped and which had moved work onto people least able to refuse it.

One woman said the new systems had made her more capable and less rested. She did not want to return to the old arrangement. She wanted the present one to pay for the competence it depended on.

The proposed budget increase would reduce funds available for another expansion. Maya showed the tradeoff. Wanjiru asked whether an expansion without enough maintenance labor deserved to be counted as capacity at all.

Maya smiled tiredly. “You know I’ve used that argument.”

“Then don’t make me start from the beginning.”

They revised the plan. The expansion became smaller. The staffing became more realistic. The output forecast fell slightly and the expected reliability improved. No single number captured whether that was better. The cooperative voted after seeing both.

At the meal afterward, Zawadi asked Maya whether she still cooked the difficult crop at home. Maya admitted she had become lazy and used an easier variety when available.

“Good,” Zawadi said. “Choice was the point.”

* * *

Jun’s first annual review at the municipal depot contained a criticism he did not expect.

His technical work was strong. His records were useful. He had improved the handover between teams. He also had a habit of taking over when a younger worker hesitated, which made the immediate task faster and the next one more dependent on him.

He wanted to explain the pressure of the schedule. His supervisor knew the schedule. He wanted to explain that he had trained people for years. The criticism did not deny that. It described what he was doing now.

He took the report home and showed Min.

She read it with an expression he recognized from difficult family conversations.

“Don’t enjoy it too much,” he said.

“I’m trying.”

The next week, he let a junior technician lead a pump repair. The work took longer. Jun asked questions instead of reaching for the tool. Twice, he had to put his own hands in his pockets to make the choice visible to himself.

The technician found a fault Jun had not yet seen. Jun felt pride and irritation arrive together. He congratulated her and kept the irritation for himself until it passed.

That evening, the Rill feed missed a scheduled transmission. He checked the expected power curve and waited for the next opportunity.

The station answered later, weakly. Its remaining life was nearly spent.

He sent the packet to the archive team and closed the display. Min had asked him to come to a small performance by a friend. He went.

The music was uneven. He applauded anyway, and for once did not check the station during the interval.

* * *

The final Rill transmission arrived several weeks later.

At first, nobody called it final. A missed signal was not proof of permanent loss. The team attempted the scheduled contacts, checked the receiving path, and compared the expected power state. Eventually the archive marked the station unresponsive with no planned recovery.

Jun received the notice at work. He finished making the machine in front of him safe, handed the task to his deputy, and went outside.

There was traffic, wind, and a person eating lunch on a bench. He stood beside the building and cried without much warning. He had left the place long ago in every practical sense. The instruments had let some part of him continue visiting.

Orla called later. They spoke about the valve, the abandoned lights, and the restaurant counter. She asked whether he still had the mural photographs. He did. She had copies too.

“Good,” she said.

He sent the photographs to the resettlement group with an invitation to share their own records. The response became larger than he expected: meals, birthdays, broken equipment, jokes, a short recording of a child reading aloud. The archive accepted the collection with the technical data.

Rill had been a place where people lived. Its final account would show more than the moment it stopped reporting.

* * *

Elena continued working under the new rules. The next review would judge them by the deliveries they enabled and the mistakes they exposed.

The name Worldship entered official use gradually. It appeared first on training certificates, then job descriptions. Engineers who had once specialized in separate systems began learning enough about neighboring systems to recognize when their success created someone else’s failure.

Maya helped design the course. She included the report from the lunar factory and the withheld food model. Amara gave permission for her correspondence to be used, with personal details removed.

“Including the part where you were wrong?” Maya asked.

“Especially that part. They can find the speeches elsewhere.”

Her grandfather’s decline did not begin with a diagnosis that divided life neatly into before and after.

He stopped using the far bed in the garden. Then he stopped carrying the watering can full. Then a neighbor began appearing with meals on days when he had always insisted on cooking for himself.

Maya noticed each change and explained it separately. The weather. A sore knee. An unusually busy week. Amara noticed too. For once, their shared ability to analyze did not make them quicker to accept the pattern.

When he fell in the kitchen, he called the neighbor first.

“You were working,” he told Maya afterward.

“I can stop.”

“I know. That’s why I didn’t call.”

She was angry with him for two days, then realized he had given her an accurate account of the problem. Every call had become an event that rearranged her life. He wanted help that did not make him feel like an emergency.

They agreed on regular visits and a local care arrangement he could argue with in person. He rejected the first proposed schedule, accepted the second, and then complained when the caregiver took a day off.

“You told her you didn’t need anyone,” Amara said.

“That’s different from saying she should disappear.”

Amara recognized the family resemblance and chose not to point it out.

* * *

Leila’s messages became part of the visiting routine.

Maya would bring the latest recording, and her grandfather would sit in the visitor chair while the garden sounds of one world mixed with the ventilation of another. Leila showed him the plants grown from the cleared seed line. He criticized their spacing, then asked her to turn the camera so he could see the leaves underneath.

The reply could not arrive during the visit. He dictated his questions anyway, slowly, with pauses while Maya wrote them down.

“Don’t make me sound polite if I wasn’t.”

“I never do.”

“You do sometimes.”

She sent his exact words. Leila answered the questions, included better images, and wrote that one criticism was wrong because the vessel’s layout imposed a constraint he had not seen. He listened to the explanation twice and conceded the point with visible reluctance.

Maya recorded the concession before he could improve it.

On a later visit, he asked whether Leila was lonely.

“Sometimes.”

“And you?”

“Sometimes.”

“That’s allowed even when you’ve chosen well.”

Maya looked at him. “Did I?”

He moved his hand over the arm of the chair. “I don’t know how to grade your life. I know you keep living it.”

It was less reassuring than she wanted and more useful than reassurance would have been.

* * *

The last argument between Amara and her father concerned a tomato.

He said the new variety had no flavor. She said he had praised the same variety a month earlier. He said that was no reason to defend this one. She cut another slice and tasted it. He was right.

“Too much water,” he said.

“Possibly.”

“Don’t do scientist possibly at me. Taste it.”

She tasted another piece. “Too much water.”

He looked satisfied. The victory exhausted him enough that he closed his eyes.

Amara sat beside him while the afternoon light moved across the floor. His hands had become thin, the veins raised beneath the skin. She remembered those hands repairing a school bag, tying a plant to a stake, lifting Maya when Maya was small enough to fit against his chest.

“I missed things,” she said.

He opened his eyes. “Yes.”

She smiled despite herself. “You could let me finish.”

“You could stop asking me to say you didn’t.”

For a while she could not speak.

Then he reached for her hand.

“You’re here now.”

She stayed until the caregiver arrived. There was work waiting when she returned home. Some of it had been handled by other people.

Some would wait until morning.

* * *

In his final week, Maya brought the coastal farm photograph and a recording from Leila. He studied the photograph longer than the recording. Plants in soil remained more interesting to him than plants in a vessel, even after he had learned to respect the latter.

He asked Maya to put the seed records in order. She expected to find a system known only to him. Instead the packets were labeled carefully, with dates, trial conditions, and notes in several hands. Leila’s earlier request for better records had become a household habit.

One envelope contained seeds marked NOT FOR PLANTING—FAILED TRIAL. Maya asked why he had kept them.

“So you don’t fall in love with the same promise twice.”

She put the envelope with the others.

He died early on a Tuesday, while the caregiver was making tea in the next room.

Maya was not there. She had been there the four nights before and had gone home to sleep because the caregiver told her to, and she would do the arithmetic of those four nights for years without once getting it to come out differently.

The next reply from Ithaca arrived two days later. She opened it after the family had been notified. Leila had filmed the underside of the leaves as requested and included a cheerful complaint about a pump.

The message still expected him to answer.

She watched it to the end. Then she watched it again, and a third time, and could not have said what she was looking for. When she finally closed it she sat in the kitchen with her coat still on until the room was cold enough to notice.

Later, when she could, she sent Leila a recording from the garden and told her what had happened. There was nothing either of them could do to make the timing kind. The delay meant that grief would arrive aboard Ithaca after it had already changed the house on Earth.

People told her he had been ninety-one. They said it kindly, and it was true, and Maya discovered she had no use for it whatsoever. She kept saying thank you.

After the funeral, they buried a portion of his ashes beneath the garden, following his instructions. Most of the mourners stayed to repair a leaking section of roof because the short rains were three weeks out and he would have been angry if they left it.

Amara and Maya washed mud from the tools together.

“He asked whether we were still on the same Earth,” Maya said.

“What did you tell him?”

“I said it depended what he meant.”

Amara smiled.

“He would have enjoyed that.”

“He pointed at the garden. Soil. Rain. Tomatoes. He said that seemed sufficient.”

They stood listening to water drip into the repaired gutter.

“Do you think we saved everyone we could?” Maya asked.

Amara took time before answering.

“No.”

It hurt less than a comforting answer would have.

“Do you think we can do better?”

Her mother set the clean spade against the wall.

“We have to keep finding out.”

CHAPTER TWENTY

Children of Motion

HOMEWORK SHEET, CLASS 4

Draw a constellation that will still look right when you are old.
(Note from your teacher: this is harder than it sounds. Do your best.)

EARTH MISSION ELAPSED TIME: YEAR 2, DAY 340

The training course began with a photograph of an empty warehouse.

Maya asked the first group of students what they knew from the image. Several said that a shipment had failed to arrive. One said the date might be false. Another asked who had taken the photograph and what they had been trying to prove.

“Good,” Maya said. “Now ask the warehouse supervisor.”

Celso joined remotely. He had agreed to teach one session and had refused to appear in promotional material for the course. He described the records that had told him his empty bay was full, and the number of times he had reported the contradiction before someone called.

A student asked whether the system had been repaired.

“Parts of it,” he said. “That’s why you’re here.”

The next session used the lunar accident. Pavel supplied the technical record. Saira described the decisions that had shortened the procedure. Maya described what she had seen and what she had only believed she remembered.

They did not ask the students to admire anyone. They asked them to identify the point at which they would have needed more information, and then to explain what they would do while waiting for it.

The exercise produced arguments. Maya liked that. She liked it less when one group found a weakness in the current regional review process that she had helped design.

They showed her the gap. She began explaining the reason for the existing rule, heard herself, and stopped.

“Write the alternative,” she said. “Include what it could break.”

* * *

After her grandfather’s house was settled, Maya and Amara disagreed about the garden.

Amara wanted to preserve it as it had been. Maya wanted to give part of the space to the neighborhood trial program and let people change what no longer worked. Both claimed to be honoring him.

Eventually they recognized that this was not a useful way to settle a practical question.

They invited the neighbors who had maintained it during his final months.

The neighbors wanted a wider path, a lower work surface, and permission to remove one bed that produced little and required too much effort. They also wanted the visitor chairs, because those had been a good idea whatever he had claimed about them.

Amara agreed to the path before she could talk herself out of it.

Maya kept one section for the seed line he had tended longest. They put the records in a weatherproof cabinet and gave several people access.

The first community workday was noisy. Someone brought a radio. Someone else objected to the music. A child dropped a packet and scattered seeds on the new path. Maya began collecting them, then laughed when she realized she could not reliably distinguish them from grit.

Amara watched from the old chair.

“He would have complained,” she said.

“About which part?”

“He’d have made a list.”

They left room in the cabinet for it.

* * *

Leila’s reply to the news of his death arrived with no attempt to make the distance smaller than it was.

She said she wished she had been there. She described the things she remembered: his suspicion of automatic watering, his eventual admission about the ventilation, the way he had asked precise questions about a place he did not expect to visit. She had cried in the growing compartment and then had to explain to a colleague why a plant could remind her of a person the colleague had never met.

Maya listened twice. Then she recorded a reply while walking through the changed garden.

She showed the wider path, the cabinet, and the visitor chairs. She told Leila about the argument with Amara and the neighbors’ unromantic solution. At the end she said, “I still want to tell you things first.”

When Leila answered, she said, “You still can. I just won’t always be the first to hear them.”

The distinction hurt. It also described a life they had begun learning how to sustain.

They no longer spoke as though a decision made before departure had settled every question that would follow. They could name love without making it a contract against change. They

could ask about new friends without treating them as evidence of betrayal. Some conversations went badly and required another exchange to repair.

Maya began inviting people to the flat again. Njeri came to dinner with two friends and pointed out that the repaired shelf was still slightly uneven. Maya told her that was a matter of interpretation. They stayed late, arguing about a film none of them had liked in the same way.

After they left, the room felt occupied even in their absence. Maya washed the dishes and sent Leila a photograph of the table.

No crisis today, she wrote. We complained about art.

* * *

The charter negotiations had continued through shortages, funerals, and ordinary elections.

The final version disappointed people who wanted a single government to end all conflicts. It also disappointed people who wanted the emergency cooperation to disappear as soon as the most dramatic danger passed. It established duties to share observations and limits on how shared systems could be commanded. It left many political questions to institutions that already existed and would continue arguing.

Elena objected to one reporting requirement because it duplicated a record her teams already produced. Marcus agreed with her, to the visible surprise of several delegates. They changed the clause so the same evidence could serve both purposes without giving the issuing office power to suppress it.

Maya argued for regional authority to resolve conflicts among crop, power, and transport schedules. A delegate asked whether that would make the global picture less consistent.

“The picture is already inconsistent with the places,” she said.

“This lets the places correct it.”

The clause passed with a requirement to publish the changes.

Wanjiru read it and asked whether the funding matched the responsibility. The first draft did not. The second did, though less generously than she wanted.

Anika sent comments from *Ithaca*. The delay meant that the discussion could not be conducted as an uninterrupted meeting. The committee left the relevant sections open long enough for the arks to answer. Some members complained that this slowed the process.

“Distance doesn’t make them stop being affected,” Maya said.

She thought of Leila waiting for a reply that would contain news of a death, and resisted the urge to turn the thought into a speech.

When the text was ready, it contained objections beside several clauses. Maya liked the visible disagreements more than she had expected. They gave future readers something to examine besides the confidence of the people who had signed.

Aster was born three days after closest approach, in a hospital whose backup generator had been serviced by a mechanic wearing an astronaut sticker on his lunch container.

The local news called her a child of the new voyage. Her parents asked them to stop filming before her mother had finished eating.

By the time Aster was two, she had learned that adults looked at screens before answering questions about the weather. She found this neither strange nor impressive. Adults looked at screens before answering most questions.

The hospital garden had been designed by a committee and improved by the people who used it.

The committee provided irrigation, accessible paths, and a planting plan. The staff added places to sit where the shade actually fell. A patient suggested moving the fragrant plants closer to the entrance. A cleaner pointed out that one bed made it difficult to reach a drain. The gardener changed it before the first official inspection and then had to explain why the completed work differed from the drawing.

Maya came to review a small connection between the garden and the hospital’s food program. She expected a tour and found a workday. Someone handed her gloves. She asked whether she was qualified. The gardener said the weeds had not requested credentials.

Aster stood beside a low bed with her father, holding a small watering container. More water reached the path than the soil. Her father tried to guide her hand, then accepted that the lesson would take time.

Maya worked nearby. The child watched her remove a weed and asked why that plant had to go while another remained. Maya began an explanation about competition, then realized the gardener was listening with amusement.

“Because we chose to grow these here,” she said more simply.

Aster considered the removed plant. “Why that one?”

“It was crowding this one.”

The child looked dissatisfied and returned to watering. Maya kept the weed in her hand for a moment before putting it in the designated container. Even a simple answer could contain a question worth remembering.

* * *

Aster next wanted to know whether the small green objects on a plant were tomatoes. Maya said they would be.

“When?”

Maya looked toward the gardener.

The gardener laughed at her and answered the child.

Before the charter reading, the communications team asked Amara to rehearse her speech on the garden’s new path.

She read the prepared text. It described survival, shared work, and a future beyond the present generation. It was accurate in its claims and too smooth in its movement from fear to hope. Maya listened from behind the camera and said nothing until her mother asked.

“It sounds finished.”

“The speech?”

“The history.”

Amara looked at the pages. She had already removed several grander sentences. She could feel the remaining urge to give the audience an ending worthy of what they had endured.

“People deserve relief.”

“They’ve had relief. They also have questions.”

Amara sat in her father’s chair. For a moment the absence was so strong that she expected him to complain about her sitting there with papers instead of helping. She read the speech again and crossed out the paragraph that treated their success as proof that the system had become wise.

Maya watched without offering a replacement immediately. Her mother had to find her own words if they were to mean anything beyond compliance with another review.

Eventually Amara asked whether the names of people harmed by the failed preparations belonged in the spoken text. Maya said some had agreed to public use and some had not. They could acknowledge the losses without treating people as material for the speech.

Amara nodded and revised again.

* * *

The morning of the reading, a problem with the camera power delayed the crew.

Maya found the correct extension cable in the cupboard and nearly laughed at the thought that this would become the detail she remembered. A ceremony about centuries depended, at that moment, on a cable long enough to reach a garden.

Neighbors arrived with chairs. Njeri helped arrange them so the path remained clear. Wanjiru brought a small tray of plants as a gift and immediately began discussing where they

should go. The gardener from the hospital attended with a colleague. Aster's parents had chosen not to bring her to the formal reading; she would have found it long and preferred being in the garden to listening to people talk about it.

Messages from the arks had been prepared in advance. Anika's remarks were short and practical. Leila appeared in a separate community recording, standing among people whose faces Maya now knew from photographs and stories. The distance was present in the fact that the exchange could not respond to the morning as it unfolded.

Maya watched Leila smile at something off camera. She felt the familiar ache and, alongside it, pleasure in recognizing the room as part of Leila's life rather than merely the place keeping her away.

Ruth stood beside her. They watched together without speaking.

* * *

That afternoon, the first Worldship charter was read in public. It established no ownership of future planets. It required shared observations, published uncertainty, independent audits, and periodic renewal of the authorities created during the emergency. Several clauses remained disputed. Those were printed with the objections attached.

Amara spoke from the garden in Nairobi.

The camera crew asked whether she wanted the old house out of frame. She asked them to leave it. A repaired roof seam was visible above her shoulder.

"When I first showed you the approaching black hole," she said, "you asked whether your children would survive. I could not answer. I know what that cost you."

She paused, looking at the lens rather than the words beside it.

"We have passed the closest approach. Earth remains with the Sun. The work that let us prepare will also have to let us live here. Some of our preparations failed. Some people paid for mistakes they did not make. Their names belong in the account of this voyage."

Maya stood behind the camera. She knew which paragraph her mother had removed that morning, the one offering a clean conclusion to a history that did not have one.

"Our Sun is traveling toward the future neighborhood of Alpha Centauri. None of us here will see that passage. We can give the people who do see it better instruments, useful records, and systems they are allowed to question. We cannot choose every answer for them."

Amara looked briefly toward the place where her father had kept his visitor chair.

"For most of my life I thought of astronauts as people who left Earth. I understood the motions of this planet long before I understood what it meant to live aboard it."

She let the prepared text disappear.

“We are eight billion astronauts. We have work to do.”

The speech ended without music. The crew began packing its equipment. Someone asked where the extension cable belonged, and Maya showed them the cupboard.

After the speech, people stayed longer than the crew.

The formal text had been read. The disputed clauses remained disputed. The garden still needed work. Wanjiru found a place for the new plants and asked Maya to bring another tool. Njeri helped a neighbor photograph the revised path for a relative who had not been able to come.

Amara took off the microphone and stood for a moment without knowing what to do next. Her public role supplied clear instructions up to the end of the recording. The ordinary afternoon required her to choose.

She asked where she could help.

Someone handed her a trowel.

Maya joined her at the bed. They worked without discussing whether the speech had been well received. The first reactions were arriving, but there would be time to read them. The soil needed attention now, and the work was pleasant enough that neither had to justify remaining with it.

“I used to think I’d get you back after the encounter,” Maya said.

Amara stopped with the trowel in the soil.

“I know.”

“Then I thought that was childish.”

“It wasn’t childish to want me.”

Maya looked at her mother. There was no calculation to make the years come out differently. There was a person beside her, older than the one she had waited for and still capable of choosing where to spend an afternoon.

“I don’t need you to stop working.”

“I need to stop using work as the answer to every question.”

They returned to planting. The conversation did not require another promise. Maya had learned to value the evidence of a repeated act more than the elegance of an intention.

* * *

They finished the next row together. The seed packet bore the neighborhood trial number in her grandfather’s handwriting. Maya placed it inside a dry jar when they finished.

Above the roof, the Sun moved toward evening. It would rise again over a world carrying its oceans, its graves, and the people still trying to care for one another.

* * *

Earlier that week, Salma had invited Maya to her class to answer questions about the sky. Maya prepared for questions about the encounter. The children wanted to talk about constellations.

At three thousand kilometers per second, the Sun crossed about a third of a light-year every thirty years. Distant stars would not seem to move for thousands of years. The nearest ones would shift within a single lifetime. Alpha Centauri was already measurably brighter than it had been at Maya's birth, and would brighten further. Sirius would slide away from the rest of its constellation. The familiar shapes, which had looked fixed for all of recorded history, would begin to bend.

Salma's homework had asked them to draw a constellation that would still look right when they were old.

"There aren't any near ones," a boy said. "I checked."

"Then use far ones," said a girl. She had drawn a pattern using only stars more than a thousand light-years away. It was faint, nearly invisible from the city, and would look the same for ten thousand years. She had named it the Patient Goat.

Another boy had drawn the Great Dog with a dotted line showing where Sirius would be when he was sixty. The bright star had drifted off its nose.

"The dog's going to lose his nose," he explained.

Maya asked whether that made him sad.

He considered it seriously. "No. The dog stays where it is. We're the ones going somewhere."

Salma caught Maya's eye over the children's heads and said nothing. On the way out, she gave Maya the Patient Goat to take home.

* * *

That evening, Maya went home to the flat and opened the window.

The street carried voices, a vehicle reversing, and music from somewhere she could not identify. The shelf held books and the photographs she had chosen to keep. The cupboard closed properly.

Nothing in the room made her life complete, and nothing required it to be on hold.

She recorded a message for Leila about the ceremony, the cable, and the weed Aster had defended. She included the parts that had made her laugh. Then she read a few pages of the book she had taken to the lake.

Before sleeping, she placed her shoes beside the bed. This time she did it because she wanted to find them easily in the morning. The habit no longer had to be a private alarm system.

There would be work tomorrow. There would also be breakfast, a call to a friend, and perhaps rain. She let the day end without checking whether the world had earned permission for it.

EPILOGUE

Another Moving Star

MAYA OKAFOR, NOTEBOOK, LAST PAGE IN USE

Check the rejected ones first.
(Mom was right about that. Don’t tell her.)

EARTH MISSION ELAPSED TIME: YEAR 3, DAY 30

The anomaly arrived in a set of rejected measurements.

Three years after closest approach, Maya was reviewing the independent navigation archive when a faint star acquired an impossible-looking history.

The current observations were straightforward enough to be checked. The difficult part was matching them to older detections.

Previous catalogs had assigned some of those detections to different sources. Their positions changed too quickly to fit the assumptions used in the matching process.

Pilot proposed a reconstruction and marked it for human review.

Maya read the reconstruction once. Then she stood up so suddenly that her chair rolled into the wall, and said a word she would never have put in a report, loudly, to an empty flat.

She sat down again. She checked the time labels. She began writing the paragraph of reasons it was probably wrong.

But for a few seconds, before any of that, she had simply been delighted, and she let herself remember it.

Maya called Luis.

“I’m not on shift,” he said.

“Neither am I.”

“That used to mean something.”

She sent him the observations, including the discarded ones. He called back before she finished making breakfast.

“Who corrected for our motion?”

“Two teams. Different methods. The independent ark reduction agrees within its errors.”

“Distance?”

“About eighty-three light-years in the current reconstruction.”

The object’s inferred space velocity was approximately 0.19c. A star moving that fast was extraordinary, but not sufficient evidence of anything artificial. Natural black-hole encounters could produce extreme stellar velocities under some conditions.

Maya had included that objection in the first paragraph of her note.

Luis reached the next page.

“These bends.”

“That’s why I called.”

The reconstructed track contained two changes of direction across the archival baseline. An unseen gravitational encounter remained a possible explanation. So did a subtle problem in the observations. The first searches had not found the mass distributions required by the simplest natural models.

No one had established intention.

Someone needed to keep looking.

“We’re seeing old light,” Luis said.

“Yes.”

“Whatever happened there happened decades before we found our black hole.”

Maya opened the time labels. She had checked them already. She checked them again.

She imagined an operations room around another star, if there was such a room. People, if they were people, watching uncertain measurements while their world moved toward a future they could not see. Or she imagined a natural encounter so unfamiliar that it only resembled a choice.

Both possibilities deserved better than a headline.

“Send it to Amara,” Luis said.

“Already did.”

“And Anika.”

“Her packet is on its way.”

The distances between the arks and Earth had grown. Their conversation would now unfold across days. Independence had become something measurable in light travel time, and they continued the exchange anyway.

Maya circulated the candidate track and the reasons it might be wrong, with the observation proposal attached.

The first independent reply did not agree with the reconstructed bends.

A team using a different archive match found a straighter path with a worse fit to two old images. Its lead scientist wrote that the simplest explanation might be an incorrect identification across the historical baseline. Maya put the result beside her own instead of waiting for a more encouraging answer.

Luis called when he saw it.

“Good,” he said.

“You could try sounding disappointed.”

“I’m too old to want an interesting mistake.”

They opened the two images. One had a faint neighboring source that might have shifted the measured position. The other had an uncertain timing record. Neither could simply be discarded without changing the question they were trying to answer.

Amara joined the review from home. She asked which assumptions the two reductions shared. Maya had already listed them, and felt an unreasonable pleasure in sending the page before being asked twice.

They assigned new work. A proposed observation would improve the present motion but could not, by itself, prove what had happened decades earlier. A better distance estimate would change the inferred speed. Additional archival images might remove the bends, strengthen them, or replace the entire reconstruction with a less dramatic one.

The investigation had no natural deadline. That made it vulnerable to being postponed behind problems that did.

Maya requested a modest observing allocation with explicit review points. She did not ask the food program to surrender power for a mystery. The astronomy teams found space within an existing schedule by replacing a lower-priority repeat observation. The displaced team objected, then accepted a later slot.

Even wonder required a calendar.

* * *

Leila’s personal reply arrived separately from *Ithaca*’s technical packet. She had seen the candidate track. She knew enough now to ask whether the bends survived changes in the archive matching, and Maya smiled at the question before reaching the rest of the message.

Yesterday I wanted to ask him what he’d think of this, Leila wrote.

I still do that.

Maya looked toward the jar containing her grandfather’s labeled seed packet. She had moved it from the garden after the community group made working copies of the record. It was no longer needed as the only safe version of anything. She kept it because she wanted it.

I would have explained the star for ten minutes, she wrote back.

Then he’d have asked whether the people got out. Even before we knew if there were people.

She paused, then added a photograph from the garden. The tomatoes were not impressive. The leaves had weather damage, and someone had planted one row too close. She could almost hear his comment.

The next technical exchange would take time. She put the observation request in the queue and closed the garden photograph.

Later, her mother sent one more note.

Start with the clocks.

Maya smiled and opened the timing files.

Outside, a child was calling for someone to come and see the Moon. The Sun carried them onward at three thousand kilometers per second. Far ahead, in light that had left before any of them understood the voyage, another star appeared to have turned.

END OF BOOK ONE

NOTE ON THE SCIENCE

This revision uses a moving, single black hole as a gravitational-assist mechanism. It replaces the original relativistic binary corridor with a more distant encounter and reduces the outgoing speed from roughly one-third of light speed to approximately one percent. Black-hole merger recoil offers a theoretical origin for a rapidly moving black hole, although the particular encounter in this story would be exceptionally improbable.

The illustrative encounter uses a black-hole mass of four million solar masses, an incoming Sun–black-hole relative speed of 3,000 kilometers per second, and a solar closest approach of about 394 astronomical units. In an idealized Newtonian two-body calculation, these parameters produce roughly a 60-degree deflection in the black hole's frame. Since the deflection changes the Sun's velocity by twice the incoming speed times the sine of half the deflection angle, a 60degree turn transfers a speed equal to the incoming relative speed: if the Sun initially rests in the chosen reference frame, it leaves that frame at about 3,000 kilometers per second. Alpha Centauri has its own motion, so its relative speed and future interception require a separate vector calculation.

This calculation treats the Sun as a point mass. It does not demonstrate the survival of Earth, the Moon, or the other planets. The adopted final Earth orbit, the individual planetary outcomes, and the trajectory toward Alpha Centauri are fictional outcomes chosen for the narrative, not results of a completed many-body simulation. A scientific validation would need initial positions and velocities, encounter orientation, planetary interactions, long-term stability calculations, and climate modeling. The late recognition of such a massive nearby object is also a narrative assumption. Seventeen years before closest approach, at 633 astronomical units per year, the object lies roughly ten thousand seven hundred astronomical units away, where its differential pull across Saturn's orbit is already of order half a percent of the Sun's. Modern planetary ephemerides are fitted at the meter level. In reality these residuals would have been an unmissable and long-standing anomaly in solar system dynamics, not a quiet folder awaiting funding.

At the nominal closest approach, an approximate radial tidal acceleration across a one-AU separation is about thirteen percent of the Sun's attraction on Earth. This is a scale comparison, not a survival probability or a disruption criterion. Encounter duration, orbital phase, and orientation strongly influence the accumulated response. The characteristic encounter time is of order a year, close to Earth's own orbital period, which is the least forgiving regime: too fast to

be adiabatic, too slow to be a clean impulse. The outcome would depend severely on Earth's orbital phase at periapsis, a quantity that is effectively random and could not usefully be predicted seventeen years in advance. The survivable band in this novel is therefore better understood as a lottery than as a target. The corresponding tide across the Sun itself is much less important relative to the Sun's own surface gravity. The story therefore makes planetary orbits and habitability the principal danger rather than the Sun being visibly torn apart.

Earth's fictional post-encounter orbit has a semimajor axis of 1.19 AU and an eccentricity of 0.27. With unchanged solar luminosity, its orbit-averaged incident flux is approximately twenty-seven percent below that of a circular one-AU orbit, but the instantaneous flux ranges from about thirty-three percent above the present value at perihelion (0.869 AU) to about fifty-six percent below it at aphelion (1.511 AU). The year lengthens to roughly 474 days. Actual climate consequences are not established by any of these numbers alone; ocean heat capacity, cloud response, ice albedo and atmospheric circulation would all matter more than the orbital elements. Regional outcomes in the story are fictional, as is the assumption that such an orbit is compatible with continuous agriculture anywhere. With Jupiter and Saturn unbound, the long-term secular stability of the remaining inner planets is left unresolved in the narrative, which is the honest position rather than a hedge. The engineering program is assumed to begin with the orbital industry of the early twenty-second century.

At 3,000 kilometers per second, traversing the present 4.37-light-year distance to Alpha Centauri takes approximately 437 years. The manuscript rounds its prospective encounter to about 440 years and explicitly accounts for the need to target the star system's future position. It does not present Alpha Centauri as a brake capable of capturing the entire Sun. Time dilation at this speed is small, and dramatic naked-eye relativistic sky distortion is not expected.

Daedalus no longer propels the Solar System by internal momentum exchanges. Its decisive interventions concern infrastructure, agriculture, evacuation, observations, and accountable decisions. Radiation and dust protection are local engineering measures, not a universal shield. These changes preserve the worldship premise while avoiding several violations of momentum conservation and scale in the first draft.

Several additions in this revision rest on simple estimates. For a point mass of four million solar masses, the angular Einstein radius for a distant background source is roughly the square root of $4GM/c^2$ divided by the distance. At about 394 AU this is a little over one degree, so at closest approach the region of visibly displaced, stretched and doubled background stars is several Moon-widths across, while the black hole's shadow at the same distance is only a few arcminutes wide. Three years earlier, at more than two thousand AU, the lensed region is about

half a degree across; earlier still it is within reach of binoculars and small telescopes. The story lets this growing ring serve as a visible clock.

A twenty-seven percent reduction in average incident sunlight corresponds to roughly sixty-four watts per square meter of absorbed solar energy averaged over the planet, compared with about 3.7 watts per square meter of forcing from a doubling of carbon dioxide. Without intervention this would very likely push Earth toward severe or runaway glaciation; the story's treatment of ocean thermal inertia, ice-albedo feedback and engineered warming gases is qualitative, and real outcomes would require full climate modeling.

At 3,000 kilometers per second the Sun travels about 0.3 light-years in thirty years. Stars within roughly ten light-years, such as Alpha Centauri and Sirius, can appear to shift by degrees within a human lifetime, depending on their direction; distant stars do not visibly move. Alpha Centauri's apparent brightness rises as the inverse square of its decreasing distance.

The orbit odds quoted in chapter openings are a fictional betting market, not a scientific estimate. That is rather the point.

Scientific background

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